SPCA111N

POSTGRADUATE COURSE

MASTER OF COMPUTER APPLICATIONS

FIRST YEAR - SECOND SEMESTER

PAPER - IX

PRACTICAL-III: OOAD LAB

INSTITUTE OF DISTANCE EDUCATION

UNIVERSITY OF MADRAS

WELCOME

Warm Greetings.

It is with a great pleasure to welcome you as a student of Institute of Distance Education, University of Madras. It is a proud moment for the Institute of Distance education as you are entering into a cafeteria system of learning process as envisaged by the University Grants Commission. Yes, we have framed and introduced as per AICTE Choice Based Credit System(CBCS) in Semester pattern from the academic year 2020-21. You are free to choose courses, as per the Regulations, to attain the target of total number of credits set for each course and also each degree programme. What is a credit? To earn one credit in a semester you have to spend 30 hours of learning process. Each course has a weightage in terms of credits. Credits are assigned by taking into account of its level of subject content. For instance, if one particular course or paper has 4 credits then you have to spend 120 hours of self-learning in a semester. You are advised to plan the strategy to devote hours of self-study in the learning process. You will be assessed periodically by means of tests, assignments and quizzes either in class room or laboratory or field work. In the case of PG (UG), Continuous Internal Assessment for 20(25) percentage and End Semester University Examination for 80 (75) percentage of the maximum score for a course / paper. The theory paper in the end semester examination will bring out your various skills: namely basic knowledge about subject, memory recall, application, analysis, comprehension and descriptive writing. We will always have in mind while training you in conducting experiments, analyzing the performance during laboratory work, and observing the outcomes to bring out the truth from the experiment, and we measure these skills in the end semester examination. You will be guided by well experienced faculty.

I invite you to join the CBCS in Semester System to gain rich knowledge leisurely at your will and wish. Choose the right courses at right times so as to erect your flag of success. We always encourage and enlighten to excel and empower. We are the cross bearers to make you a torch bearer to have a bright future.

With best wishes from mind and heart,

DIRECTOR

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MCA

FIRST YEAR - SECOND SEMESTER

PRACTICAL-III: OOAD LAB

SYLLABUS

Objective of the course: This course gives training to understand the Object-based view of Systems. To develop robust object-based models for Systems. To inculcate necessary skills to handle complexity in software design

Course Outcomes: After successful completion of this course, the students should be able to analyze and model software specifications, Ability to abstract object-based views for generic software systems. Ability to deliver robust software components.

1. Student information system.
2. Stock Maintenance System.
3. Banking system.
4. Online course reservation system.
5. Exam Registration.
6. Employee Management System.
7. Project Tracking System.
8. Library Information System.
9. E-ticketing
10. E-book management system.
11. Recruitment system.
12. Conference Management System.
13. BPO Management System.
14. Credit card processing.
15. Gas Booking System.

MCA

FIRST YEAR - SECOND SEMESTER

PRACTICAL-III: OOAD LAB

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INSTITUTE OF DISTANCE EDUCATION
UNIVERSITY OF MADRAS

CHENNAI - 600 005

RECORD OF PRACTICALS

M.C.A. (First Year)

20____ - 20____

Practical - III

OOAD LAB

NAME :

ENROLMENT NUMBER :

GROUP NO. :

INSTITUTE OF DISTANCE EDUCATION
UNIVERSITY OF MADRAS

CHENNAI - 600 005.

Certified that this is the Bonafide Record of work done by ____________________ with Enrolment Number ______________________ of First Year M.C.A. Degree Course in the Institute of Distance Education, University of Madras during the year ____________ in respect of **OOAD LAB**

Date: **Co-ordinator**

Submitted for First Year M.C.A. Degree Course Practical Examination held on ____________________ at ____

Date: **Examiners**

1. **Name:**

   Signature:

2. **Name:**

   Signature:

LESSON - 1

STOCK MAINTAINENCE SYSTEM

AIM:

To draw the diagrams [use case, class, activity, sequence, collaboration, state-chart, component and deployment] for the Stock maintenance system.

PROJECT DESCRIPTION:

StarUML software is designed for supporting the computerized stock maintenance system. In this system, the customer can place order and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer.

USE CASE DIAGRAM

This diagram will contain the actors, use cases which are given below

- **Actors**: Admin, Supplier, Shopkeeper, Customer.
- **Use case**: Product details, Purchase details, Sales details, Stock details, Purchase the product, Supply the product, Product delivery.

CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBUTES	OPERATIONS
Product Details	Code, Product Name, Opening Stock. Price	Add(), Delete(), Update(), Clear(), Exit().
Purchase Details	Date, Code, Name, Qty, Price	Purchase-Save(), Purchase Delete(). Pedit(), Exit().
Sales Details	Date, Customer Name, Product code, Price, Qty	Add Sales(), Exit().
Stock Details	Date, Date2	View Stock(), Cancel().

ACTIVITY DIAGRAM

This diagram will have the activities as Start point, End point, Decision boxes as given below:

- **Activities**: Login, Manage Stock, Manage Product, Manage Customer. Manage Quality, Manage Bill, Logout.
- **Decision box:** Valid or not

SEQUENCE DIAGRAM:

This diagram consists of the objects, messages and return messages. A Sequence diagram simply depicts interaction between objects in a sequential order.

Object: Customer, Supplier, Admin, Database.

COLLABORATION DIAGRAM:

This depicts the relationships and interactions among software objects. They are used to understand the object architecture within a system rather than the flow of a message as in a sequence diagram.

STATE DIAGRAM:

State Diagram is used to capture the behavior of a software system. UML State machine diagrams can be used to model the behavior of a class, a subsystem, a package, or even an entire system.

COMPONENT AND DEPLOYMENT DIAGRAMS

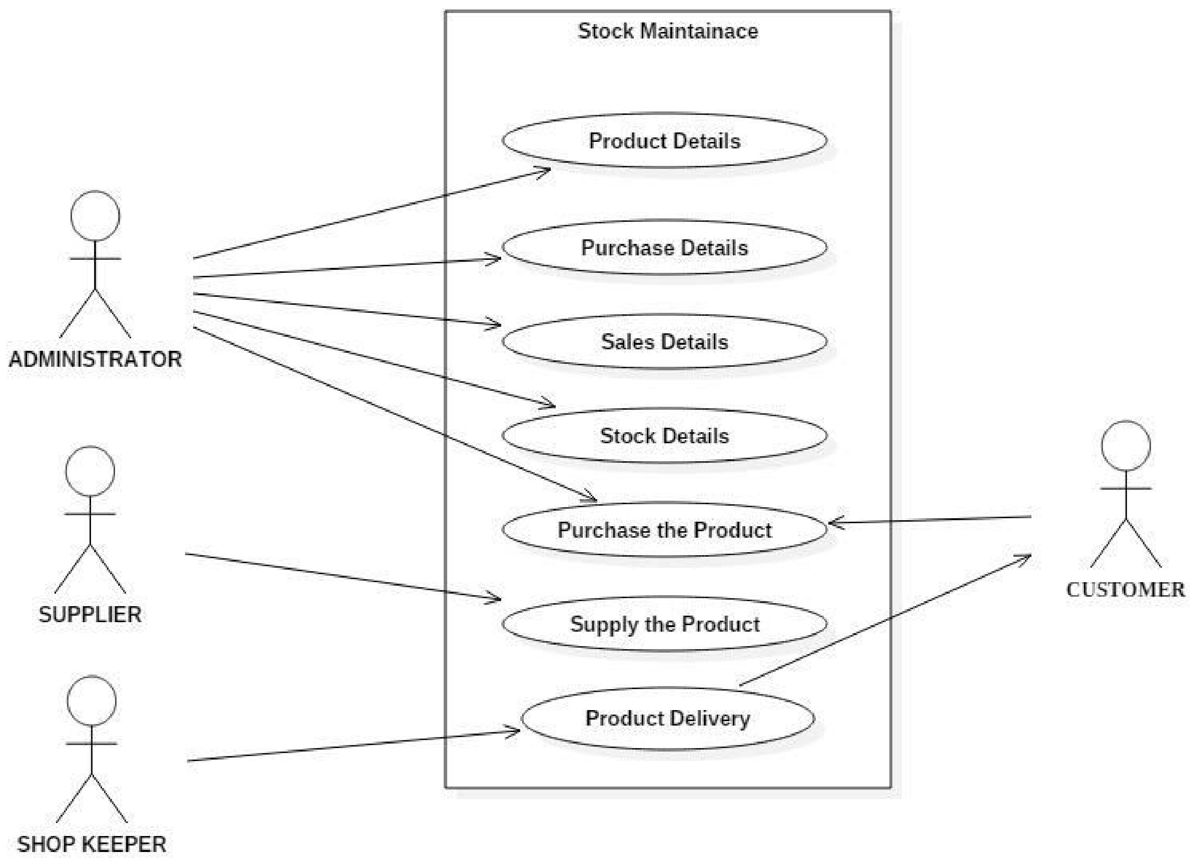
COMPONENT DIAGRAM:

- Component diagrams are used to visualize the organization and relationships among components in a system. These diagrams are also used to make executable systems.

DEPLOYMENT DIAGRAM:

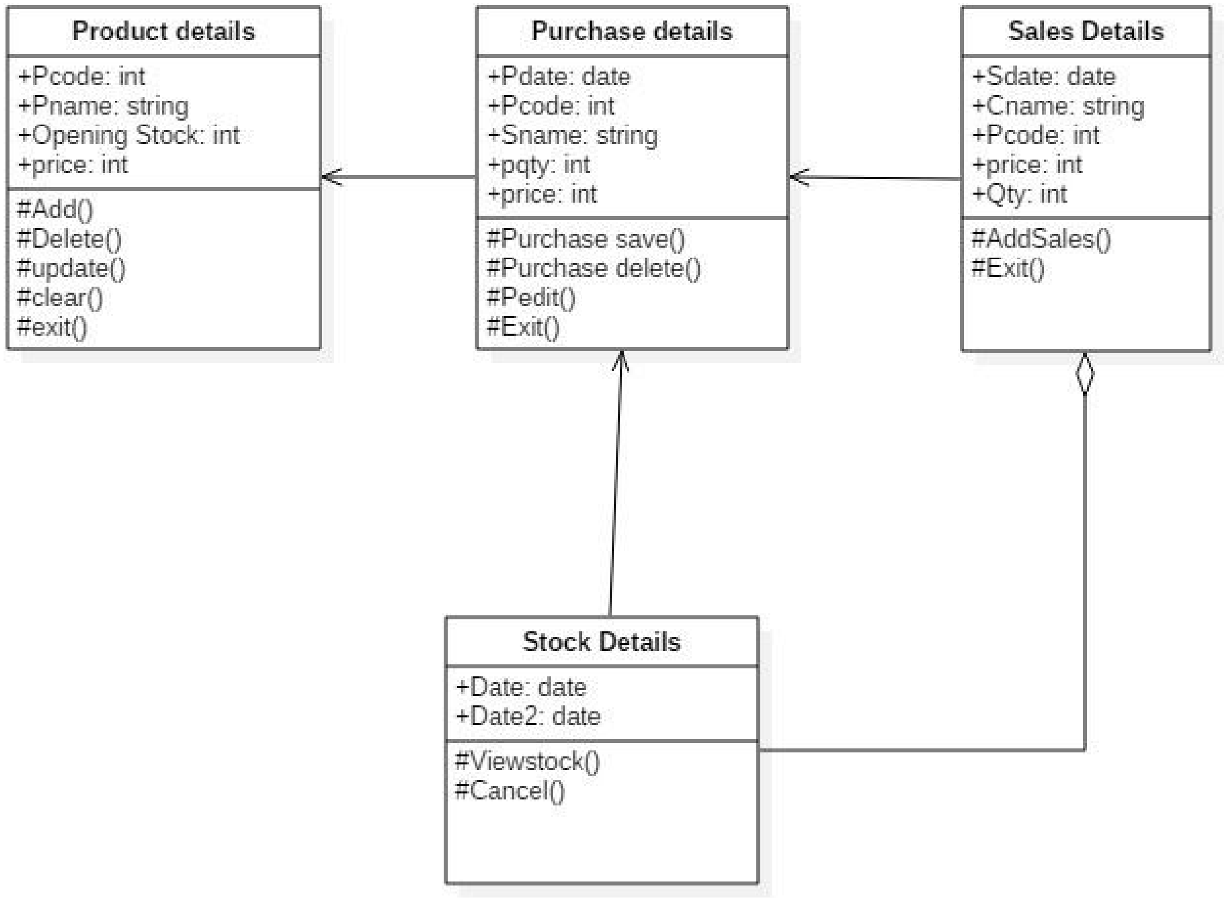
- Deployment diagrams are used to visualize the topology of the physical components of a system, where the software components are deployed.

USE CASE DIAGRAM:

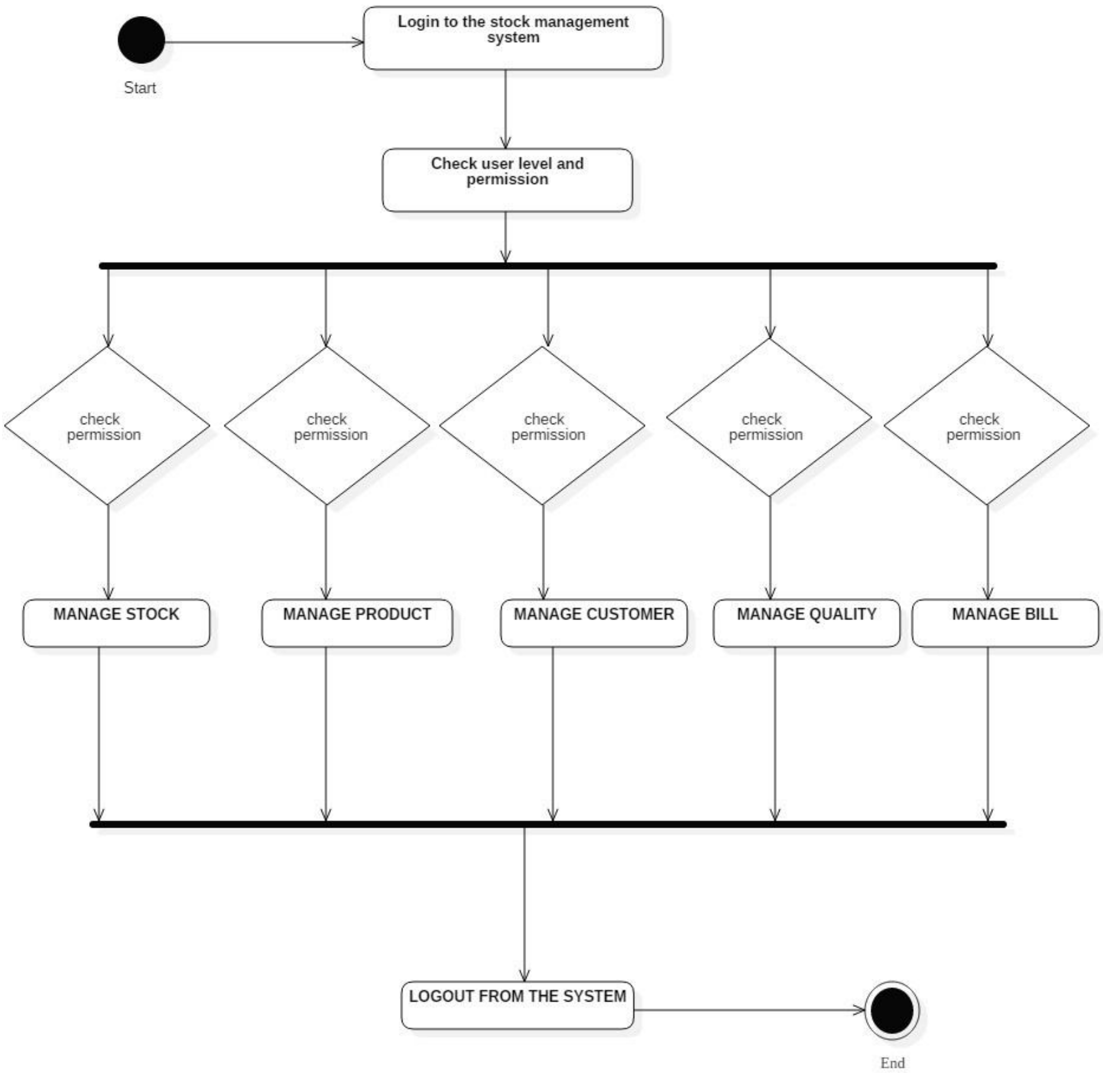


4

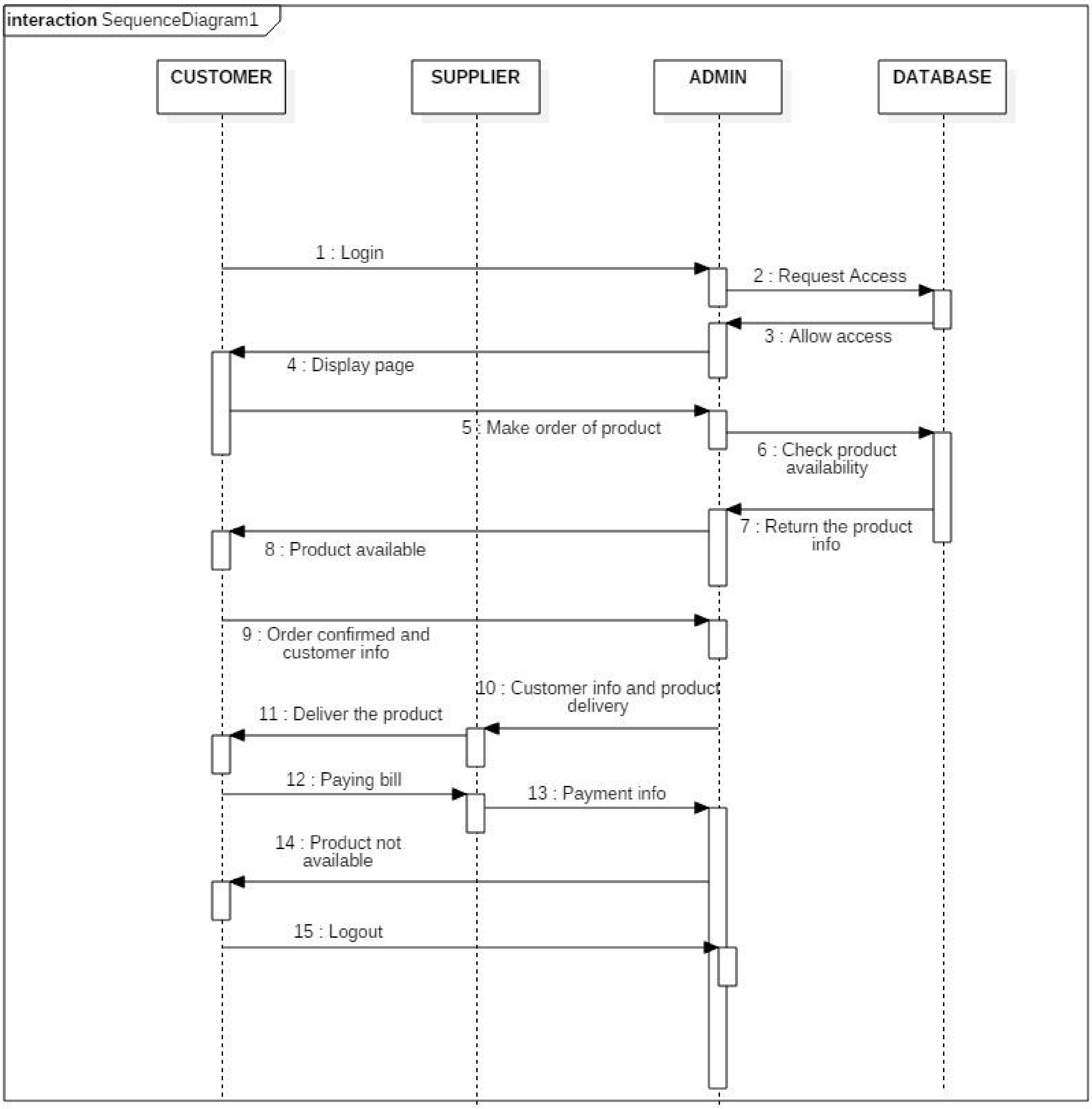
CLASS DIAGRAM:



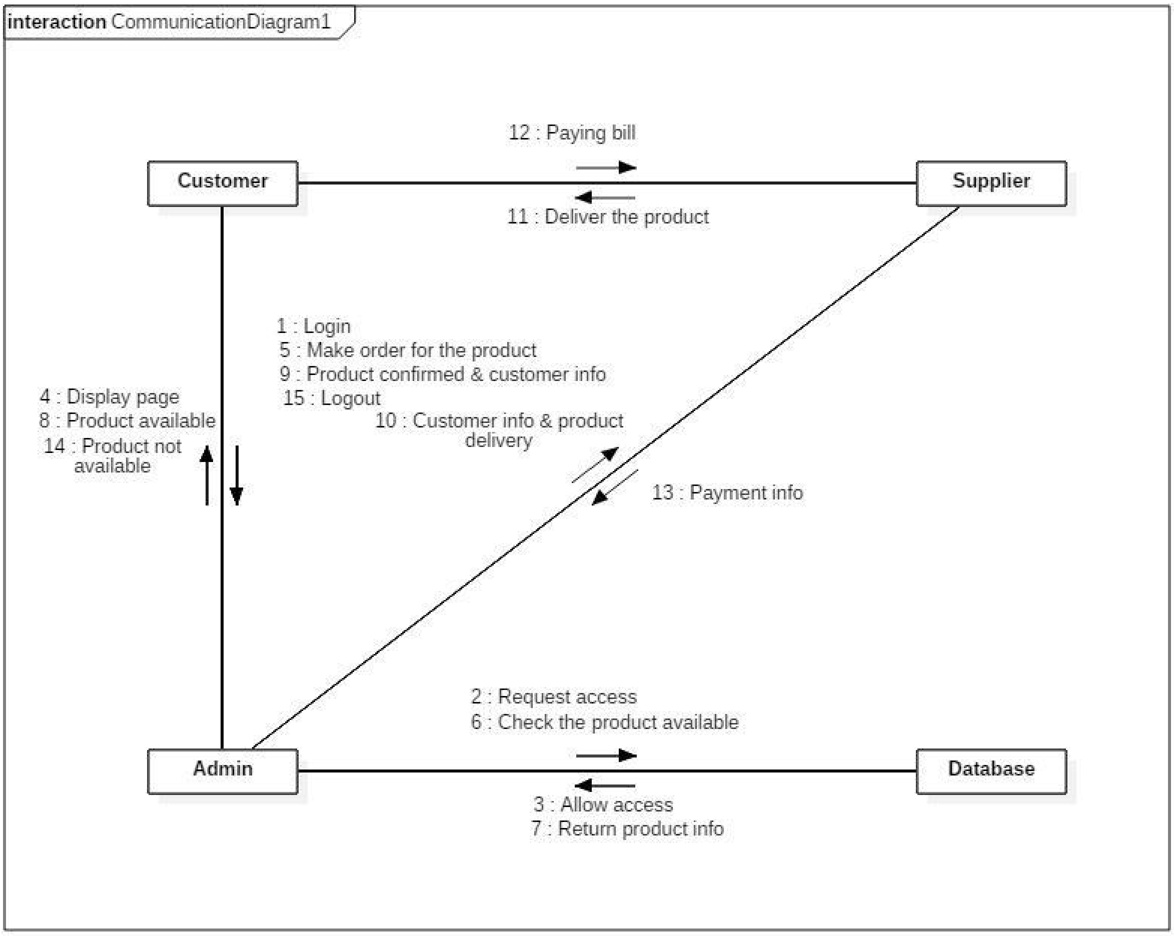
ACTIVITY DIAGRAM:



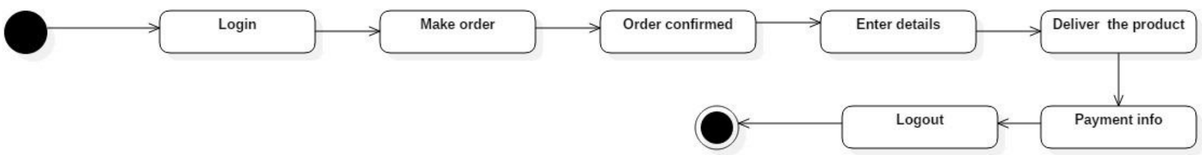
SEQUENCE DIAGRAM:



COLLOBORATION DIAGRAM:



STATE CHART DIAGRAM:



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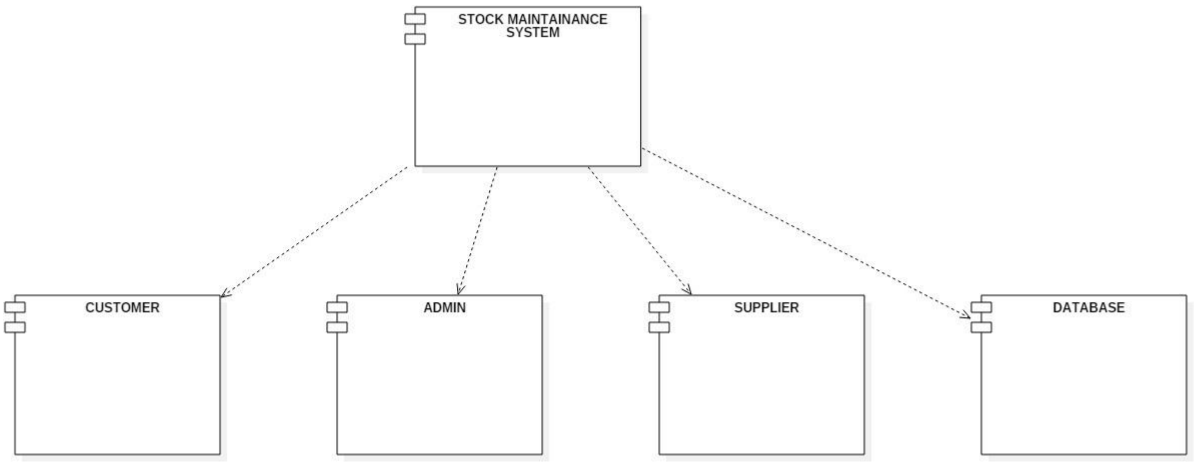
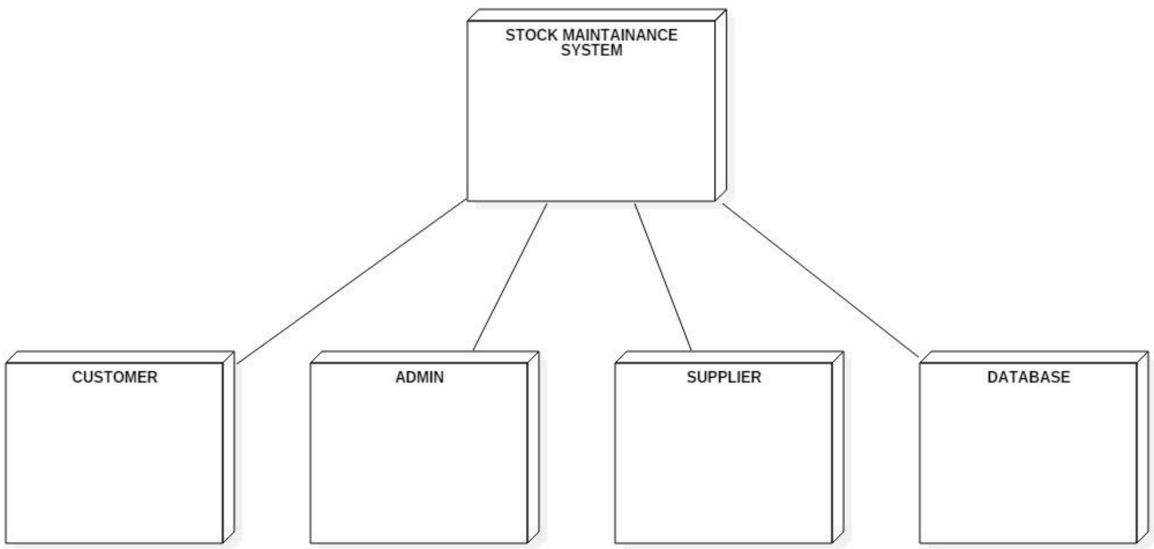
COMPONENT & DEPLOYMENT DIAGRAM

COMPONENT:



DEPLOYMENT:



LESSON: 02

BANKING MANAGEMENT SYSTEM

AIM:

To draw the diagrams [use case, activity, class, sequence, collaboration, state-chart, component and deployment] for the Banking Management System

PROJECT DESCRIPTION:

StarUML software is designed for supporting the computerized banking management system. In this system, the customer can transaction and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer.

USE CASE DIAGRAM:

This diagram will contain the actors, use cases which are given below

- **Actors**: Customer, Manager, CashDispencer, Clerk, Loan officer.
- **Usecase**: Open_account, Withdraw_cash, Clear_checks, Loan_application, Get_report

ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

- **Activities**: Correctaccno, Cash Deposit, Cash withdraw, Low balance.
- **Decisionbox:** Fill details in bank slip,check accno, withdraw, Deposit, Update account, withdraw cash, check account balance

CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBUTES	OPERATIONS
Customer	Customer name, Account no, Address, Phoneno.	Withdraw Amount(), Deposit amount()
Bank Employee	Emp Name, Emp position	Deposit amount(), Withdraw amount(), Check balance().
Account	Acc no, Balance, Customer name	Update account()

SEQUENCE DIAGRAM:

This diagram consists of the objects, messages and return messages. A Sequence diagram simply depicts interaction between objects in a sequential order.

Object: Bank, Customer, Saving Account, Checking Account

COLLABORATION DIAGRAM:

This depicts the relationships and interactions among software objects. They are used to understand the object architecture within a system rather than the flow of a message as in a sequence diagram.

STATE DIAGRAM:

State Diagram is used to capture the behavior of a software system. UML State machine diagrams can be used to model the behavior of a class, a subsystem, a package, or even an entire system.

COMPONENT AND DEPLOYMENT DIAGRAMS

COMPONENT DAIDRAM:

- Component diagrams are used to visualize the organization and relationships among components in a system. These diagrams are also used to make executable systems.

DEPLOYMENT DIAGRAM:

- Deployment diagrams are used to visualize the topology of the physical components of a system, where the software components are deployed.

NODE 1: Customer Client

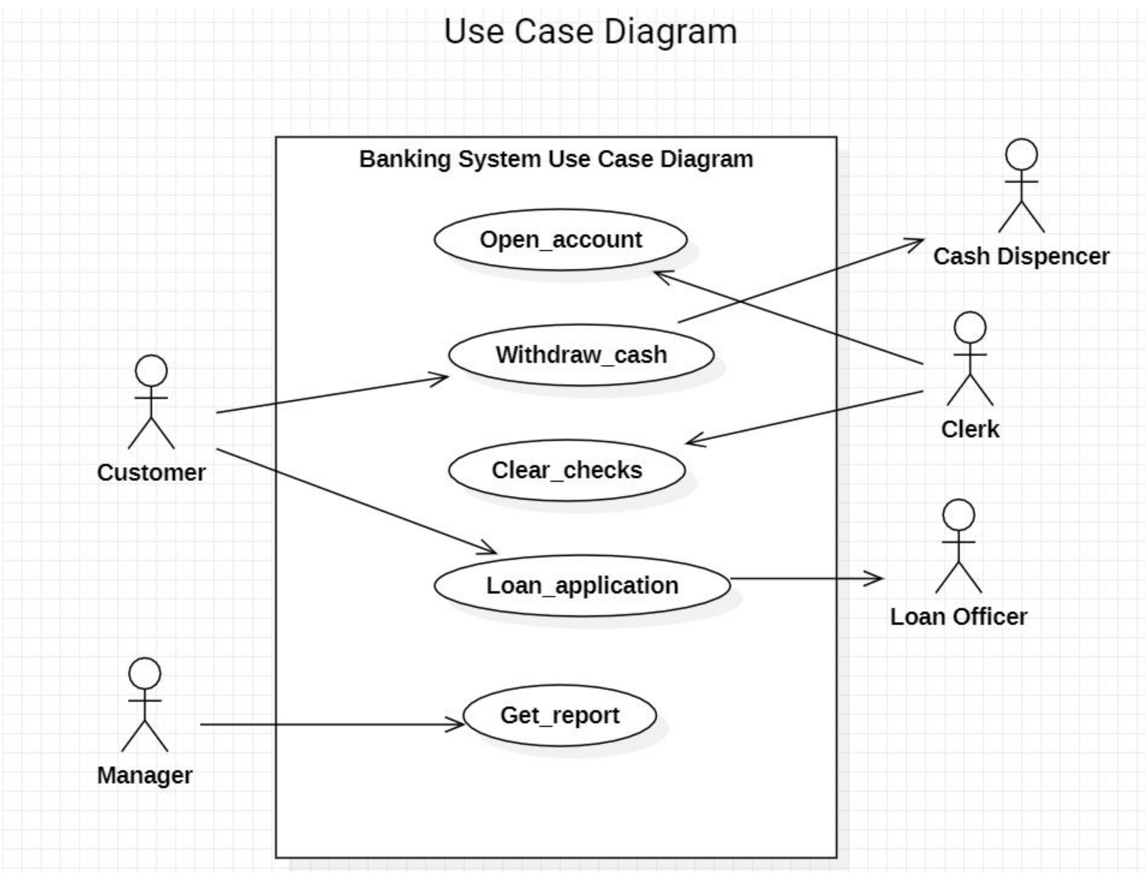
1. Customer Forms
2. Controllers

NODE 2: Banking Client.

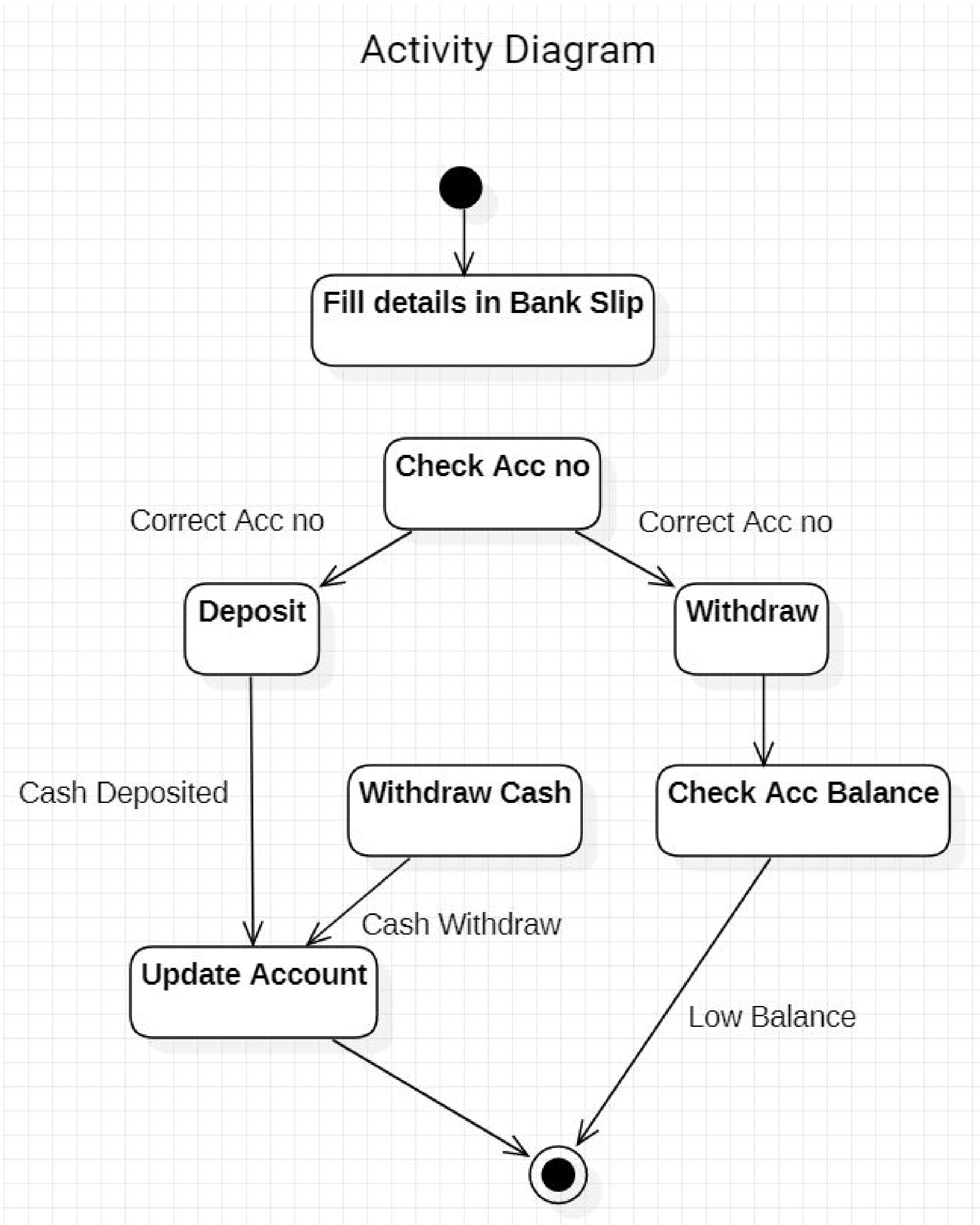
1. BankingForms.
2. Controllers.

NODE 3: Bank Server. 1.CustomerBusinessObject. 2.BankingBusinessObject

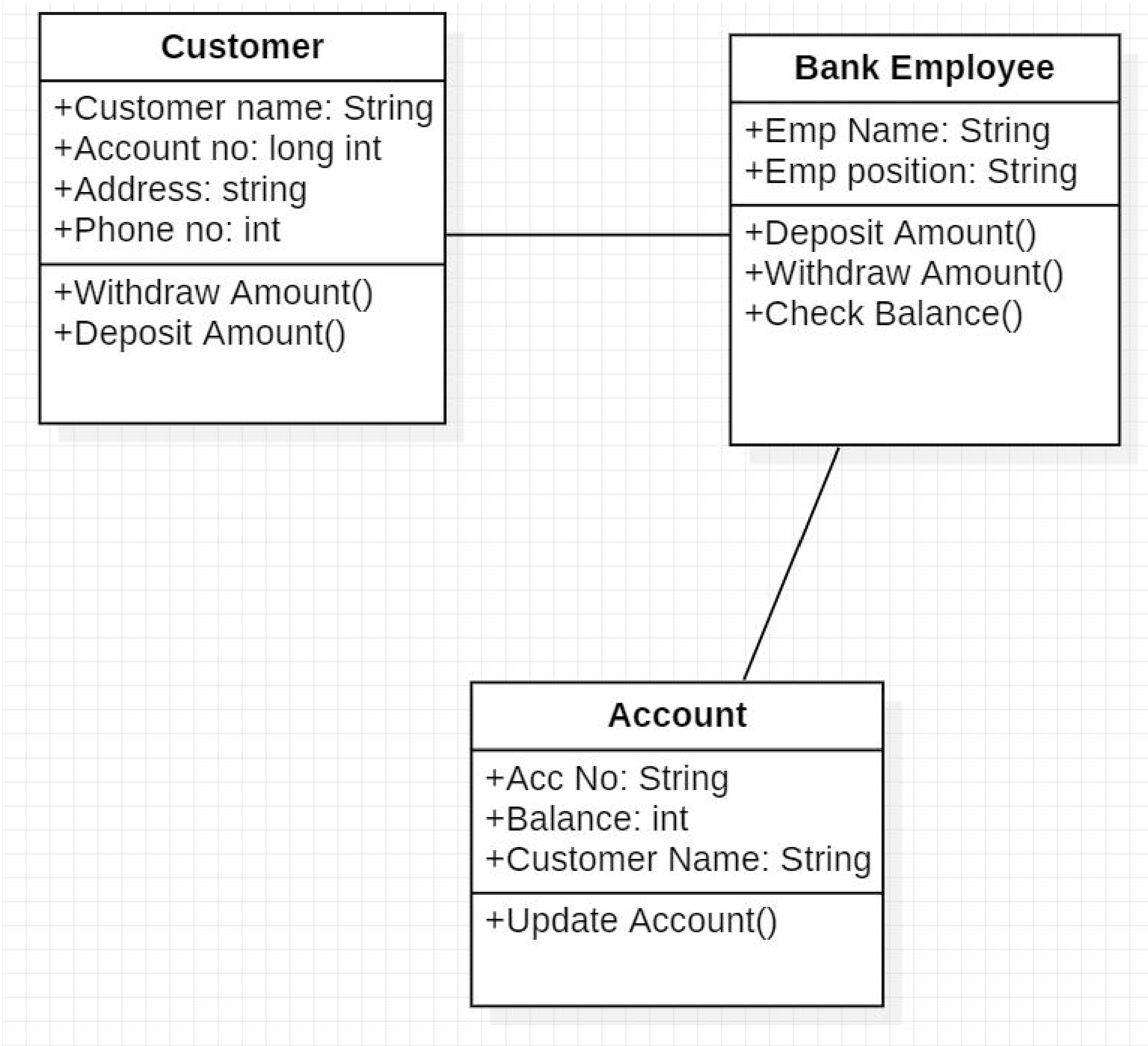
USECASE DIAGRAM:



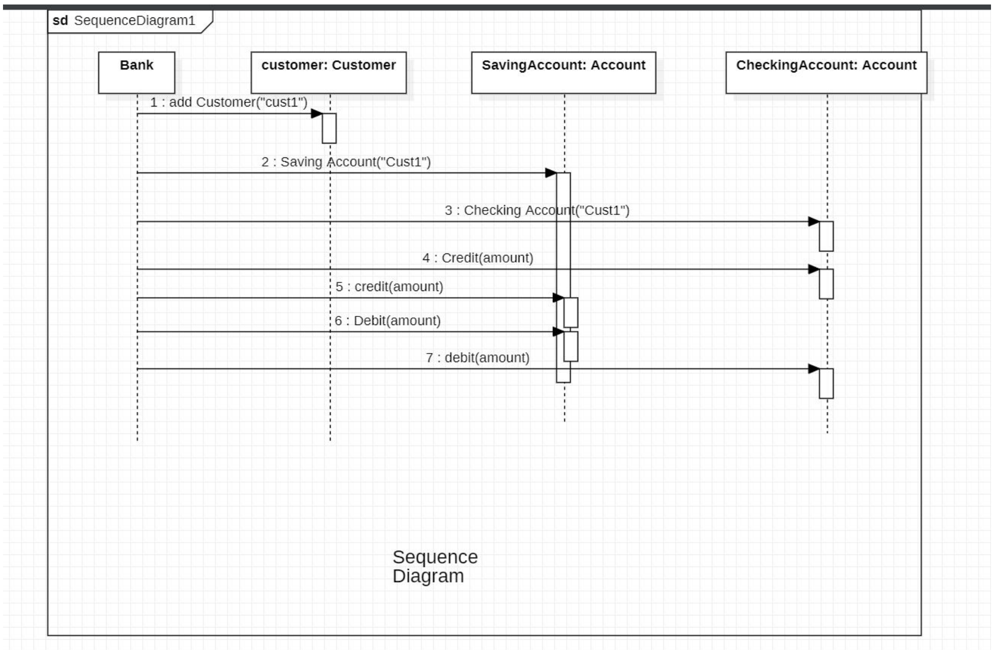
ACTIVITY DIAGRAM:



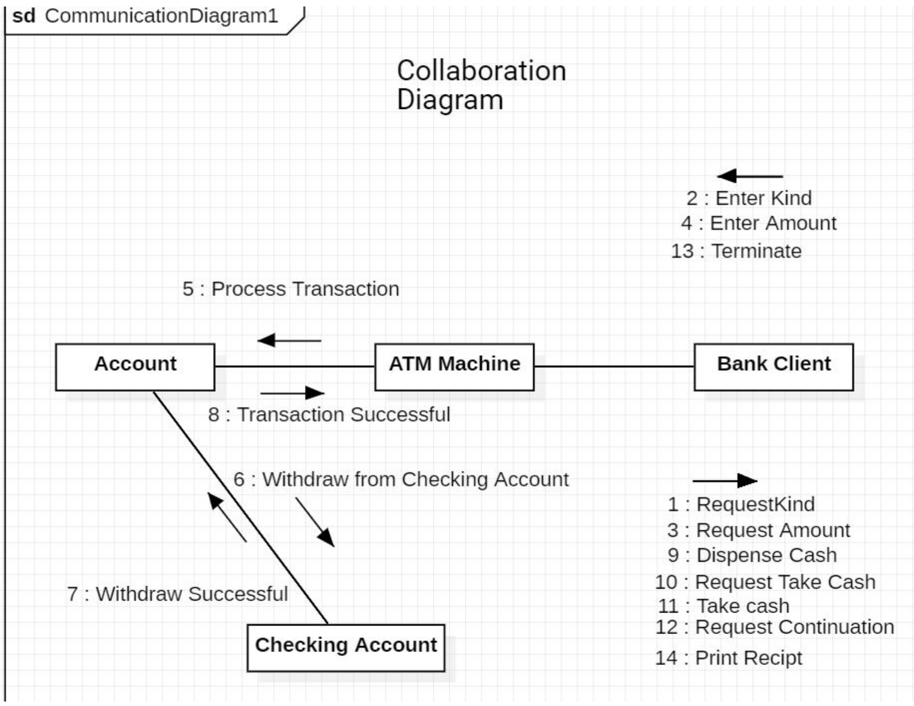
CLASS DIAGRAM:



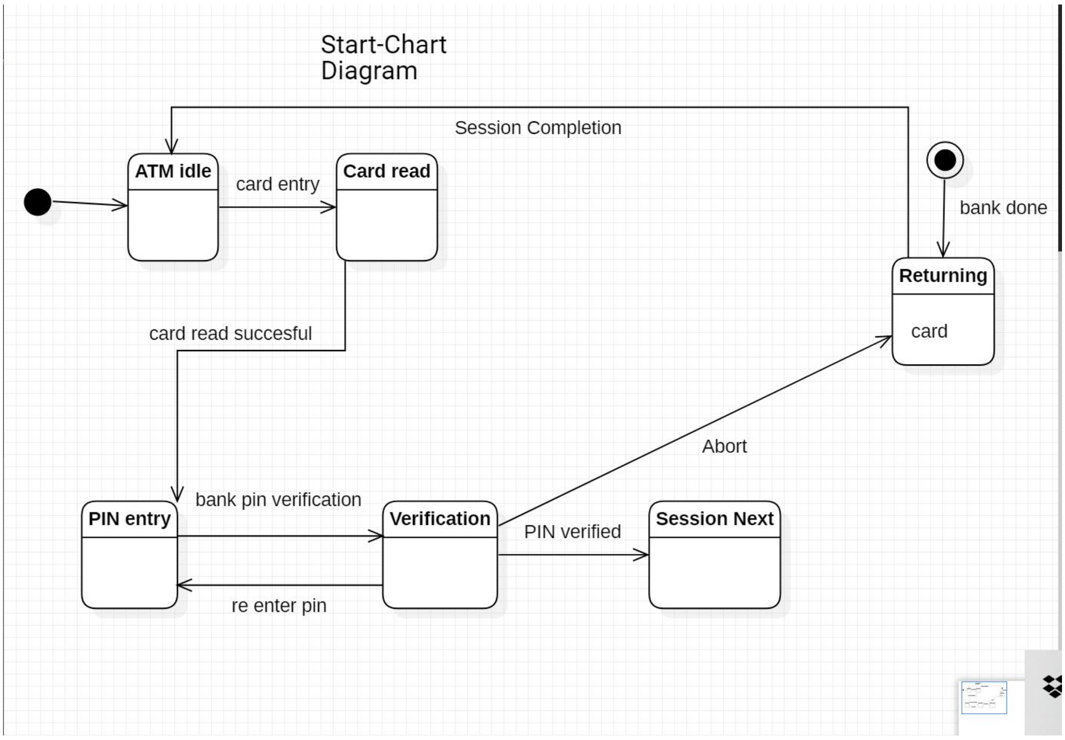
SEQUENCE DIAGRAM:



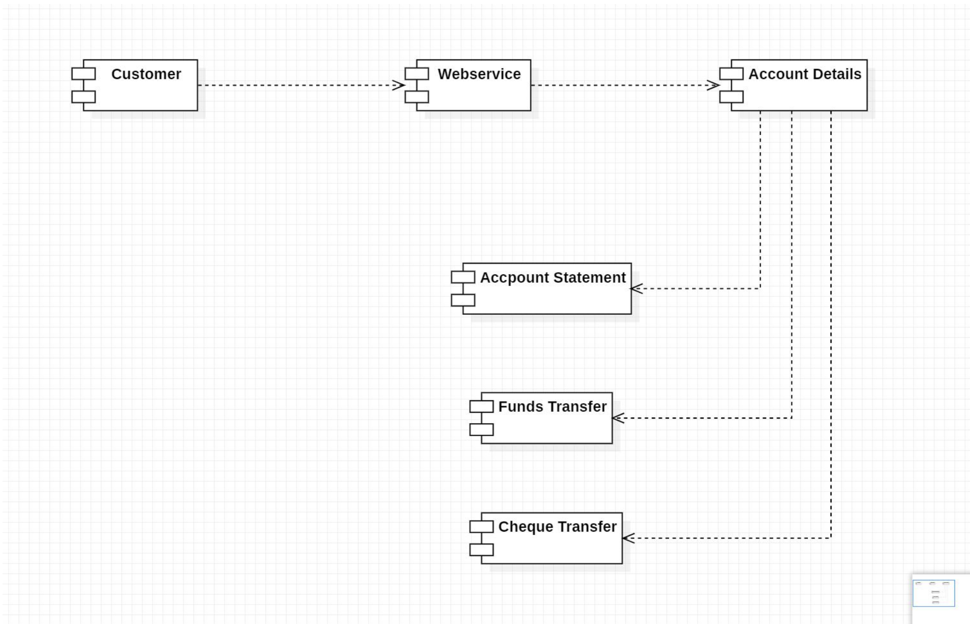
COLLABRATION DIAGRAM:



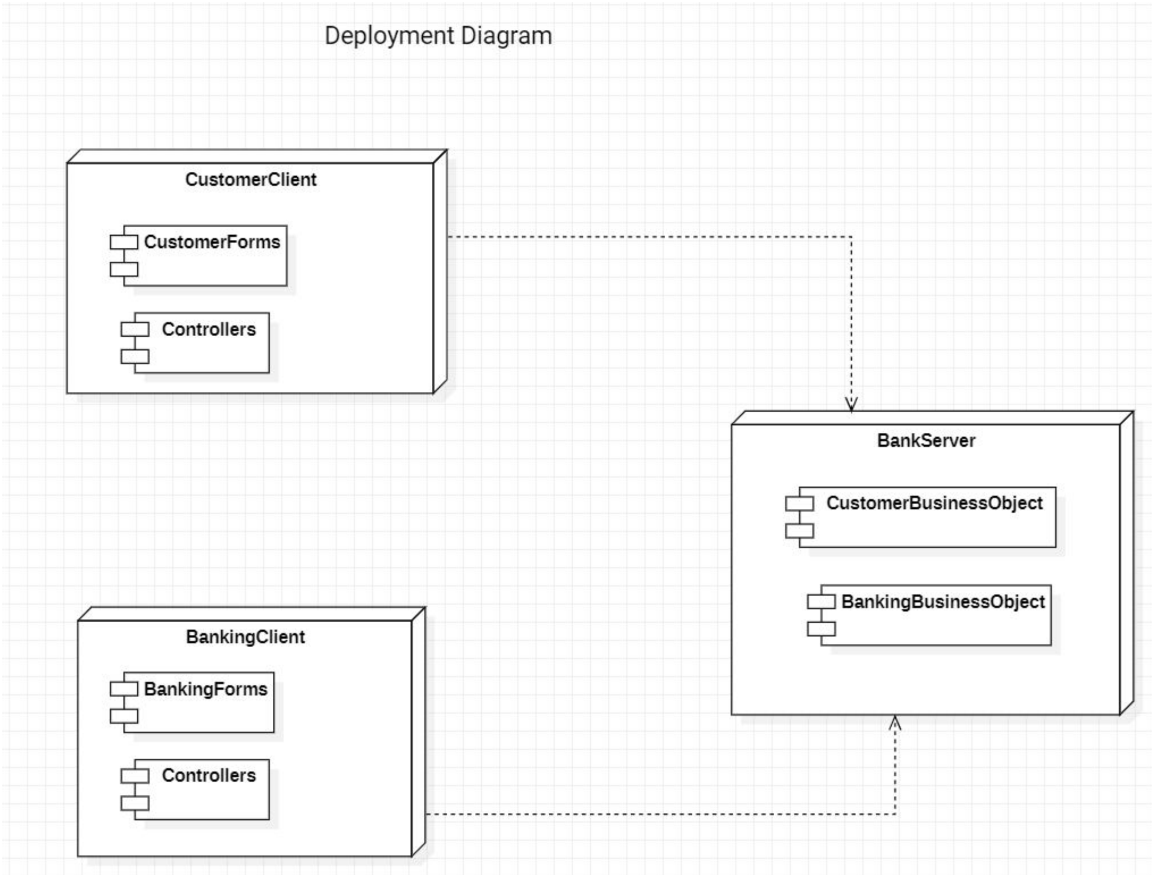
STATE CHART DIAGRAM:



COMPONENT DIAGRAM:



DEPLOYEMENT DIAGRAM:



LESSON: 03

ONLINE COURSE RESERVATION SYSTEM

AIM:

To draw the diagrams [use case, class, activity, sequence, collaboration, state-chart, component and deployment] for the Online Course Reservation system.

PROJECT DESCRIPTION:

- The system is built to be used by students and managed by an administrator. The student and employee have to login to the system before any processing can be done.
- The student can see the courses available to him/her and register to the course he/she wants. The administrator can maintain the course details and view all the students who have registered to any course
- Course Reservation System is an interface between the Student and the Registrar responsible for the issue of Course. It aims at improving the efficiency in the issue of Course and reduces the complexities involved in it to the maximum possible extent.
- If the entire process of 'Issue of Course' is done in a manual manner then it would takes several months for the course to reach the applicant. Considering the fact that the number of applicants for course is increasing every year, an Automated System becomes essential to meet the demand.
- So this system uses several programming and database techniques to elucidate the work involved in this process

USE CASE DIAGRAM:

This diagram will contain the actors, use cases which are given below

- **Actors**: Student, Registrar.
- **Use case**: Login & logout, view course details, Reserve of course, fees payment check profile status.

CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBUTES	OPERATIONS
Student	St name, St id, St qualification, St address, St mobile no, St email id, St dob, St sex	addStDetails() modifyStDetails() delStDetails() reserveCourse()
Course Catalogue	Course id, Course name, Course duration, Course fee, Course elgibility criteria, Total noofseats, Course availseats	addCourse() updateCourse () delCourse()
Reserve Course	St id, Course id, Date, Amtpaid, Reg id, DDno	getCourseDetails() checkEligibility() confirmRegistration()

ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

- **Activities**: Login, View course, Select course, Fill up the form, submit form, Eligibility confirmation, fees payment, course is registered.

SEQUENCE DIAGRAM:

This diagram consists of the objects, messages and return messages. A Sequence diagram simply depicts interaction between objects in a sequential order.

Object: Student, Reserve course (Registrar), Course catalog.

COLLABORATION DIAGRAM:

This depicts the relationships and interactions among software objects. They are used to understand the object architecture within a system rather than the flow of a message as in a sequence diagram.

STATE CHART DIAGRAM:

State Diagram is used to capture the behavior of a software system. UML State machine diagrams can be used to model the behavior of a class, a subsystem, a package, or even an entire system.

COMPONENT AND DEPLOYMENT DIAGRAMS

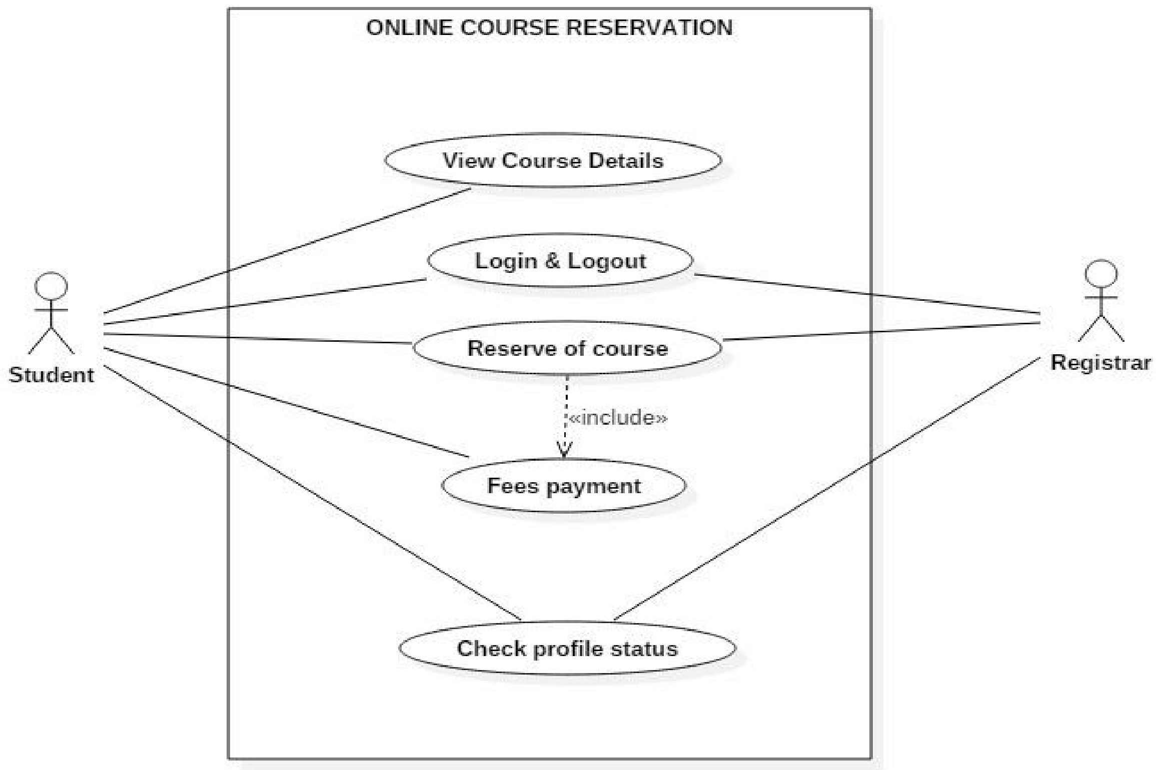
COMPONENT DIAGRAM:

- Component diagrams are used to visualize the organization and relationships among components in a system. These diagrams are also used to make executable systems.

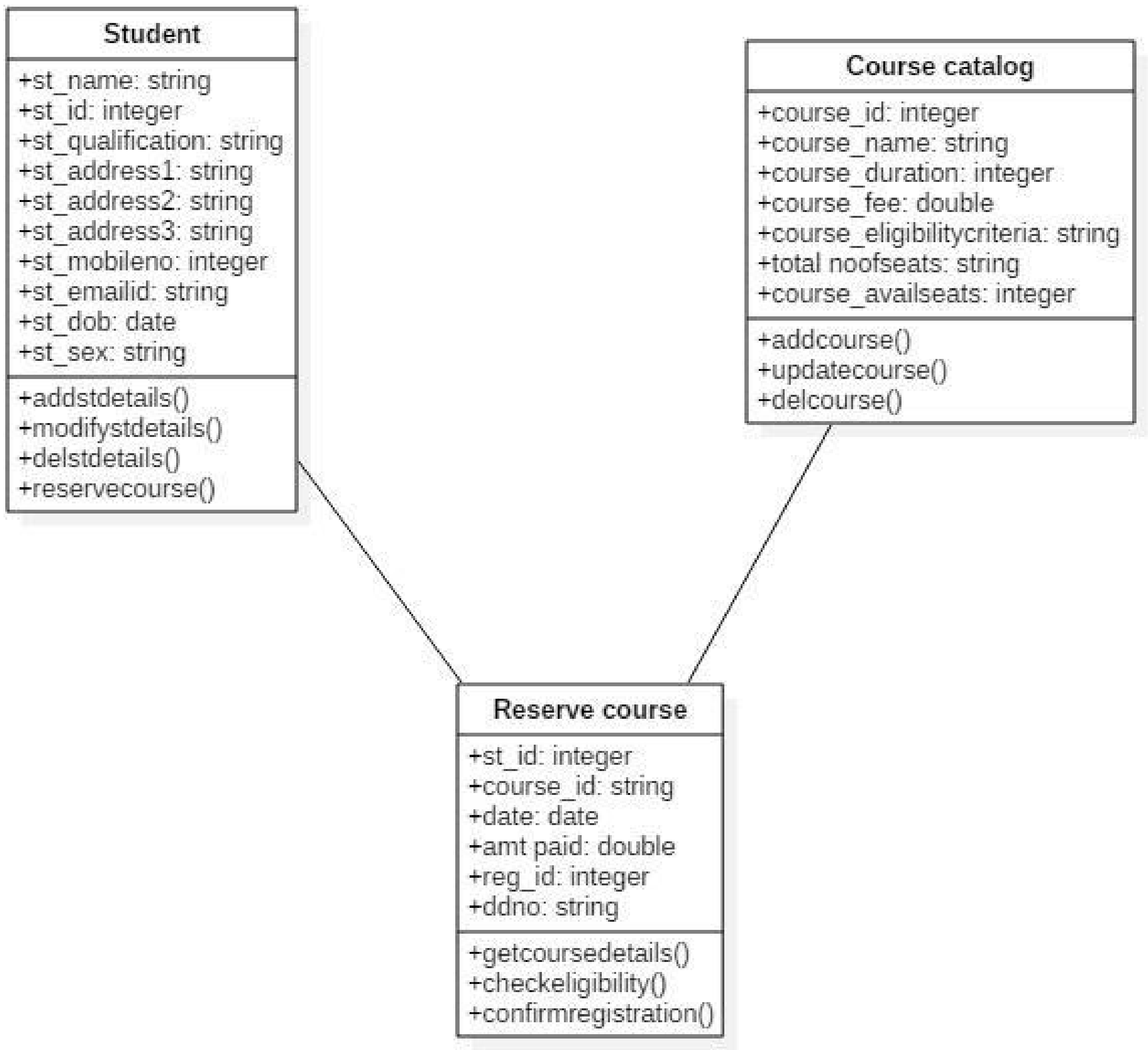
DEPLOYMENT DIAGRAM:

- Deployment diagrams are used to visualize the topology of the physical components of a system, where the software components are deployed.

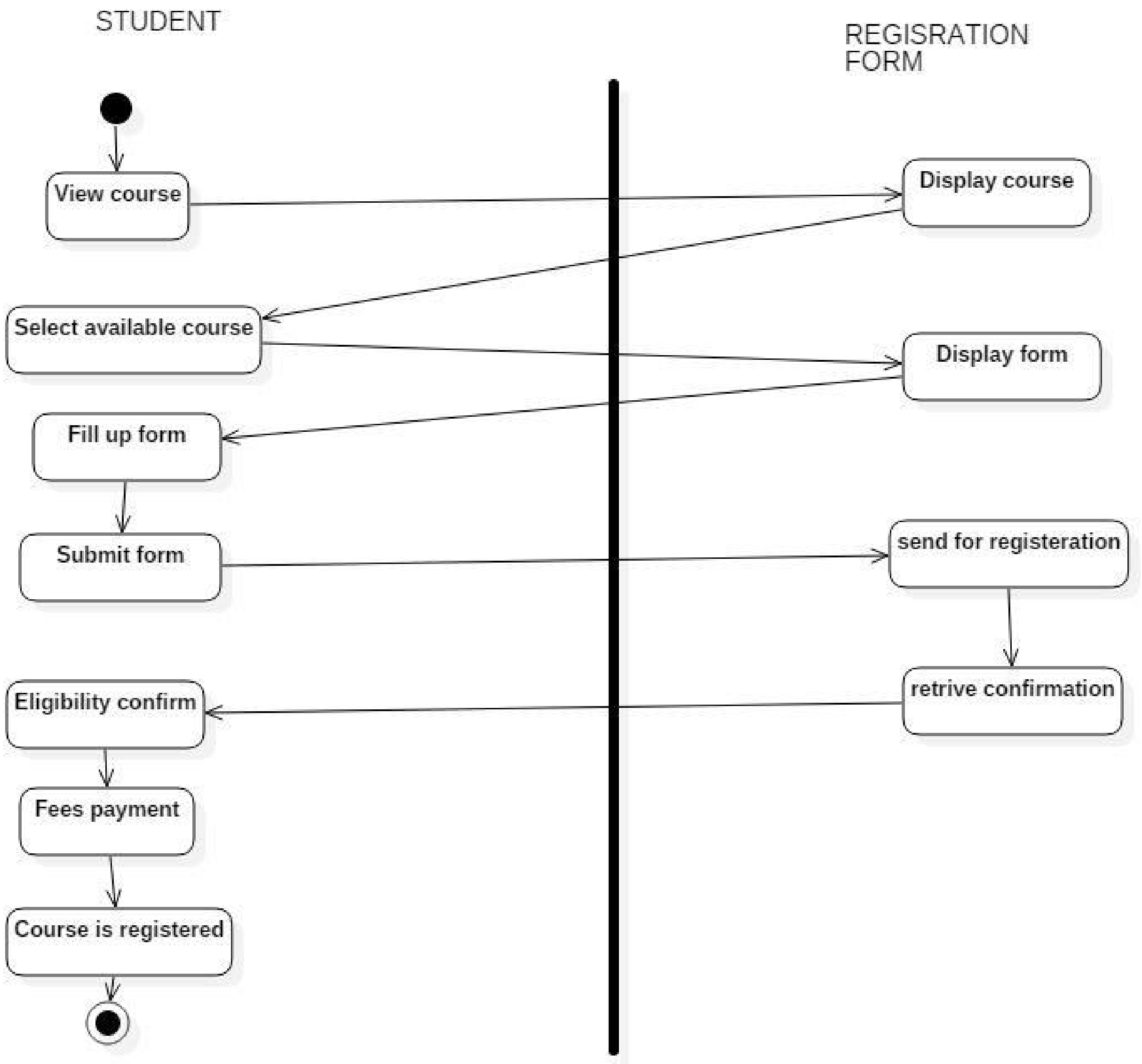
USE CASE DIAGRAM:



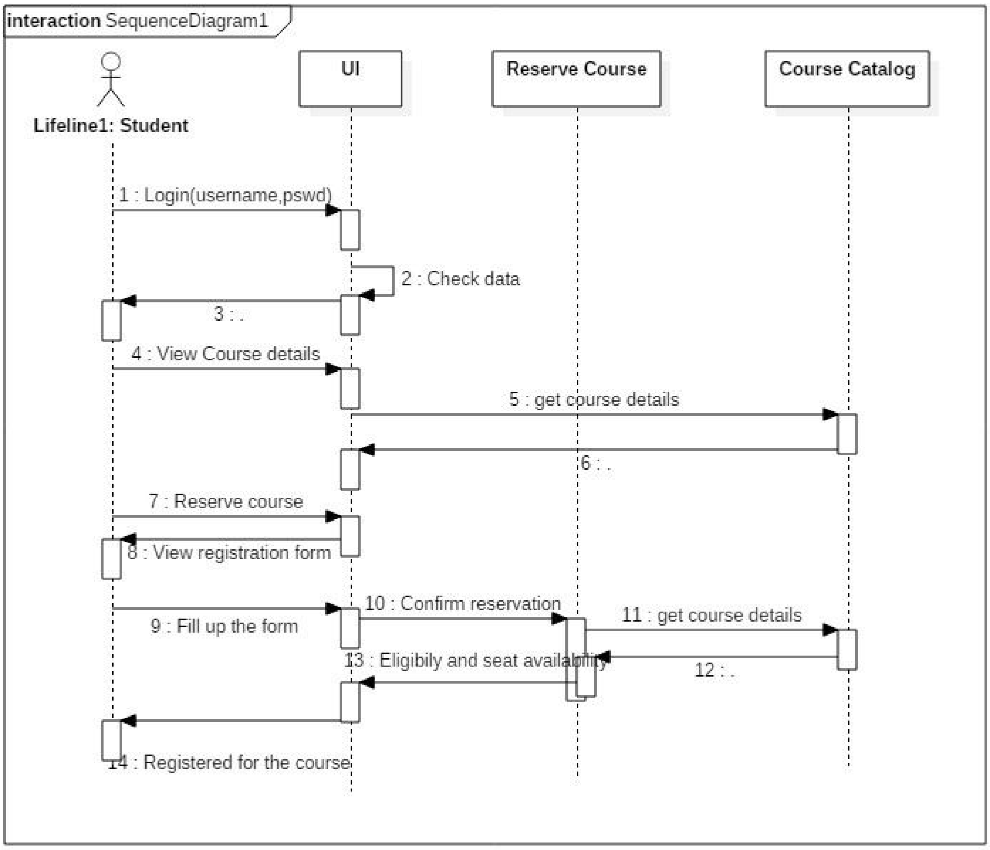
CLASS DIAGRAM:



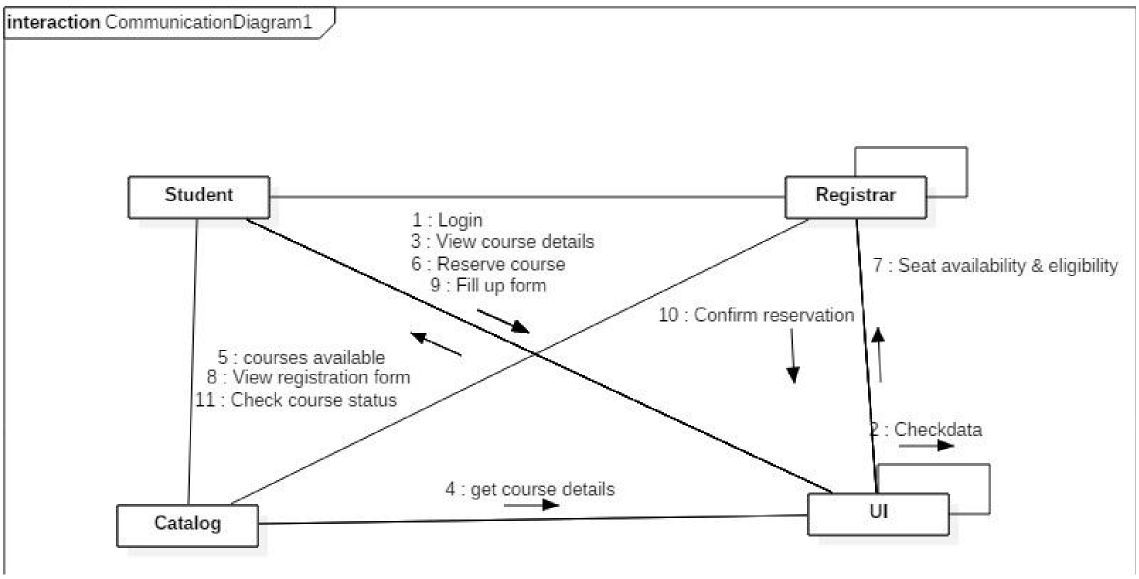
ACTIVITY DIAGRAM:



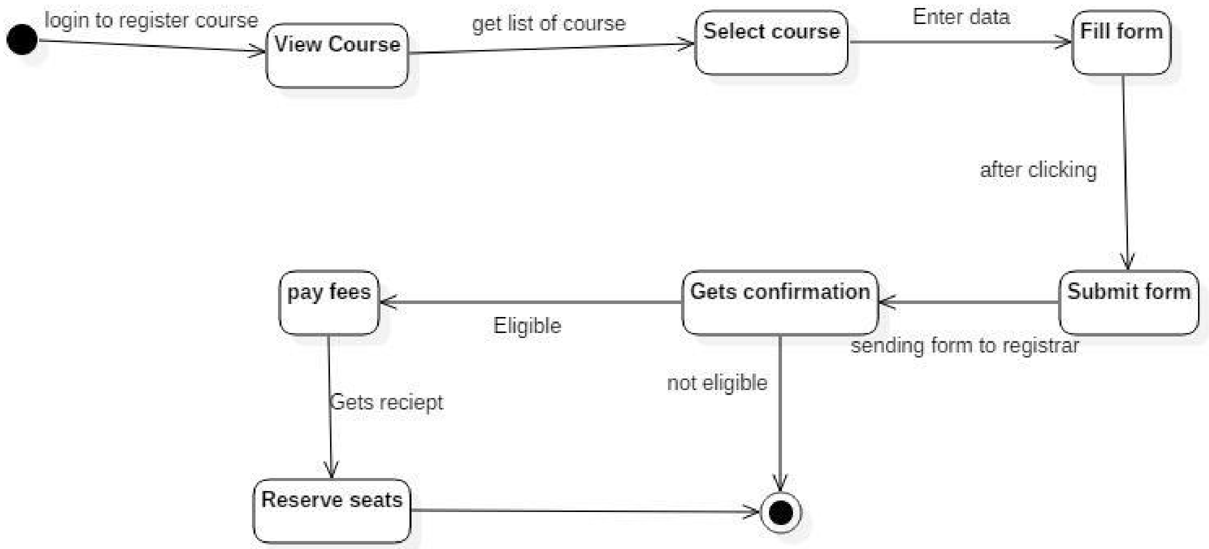
SEQUENCE DIAGRAM:



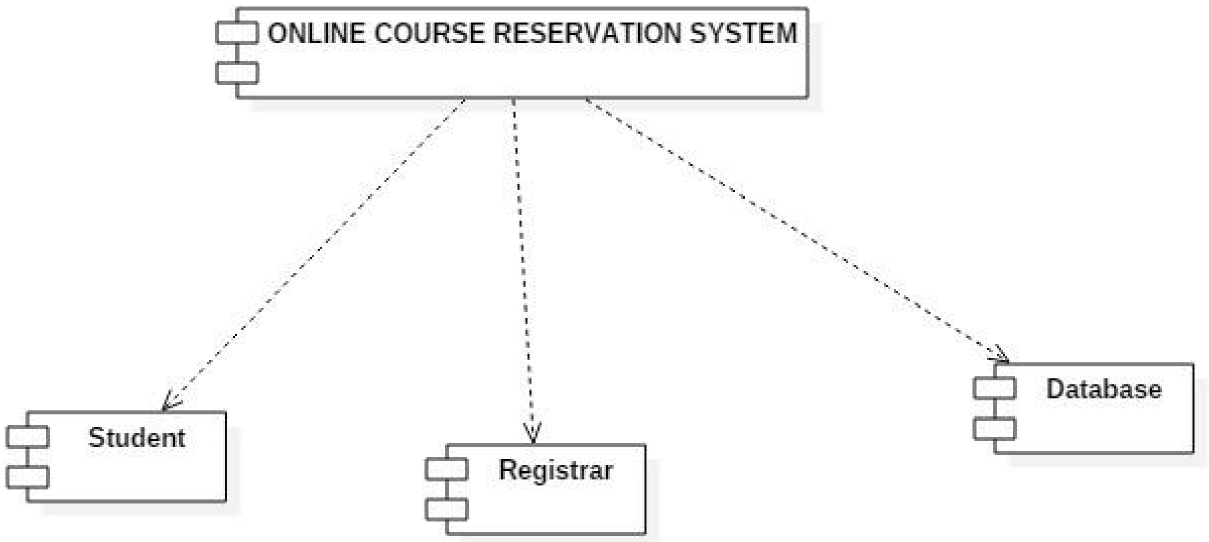
COLLOBORATION DIAGRAM:



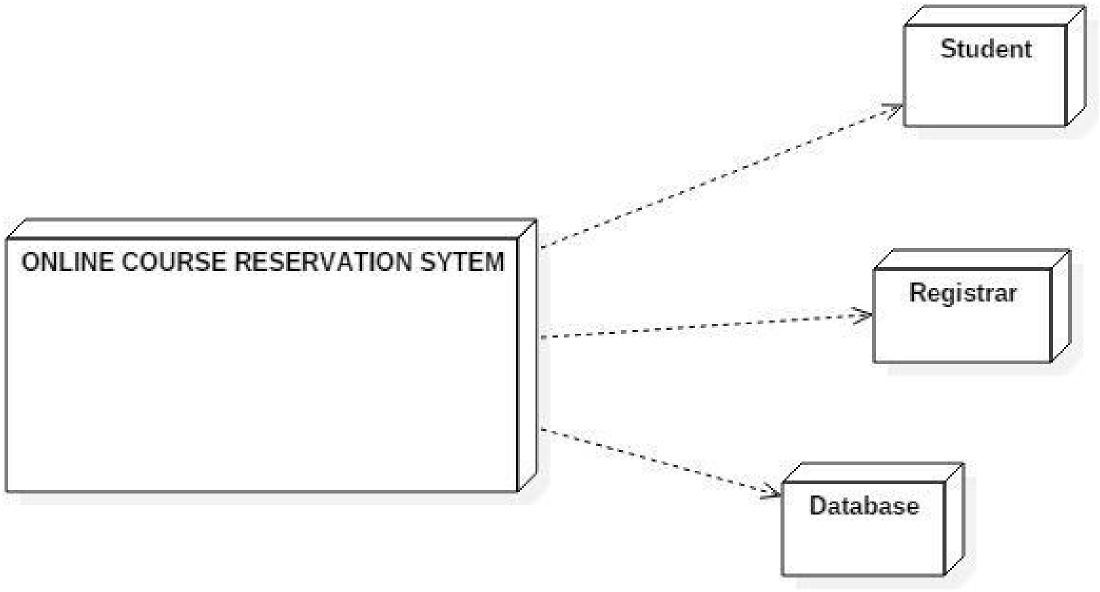
STATE CHART DIAGRAM:



COMPONENT & DEPLOYMENT DIAGRAM COMPONENT:



DEPLOYMENT:


Student
ONLINE COURSE RESERVATION SYTEM
Registrar
Database

LESSON: 04
EMPLOYEE MANAGEMENT SYSTEM

AIM:

To draw the diagrams [use case, activity, class, sequence, collaboration, state-chart, component and deployment] for the Stock maintenance system.

PROJECT DESCRIPTION:

The project is designed to fullfill requirements of employees, saving and retrieving information, attendance and salary expectations

processing. The system provides a recruiting committee part to process applicant application, an admin part to register employees, a manager part to manage staff data, an employee part to mark attendance or see his/her details/status and a finance part to control payment of salary.

Basic Features:

1. Online advertising
2. Accepting applications
3. Registering new employees
4. Creating accounts and attendance for employees.
5. Employees can check their details/status and mark attendance.
6. Employee, Manager or administrator see their performances.
7. Calculating net salary and delivering according to attendance

USE CASE DIAGRAM:

This diagram will contain the actors, use cases which are given below

• **Actors**:

1. Employee
2. Manager

3. Admin
4. Finance

• **Use case**:

1. Apply For Job
2. Test & Interview
3. Inform Result
4. Register & Create Account for New Employee
5. Update Employee details
6. View Employee Details
7. Calculate Net salary
8. Approve Net Salary
9. Pay Net Salary

ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

• **Activities**:

1. Apply For Job
2. Analyzing Application
3. Inform Applicant
4. View Response
5. Test & Interview
6. Informed Applicant
7. Provide Requests
8. Register new Employee
9. Create Account 10.Create Attendance 11.Show Basic Salary 12.Overtime & Absentee 13.Showing Net Salary 14.Pay Net Salary

• **Decision box:** Valid or not

CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBUTES	OPERATIONS
RecruitingCommitee	Recruiting details	..()
Admin	Name, Address, Phone	..()
Manager	Name , Address, Phone	..()
Employee	Name, address, phone, age.	..()
Finance	Name, Address, phone.	Net salary (), Date()
Net salary	Basic salary, absentees, overtime.	..()
Attendence	Name, Date&time	..()

SEQUENCE DIAGRAM:

This diagram consists of the objects, messages and return messages. A Sequence diagram simply depicts interaction between objects in a sequential order.

STATE DIAGRAM:

State Diagram is used to capture the behaviour of a software system. UML State machine diagrams can be used to model the behaviour of a class, a subsystem, a package, or even an entire system.

State

- Analysing Application
- Inform Applicant
- View Response

- Test & Interview
- Informed Applicant
- Provide Requests
- Register new Employee
- Create Account
- Create Attendance
- Show Basic Salary
- Overtime & Absentee
- Showing Net Salary
- Pay Net Salary

COLLABORATION DIAGRAM:

This depicts the relationships and interactions among software objects. They are used to understand the object architecture within a system rather than the flow of a message as in a sequence diagram.

COMPONENT AND DEPLOYMENT DIAGRAMS

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DEPLOYMENT DIAGRAM:

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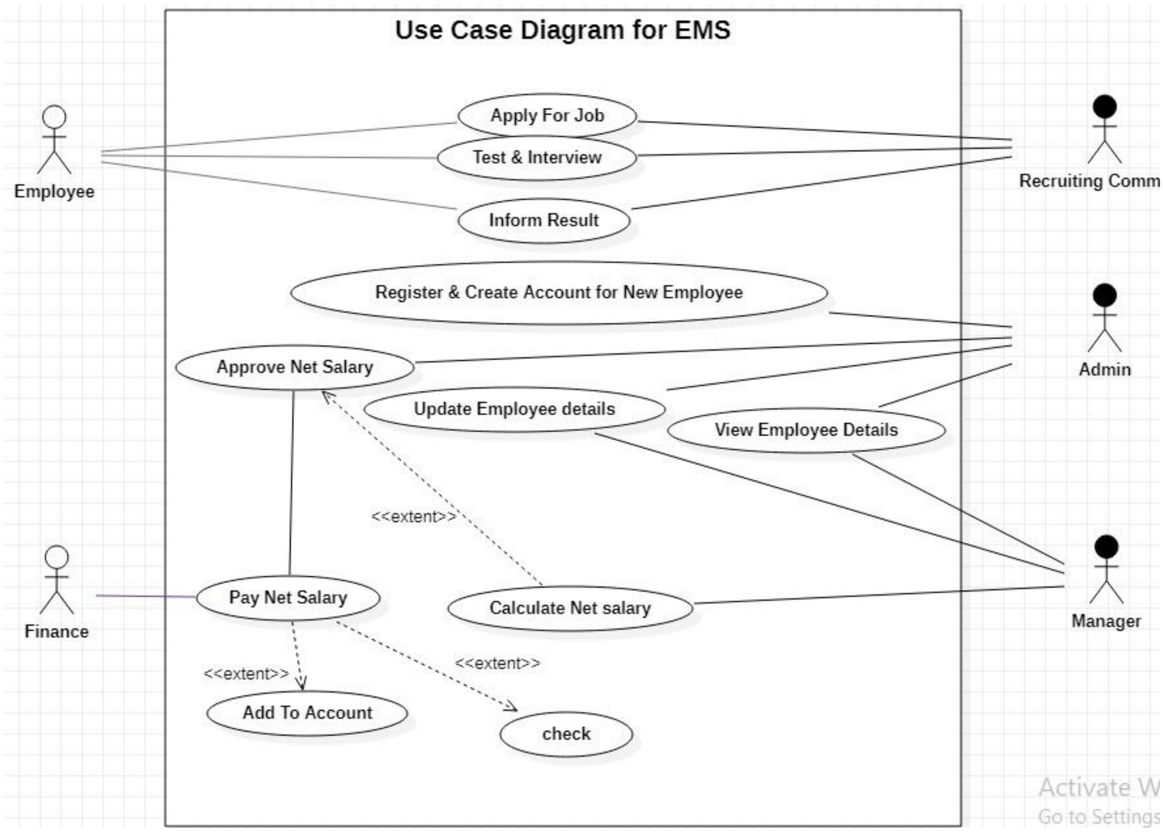
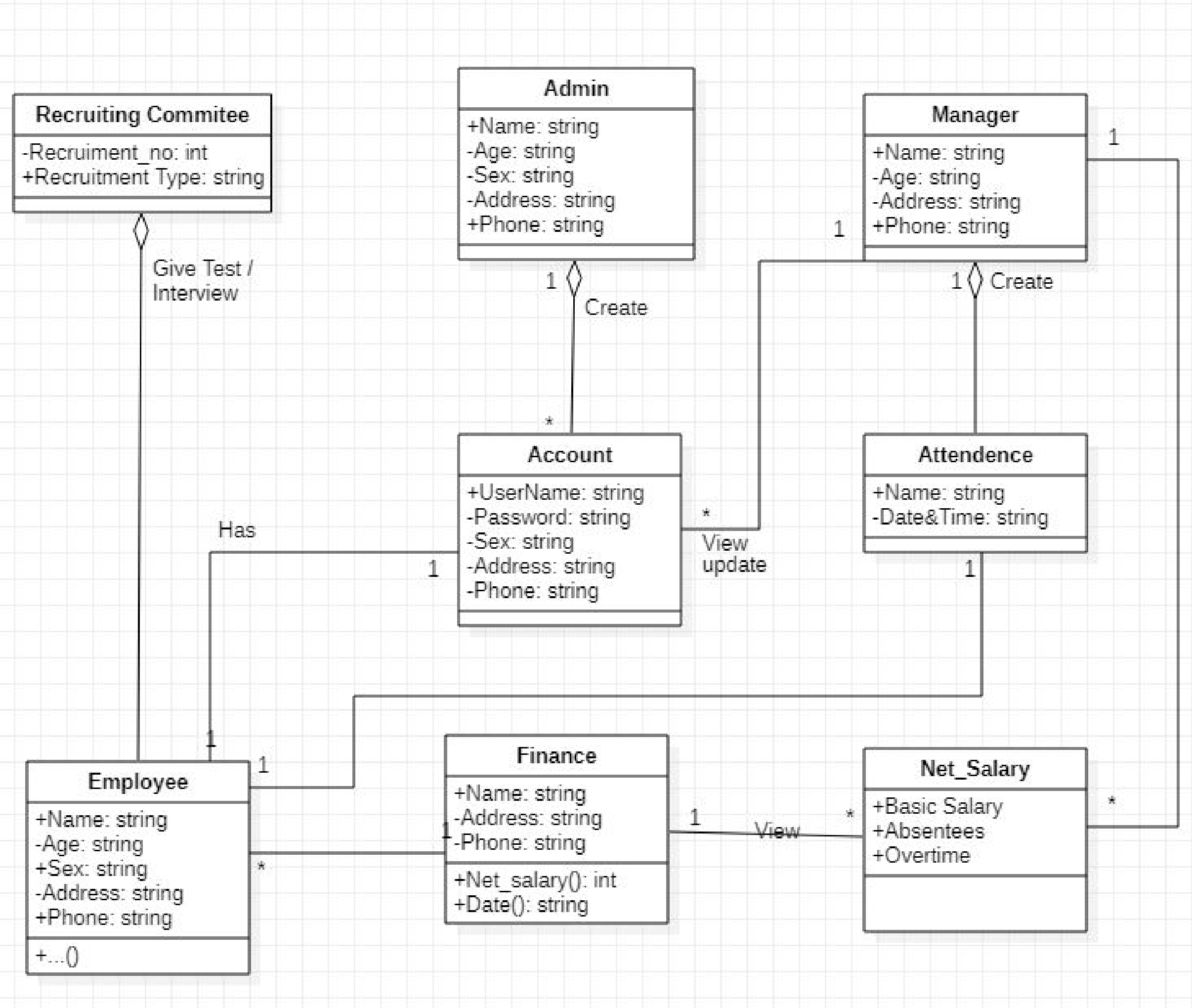
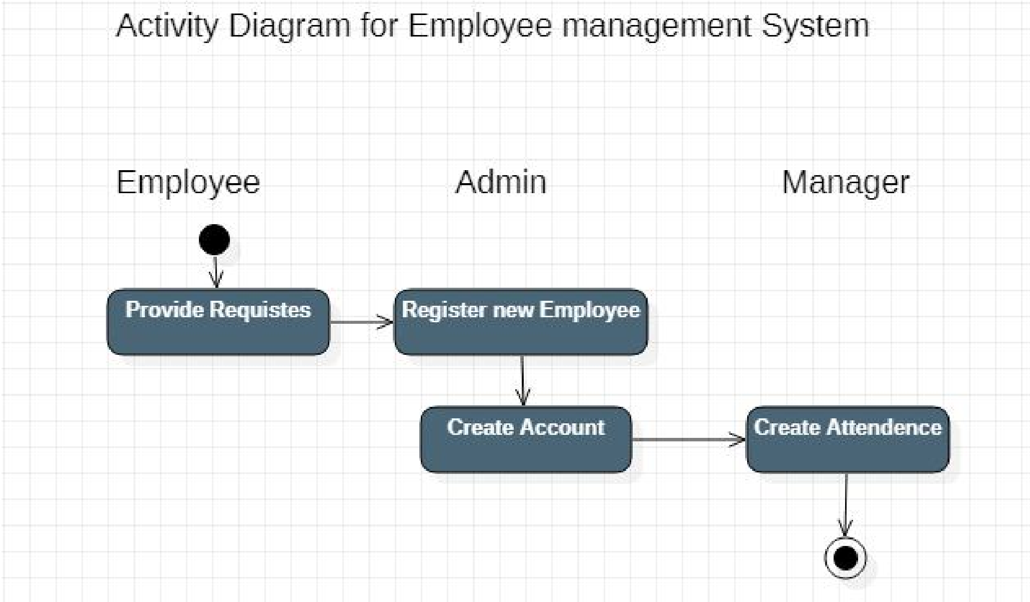
USE CASE DIAGRAM:

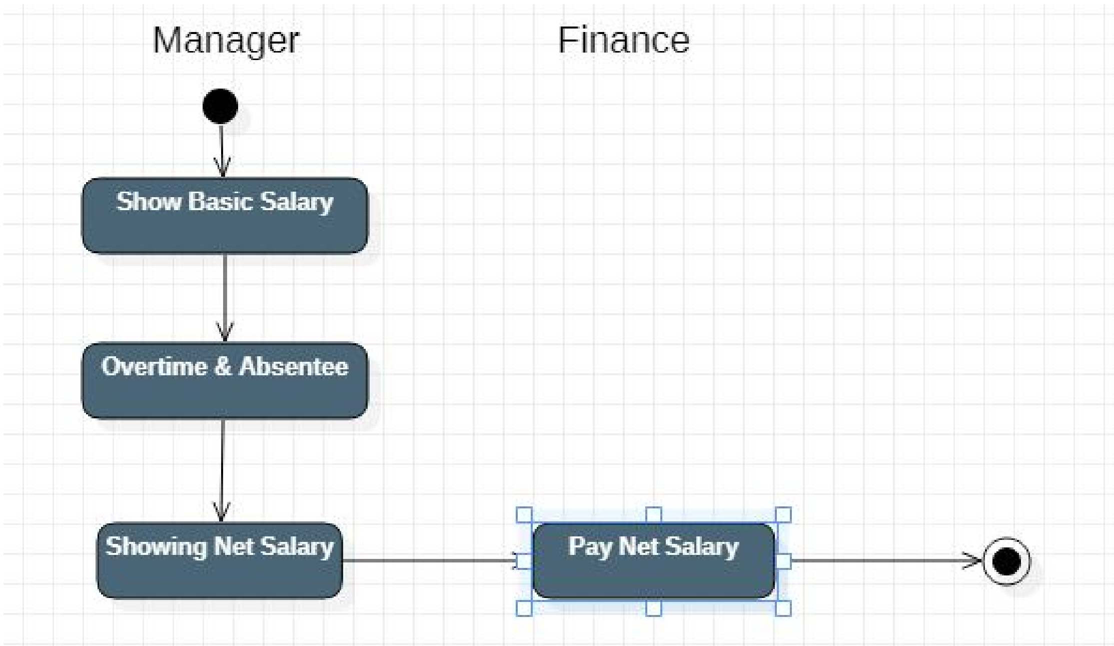


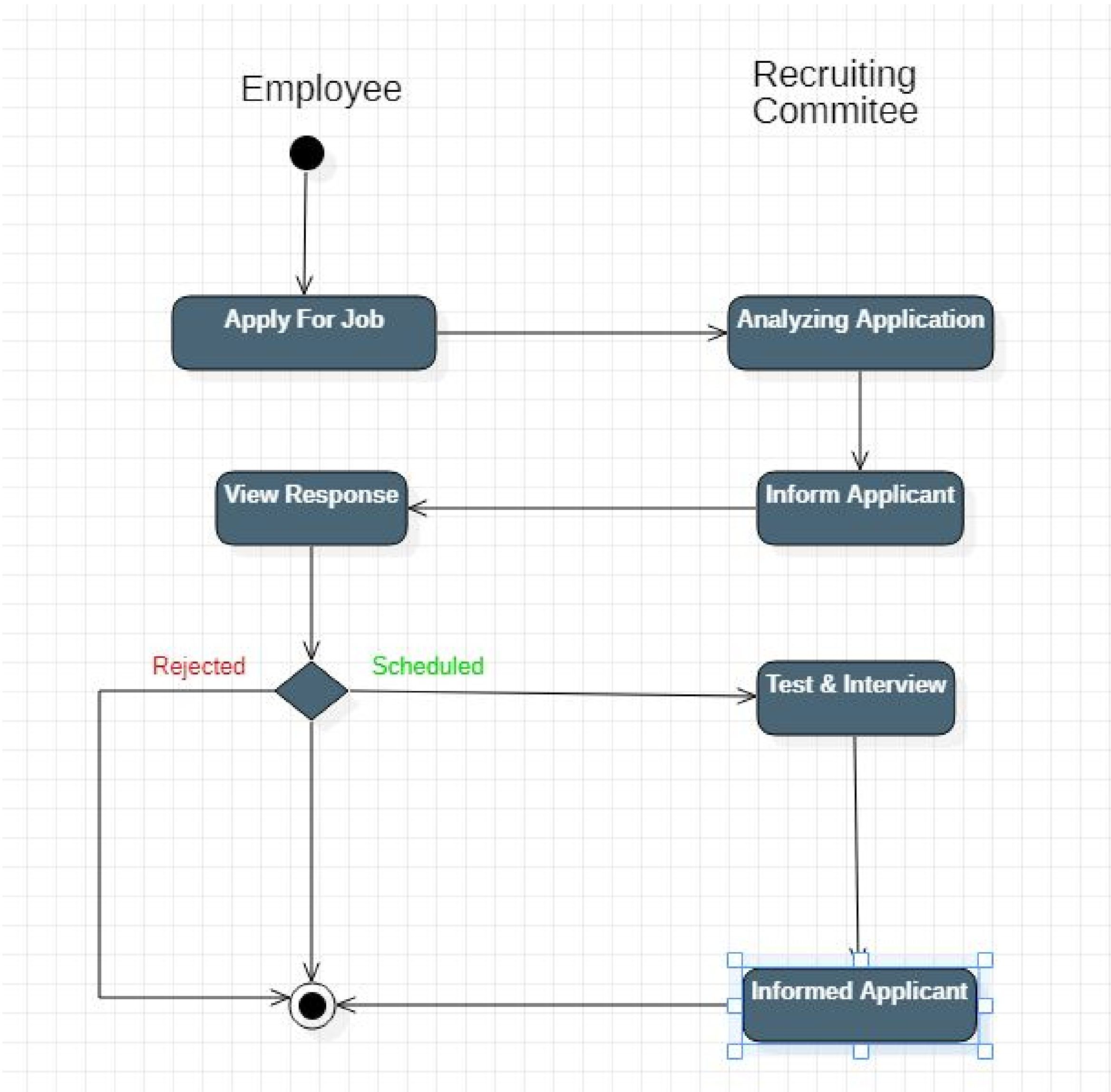
CLASS DIAGRAM:



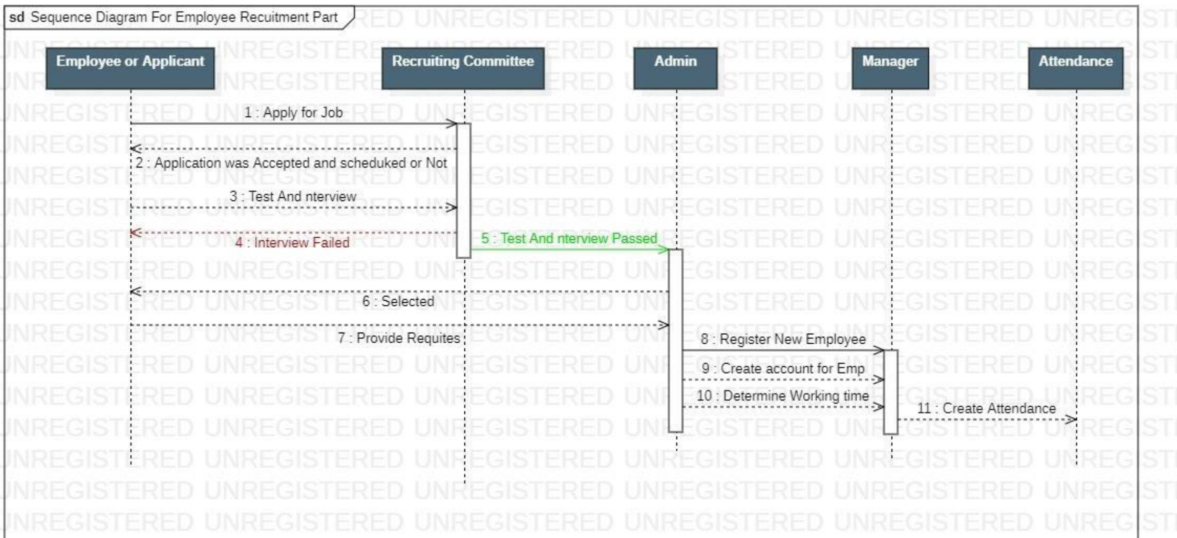
ACTIVITY DIAGRAM:



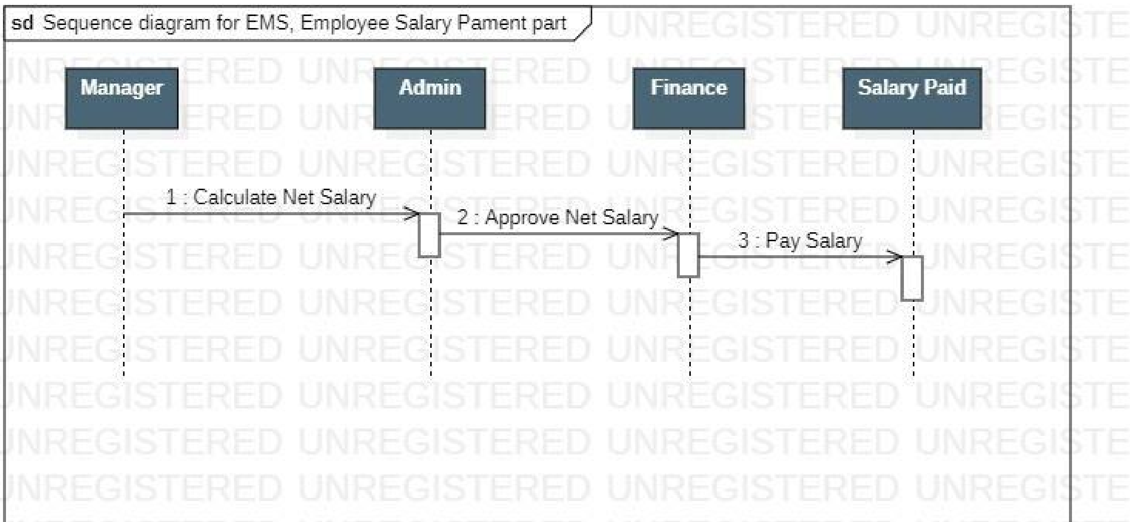



Employee
Recruiting Commitee
Apply For Job
Analyzing Application
Inform Applicant
View Response
Rejected
Scheduled
Test & Interview
Informed Applicant

SEQUENCE DIAGRAM:

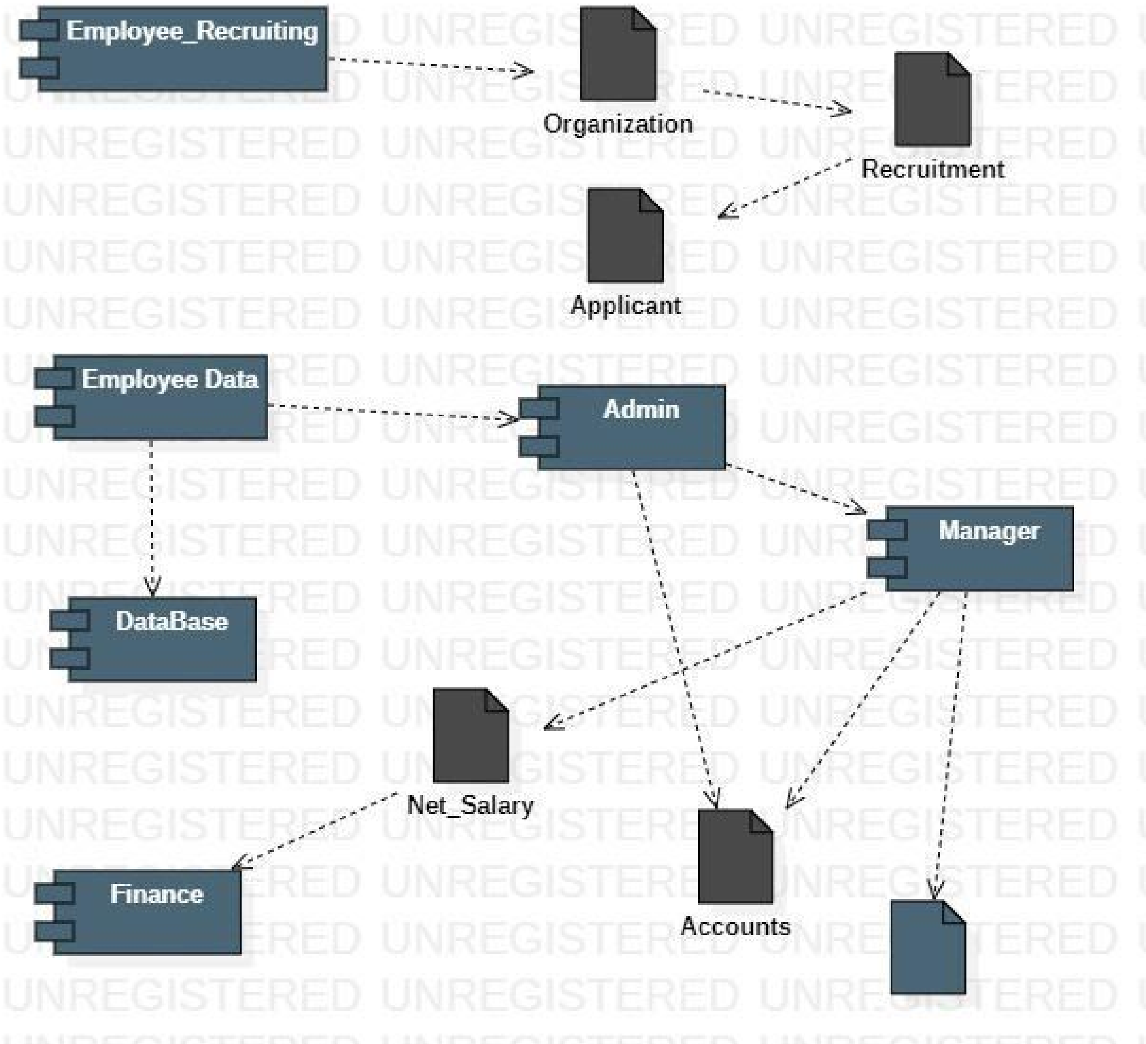


Sequence Diagram For EMS,Employee Recuitment Part



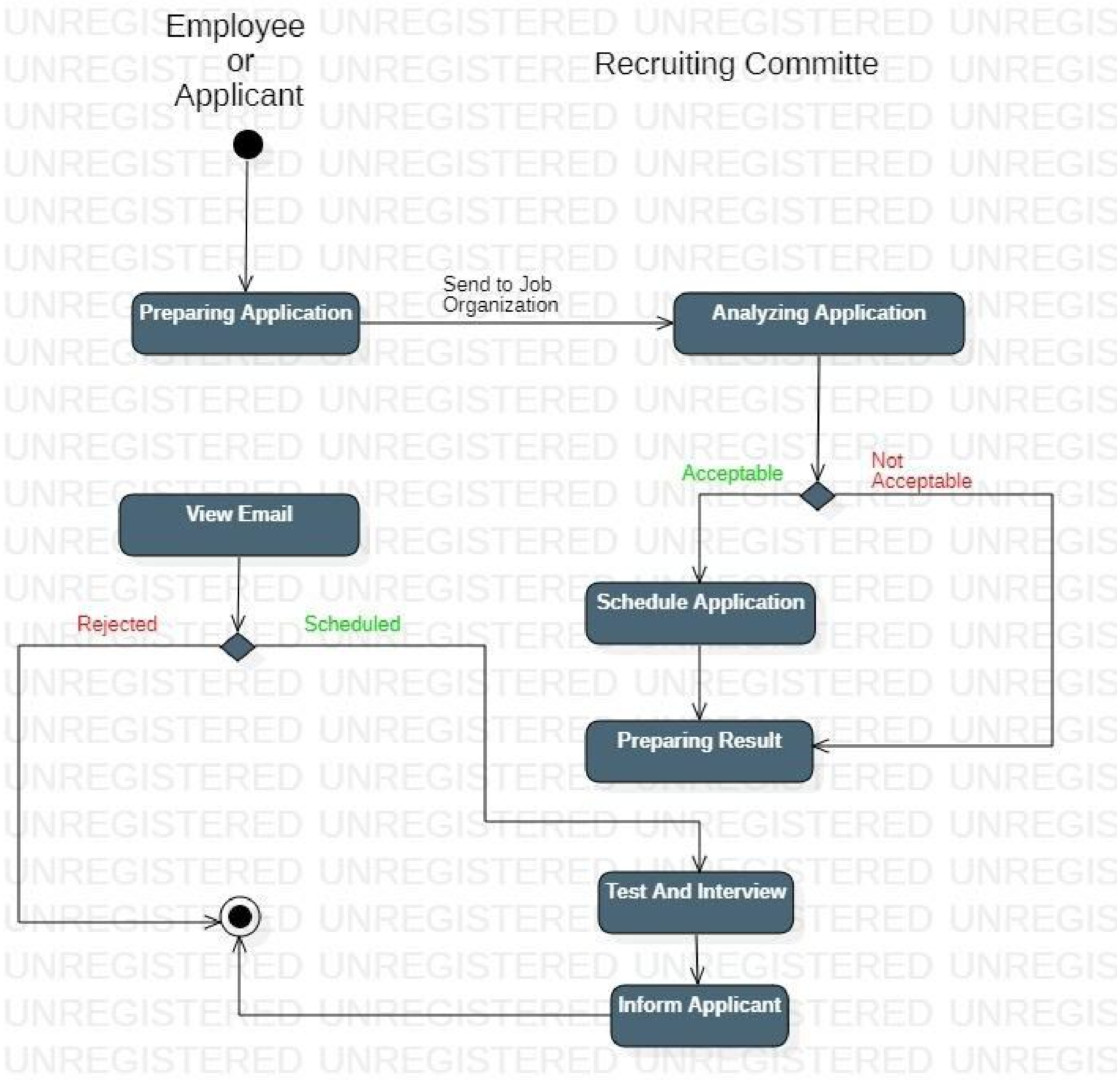
Sequence diagram for EMS, Employee Salary Pament part

COLLABORATION DAIDRAM:

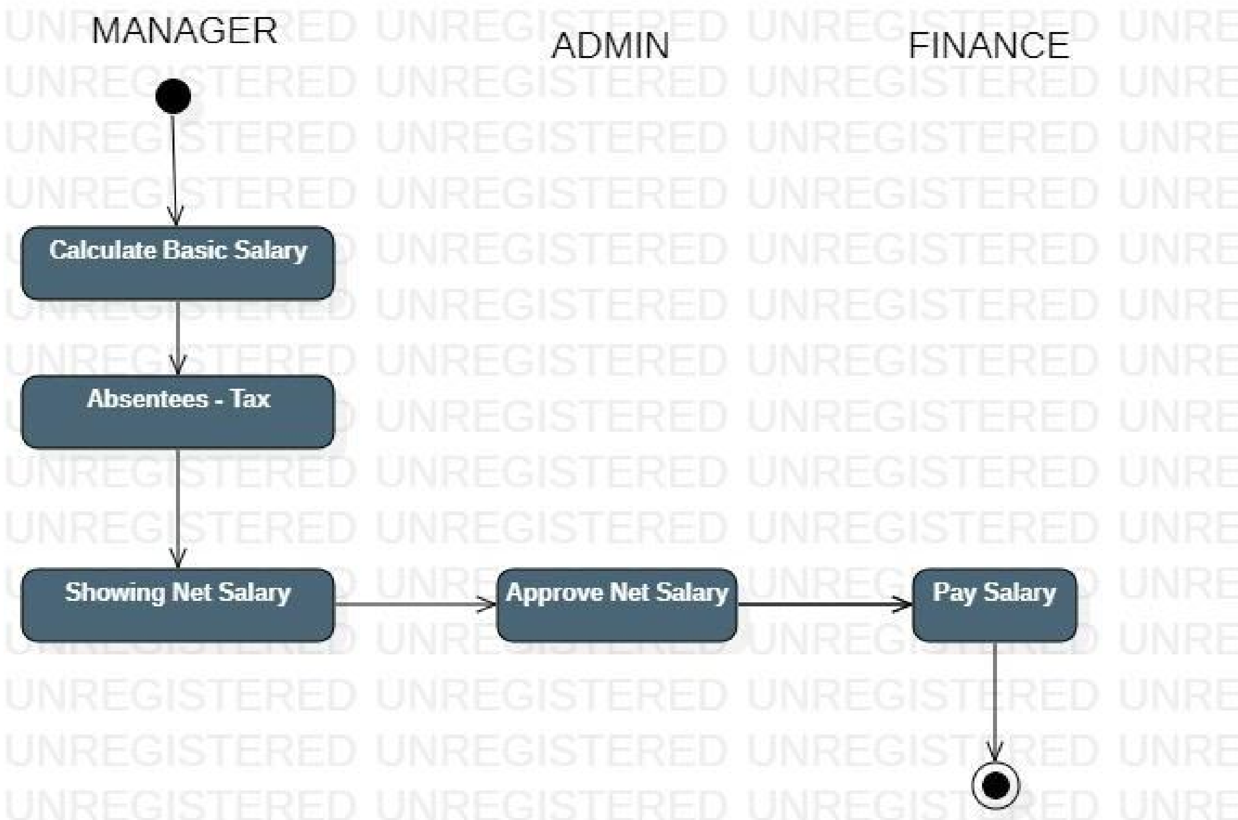


Component Diagram For EMS

STATE DIAGRAM:

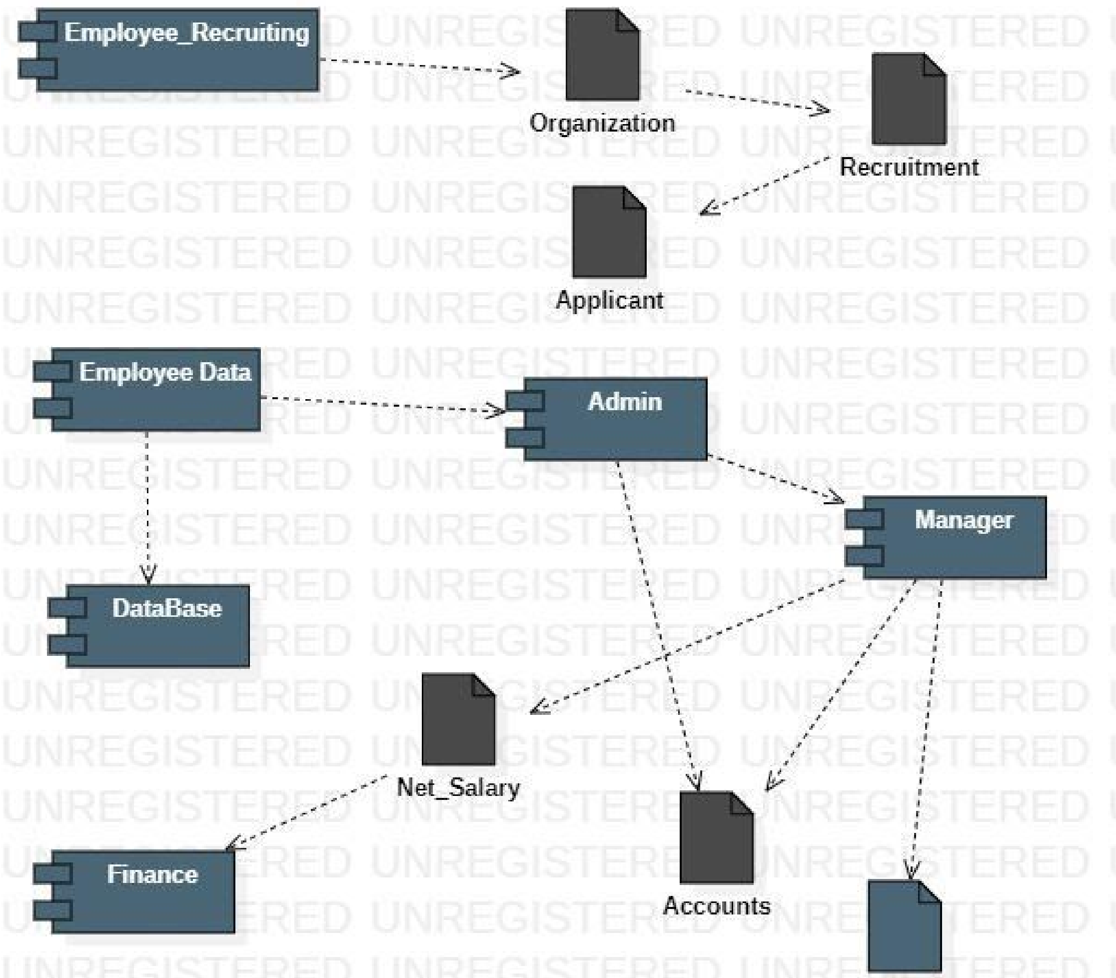


State Chart diagram for EMS, Employee Recruiting part



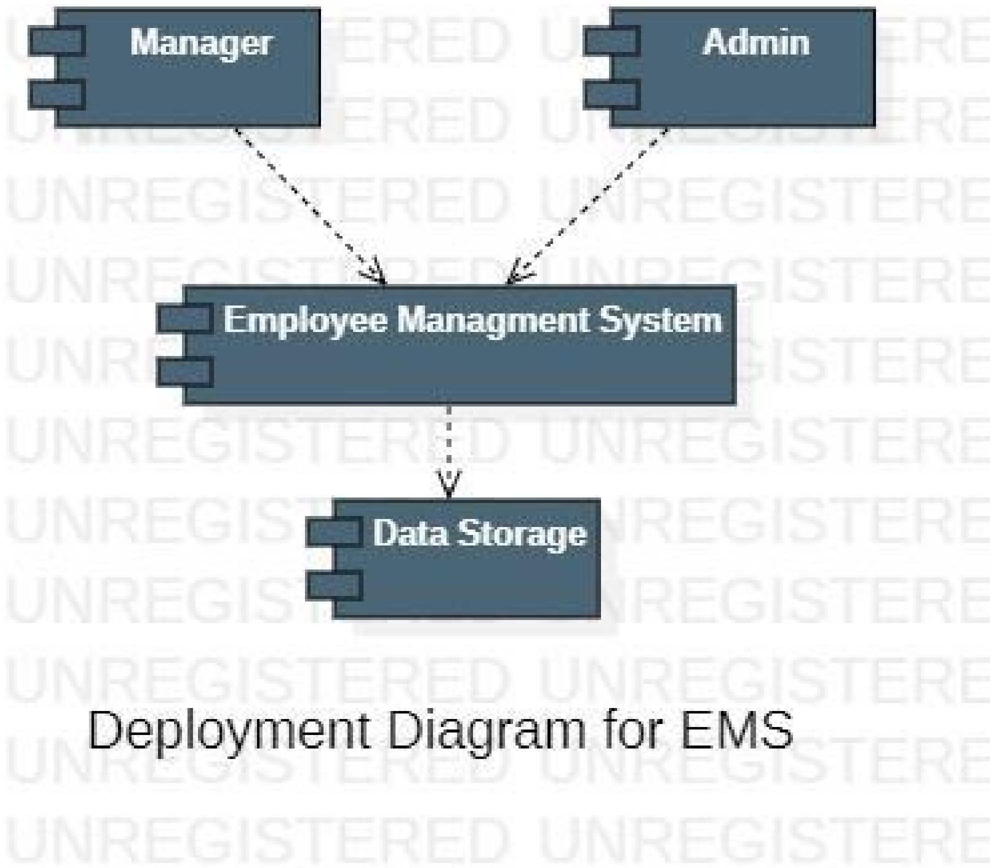
State Chart diagram for EMS, Employee Salary payment part

COMPONENT DAIDRAM:



Component Diagram For EMS

DEPLOYMENT DIAGRAM:



Deployment Diagram for EMS

LESSON - 05

LIBRARY INFORMATION SYSTEM

AIM:

To draw the diagrams [use case, class, activity, sequence, collaboration, state-chart, component and deployment] for the Library information system.

PROJECT DESCRIPTION:

The library management system is a software system that issues books and magazines to registered students only. The student has to login after getting registered to the system. The borrower of the book can perform various functions such as searching for desired book, get the issued book and return the book.

USE CASE DIAGRAM:

This diagram will contain the actors, use cases which are given below

- **Actors**: Librarian, Member.
- **Use case**: Register Member, Enquiry, Verify Member, Issue book, Check availability of book, Return book, Calculate Fine, Maintaining Book.

CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBUTES	OPERATIONS
Book	Bookid, author, name, price, rackNo, status, edition, date of purchase.	displayBookdetails(), updatestatus().
Librarian	Name, Password.	Searchbook(), verifyMember(), issueBook(), Calculatefine(),createbill(), returnbook()

Transaction	Transid, memberid, bookid, dateofIssue, dueDate.	createTransaction(), deleteTransaction(), retrieveTransaction().
Bill	billNo, date, memberid,amount.	billCreate(), billUpdate().
Member Record	Memberid, type, dateofMembership, noofbookissued, maxbooklimit, name, address, phoneNo.	retreiveMember(), increasebookissued(), decreasebookissued(), paybill().

ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

- **Activities**: Enquiry about books, Check availability of books, Validate member, Book not available, Register member, Check number of books issued to member, Book not issued, Issue book, Add member, book and issue detail, update book status.
- **Decision box:** Valid or not

SEQUENCE DIAGRAM:

This diagram consists of the objects, messages and return messages. A Sequence diagram simply depicts interaction between objects in a sequential order.

Object: Librarian, Book, Member record, Transaction.

COLLABORATION DIAGRAM:

This depicts the relationships and interactions among software objects. They are used to understand the object architecture within a system rather than the flow of a message as in a sequence diagram.

State Diagram:

State Diagram is used to capture the behaviour of a software system. UML State machine diagrams can be used to model the behaviour of a class, .

a subsystem, a package, or even an entire
system

COMPONENT AND DEPLOYMENT DIAGRAMS

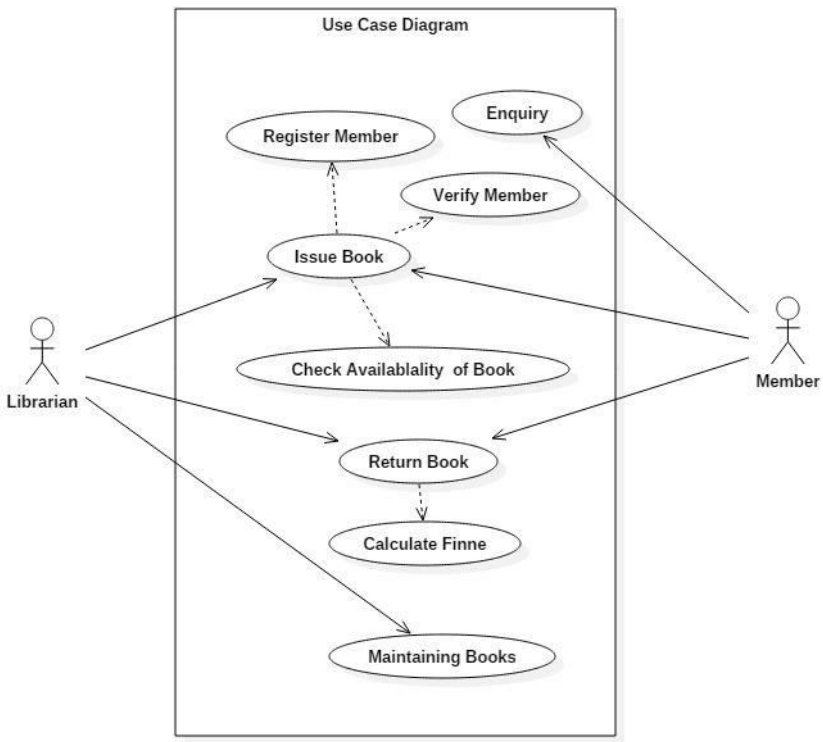
COMPONENT DIAGRAM:

- Component diagrams are used to visualize the organization and relationships among components in a system. These diagrams are also used to make executable systems.

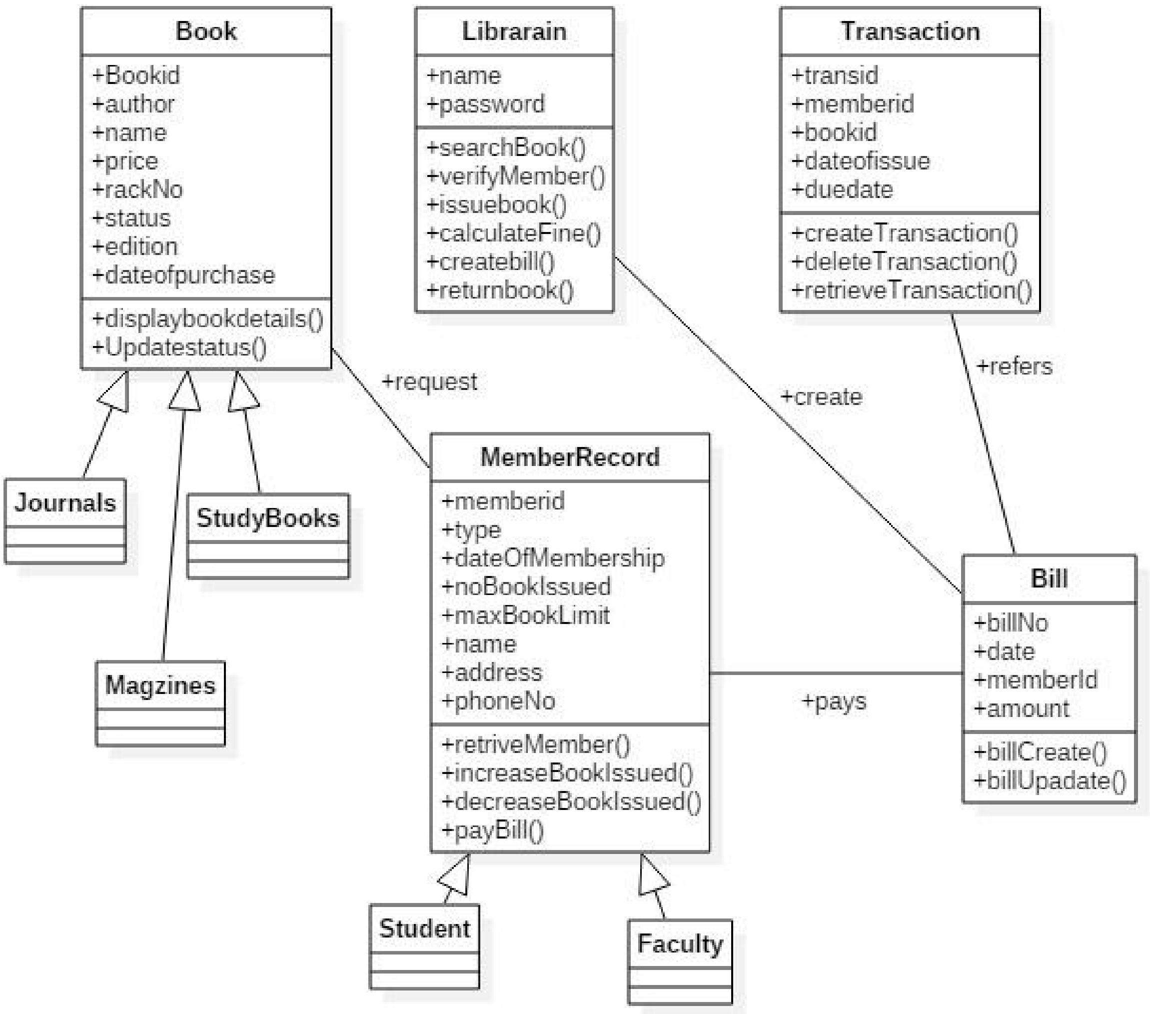
DEPLOYMENT DIAGRAM:

- Deployment diagrams are used to visualize the topology of the physical components of a system, where the software components are deployed.

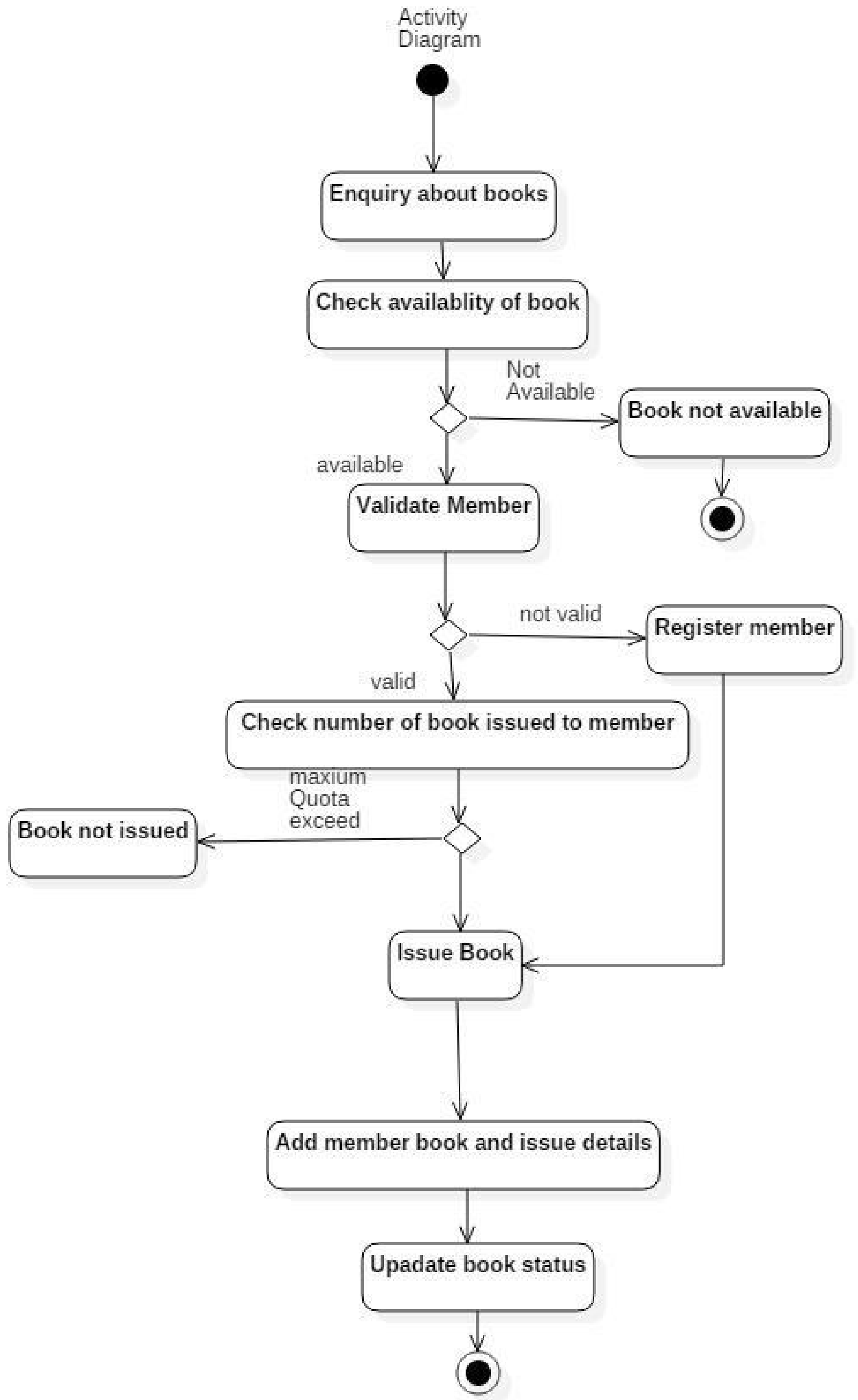
USE CASE DIAGRAM:



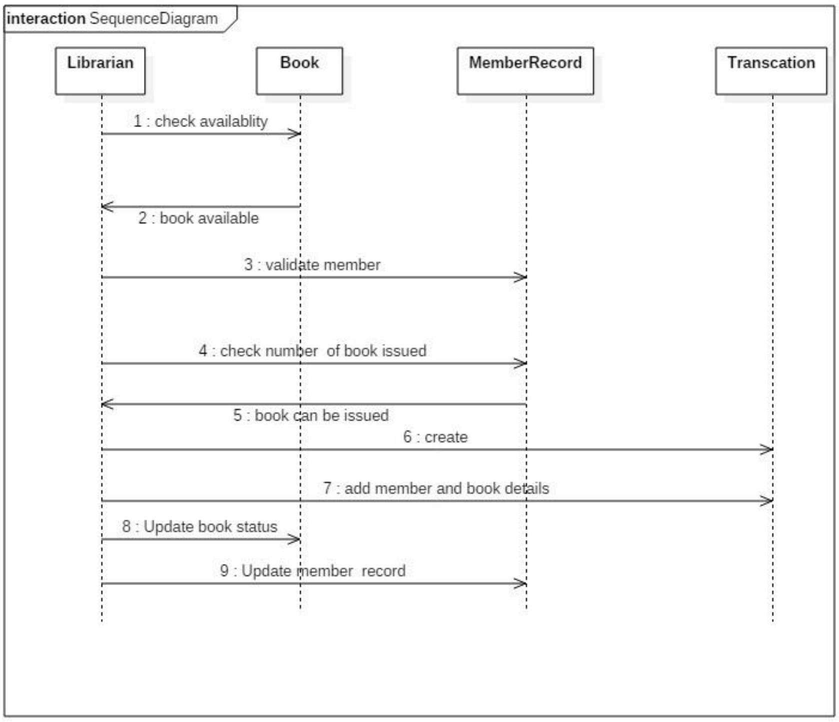
CLASS DIAGRAM:



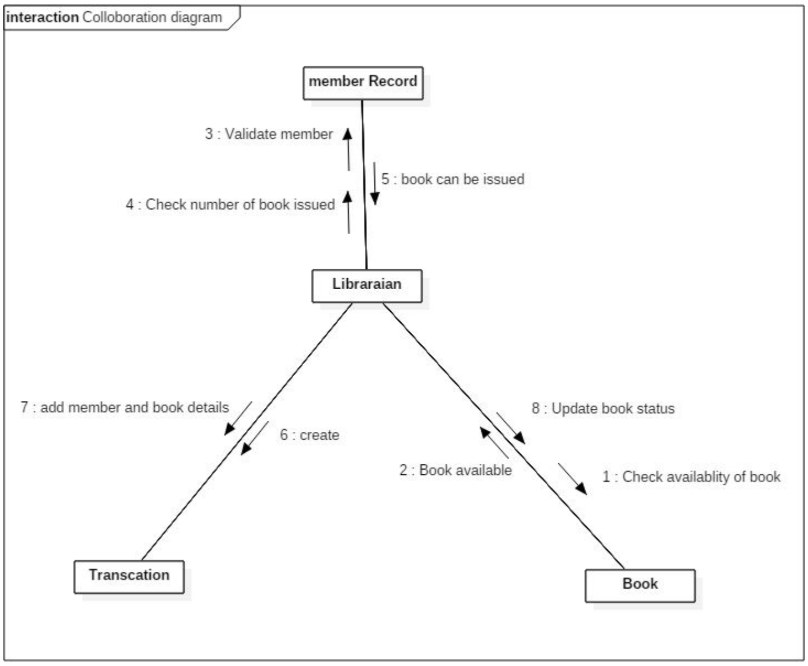
ACTIVITY DIAGRAM:



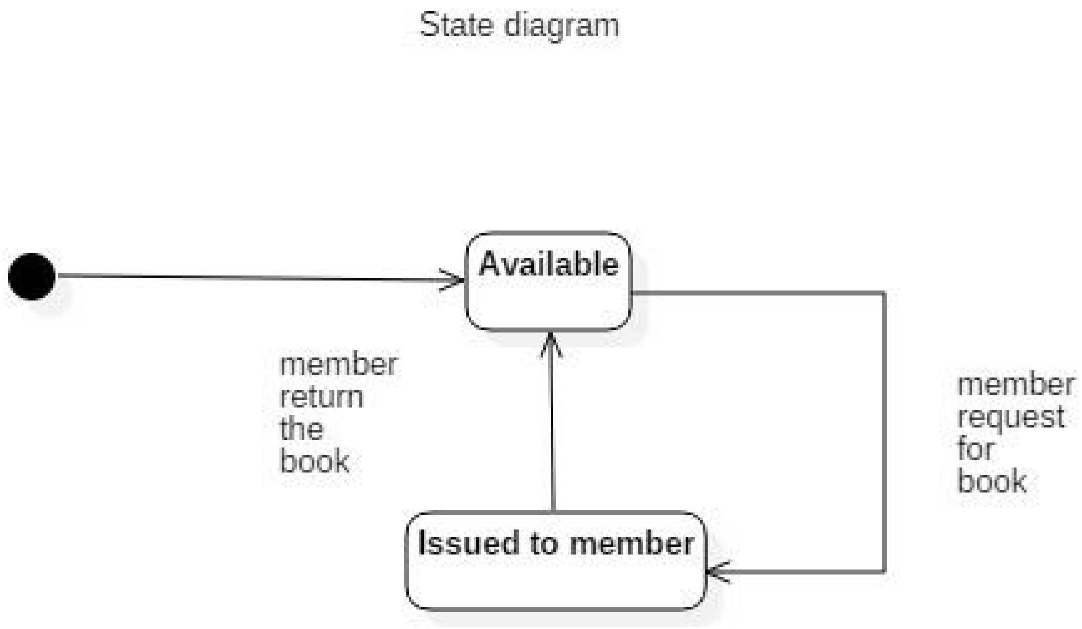
SEQUENCE DIAGRAM:



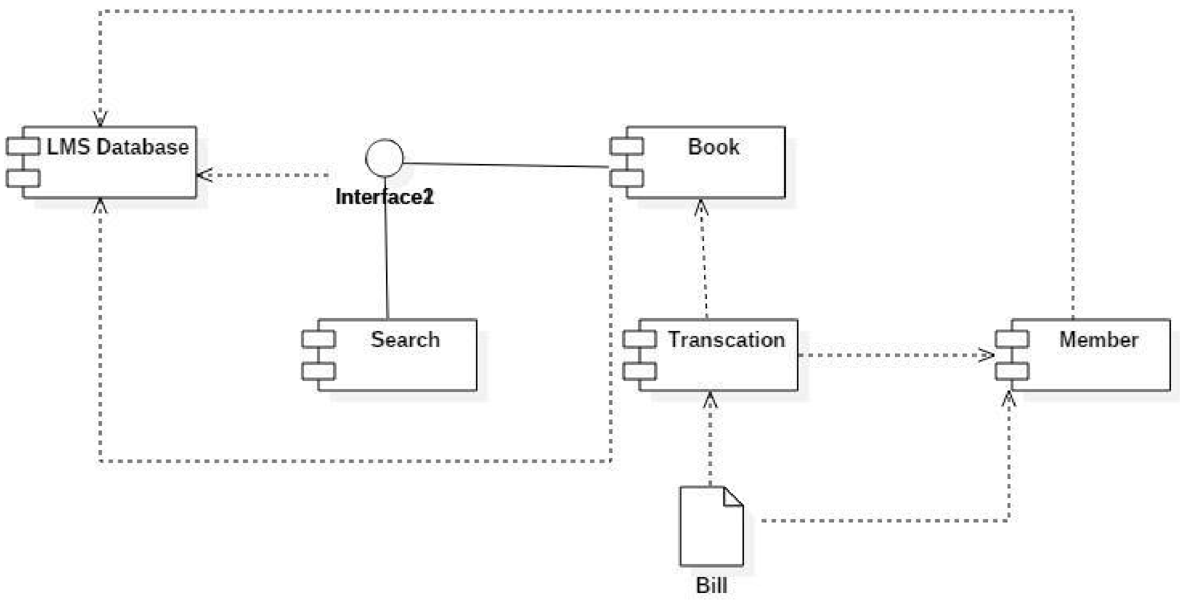
COLLOBORATION DIAGRAM:



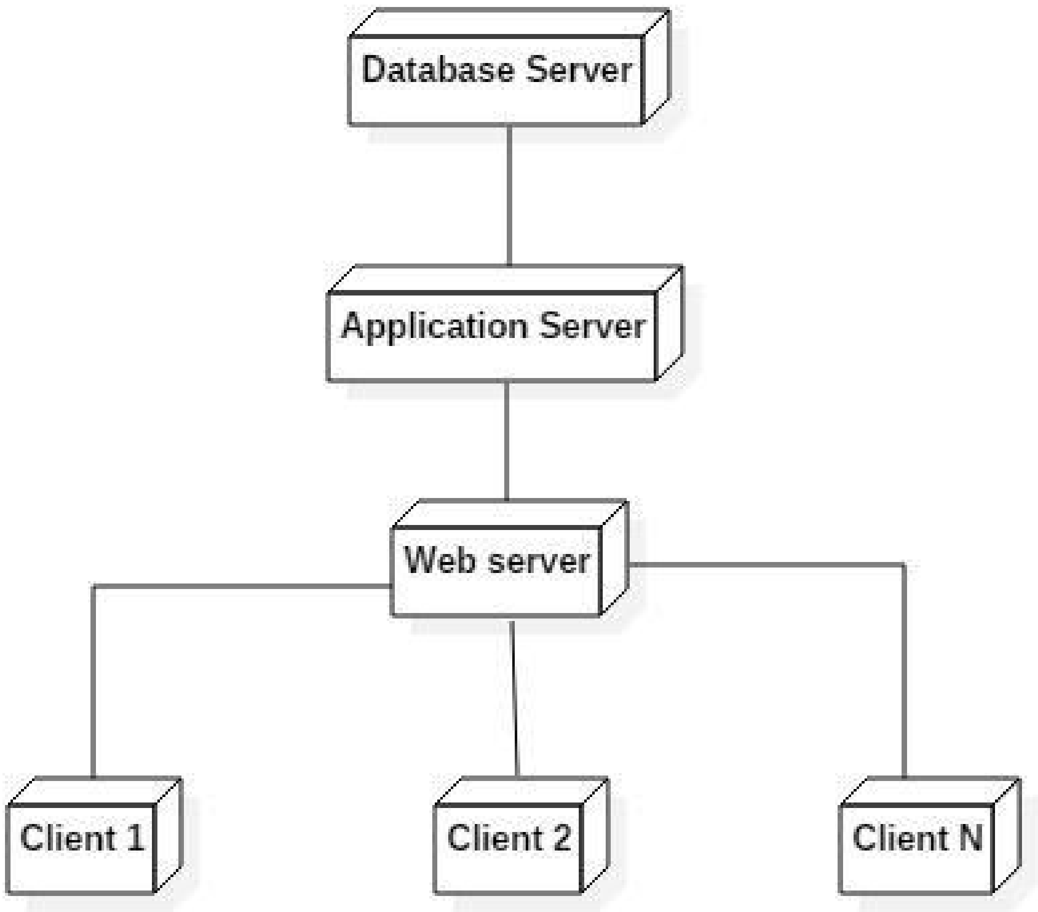
STATE CHART DIAGRAM:



COMPONENT & DEPLOYMENT DIAGRAM COMPONENT:



DEPLOYMENT:


Database Server
Application Server
Web server
Client 1
Client 2
Client N

LESSON: 06

STUDENT INFORMATION SYSTEM

AIM:

To draw the diagrams [use case, class, activity, sequence, collaboration, state-chart, component and deployment] for the Student Information System.

PROBLEM STATEMNET:

A Student Information System is a software application for educational establishments to manage student database. It can be used by educational institutes or colleges to maintain the records of students easily. It provides capabilities for entering student exam marks, attendance, uploading time table, announcements etc.

- **Admin:**
  - i. Add new teacher
  - ii. Remove teacher
  - iii. Add new student
  - iv. Modify student
  - v. Remove student
  - vi. Add class schedule
  - vii. Add announcements
- **Staff:**
  - i. Prepare attendance
  - ii. Declare student marks
- **Student:**
  - i. View attendance
  - ii. View marks
  - iii. View time table

USE CASE DIAGRAM:

This diagram will contain the actors, use cases which are given below

- Actors: Admin, Staff and Student.
- Use case: Login, view attendance, view marks, view time table, mark attendance, declare marks, view announcements, add staff, remove staff, add student, modify student, remove student, add class schedule, add announcements.

ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

• Activities:

i. Registration and login

ii. Student profile, add, modify or remove student

iii. Staff, add, modify or remove staff

iv. Attendance, mark or view attendance

v. Mark, declare and view mark

vi. Announcements, add or view announcements

vii. Time table, modify or view time table

• Decision box: Password Valid or not

CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBUTES	OPERATIONS
Admin	User ID and Password	Add new teacher Remove teacher Add new student Modify student Remove student

		Add class schedule Add announcements
Staff	User ID and Password	Prepare attendance Declare student marks
Student	User ID and Password	View attendance View marks View time table

SEQUENCE DIAGRAM:

This diagram consists of the objects, messages and return messages. A Sequence diagram simply depicts interaction between objects in a sequential order.

Object: Student, Login, View profile, Update profile, View Result and Database.

COLLABORATION DIAGRAM:

This depicts the relationships and interactions among software objects. They are used to understand the object architecture within a system rather than the flow of a message as in a sequence diagram.

Object: Student, Staff, Admin, System and Database.

STATE DIAGRAM:

State Diagram is used to capture the behaviour of a software system. UML State machine diagrams can be used to model the behaviour of a class, a subsystem, a package, or even an entire system.

- First state is to login
- The next two states are to enter and view student details
- Then we have to enter the marks
- That entered marks will be updated and stored
- The final state is to logout

COMPONENT AND DEPLOYMENT DIAGRAMS

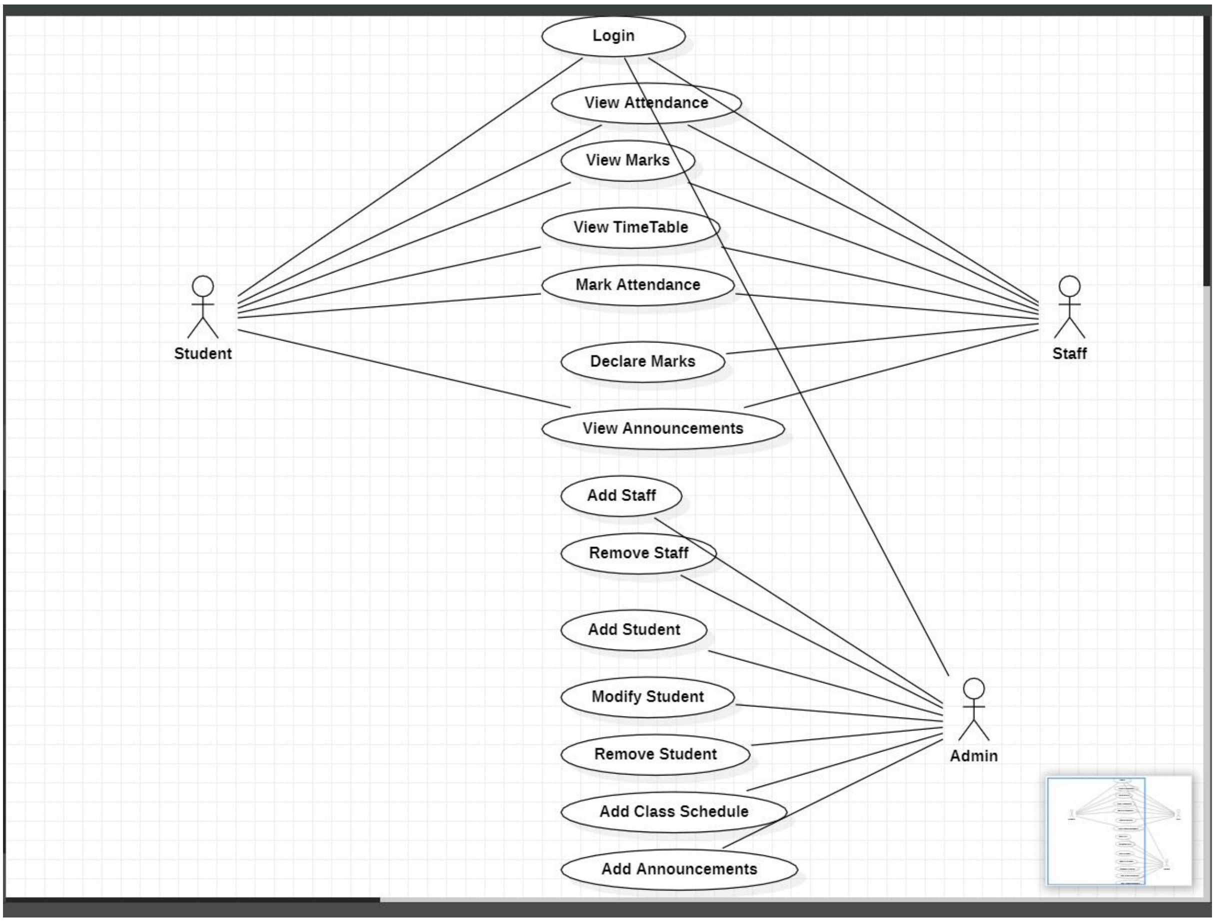
COMPONENT DAIGRAM:

- Component diagrams are used to visualize the organization and relationships among components in a system. These diagrams are also used to make executable systems.

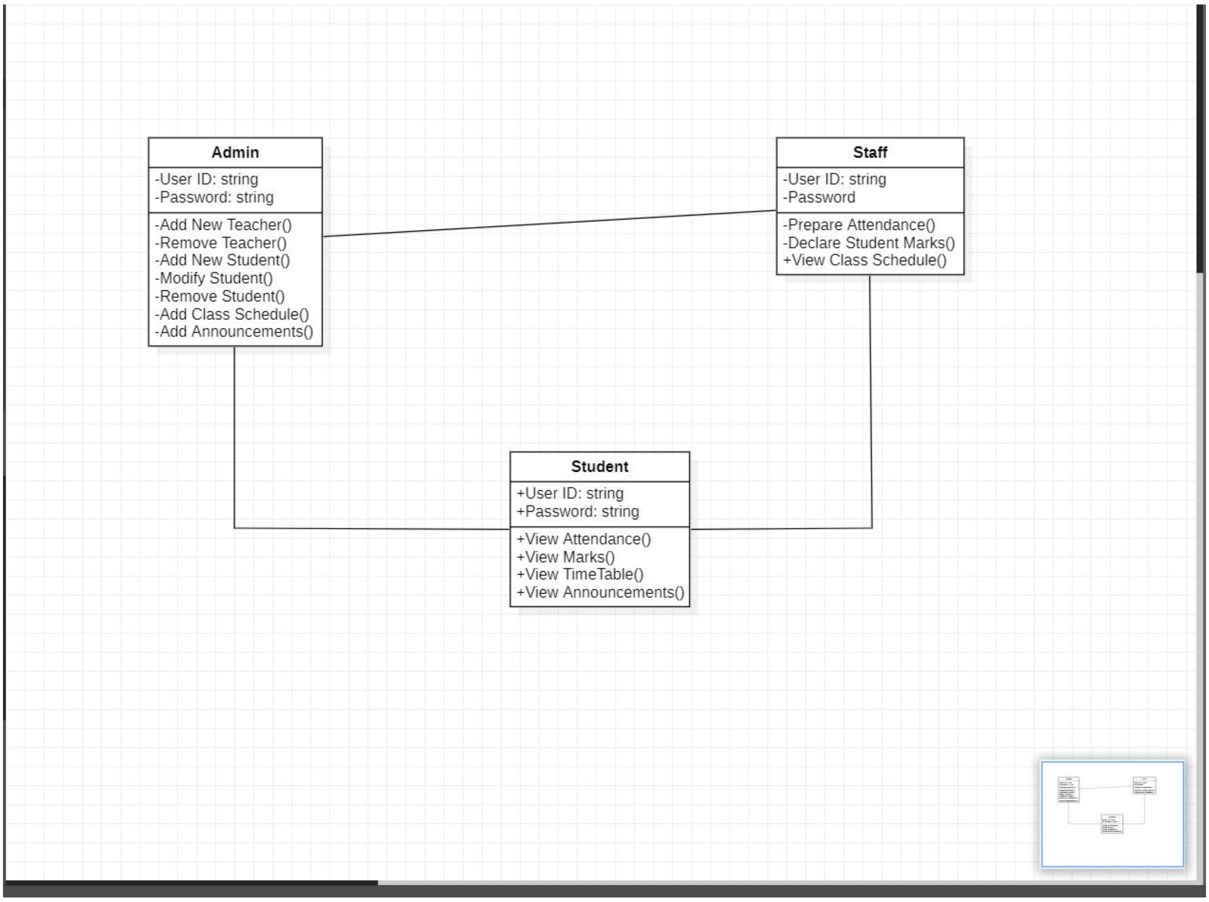
DEPLOYMENT DIAGRAM:

- Deployment diagrams are used to visualize the topology of the physical components of a system, where the software components are deployed.

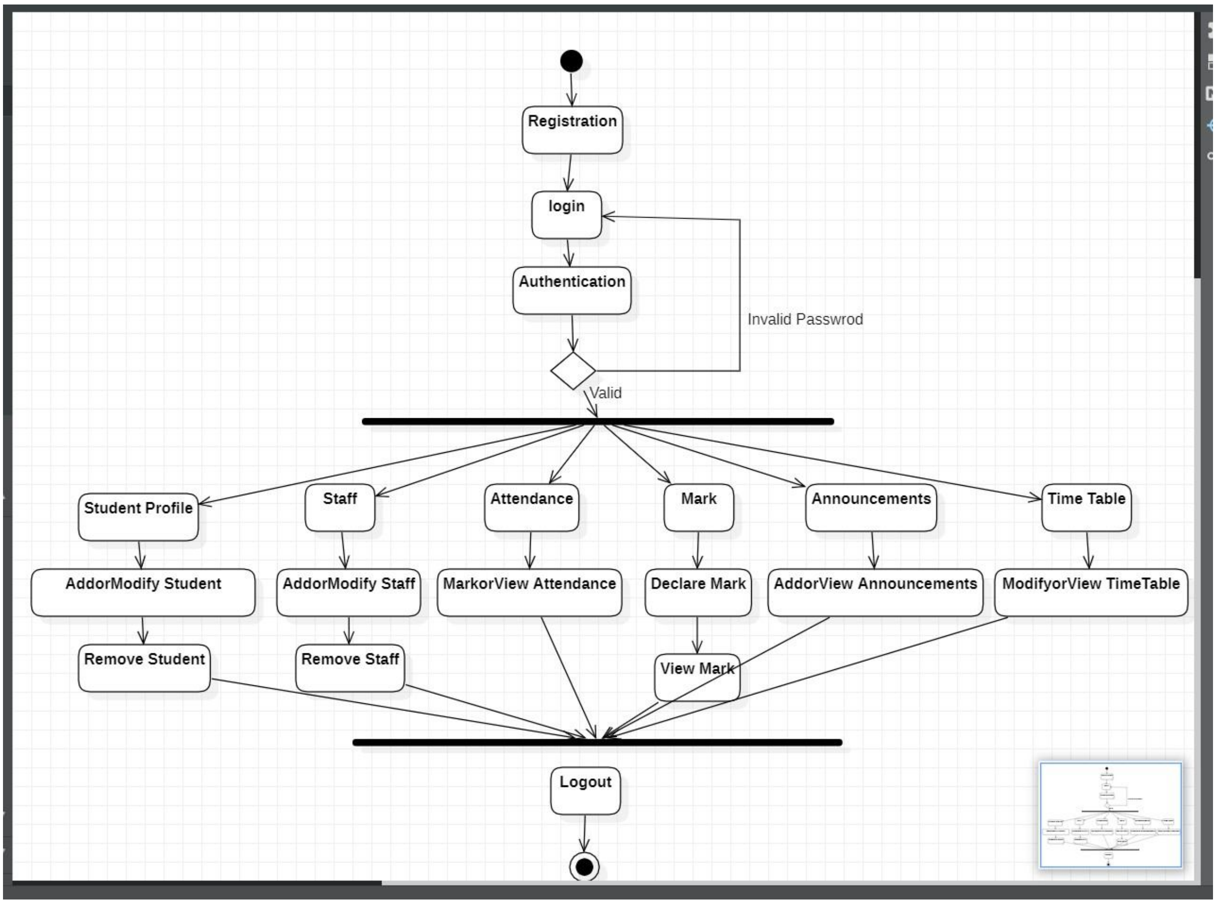
USE CASE DIAGRAM:



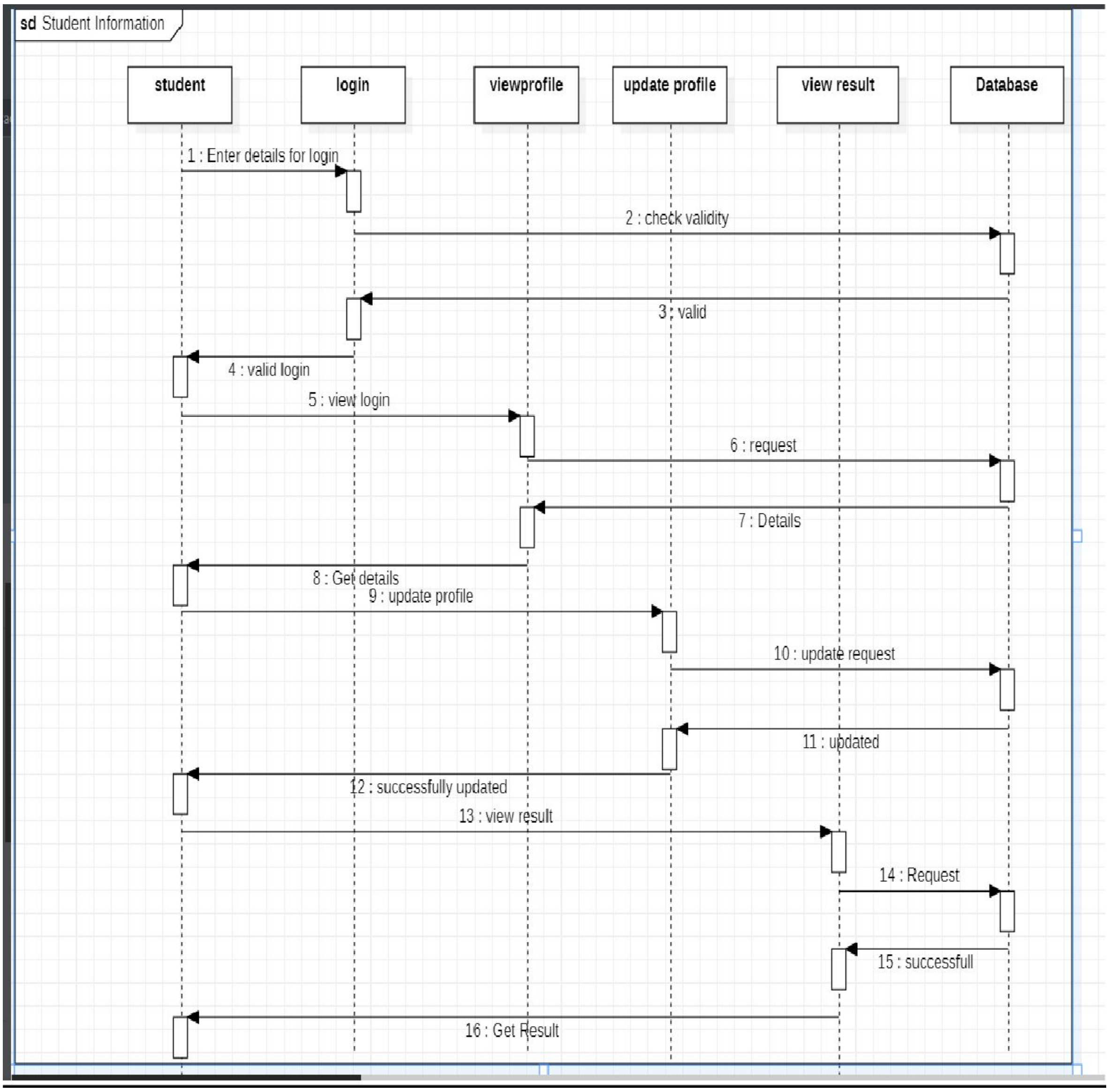
CLASS DIAGRAM:



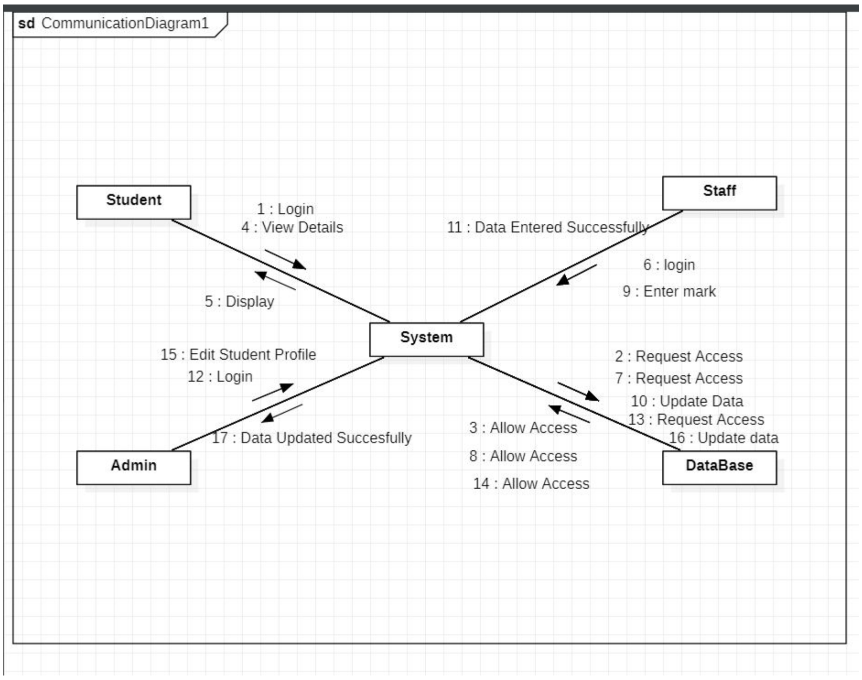
ACTIVITY DIAGRAM:



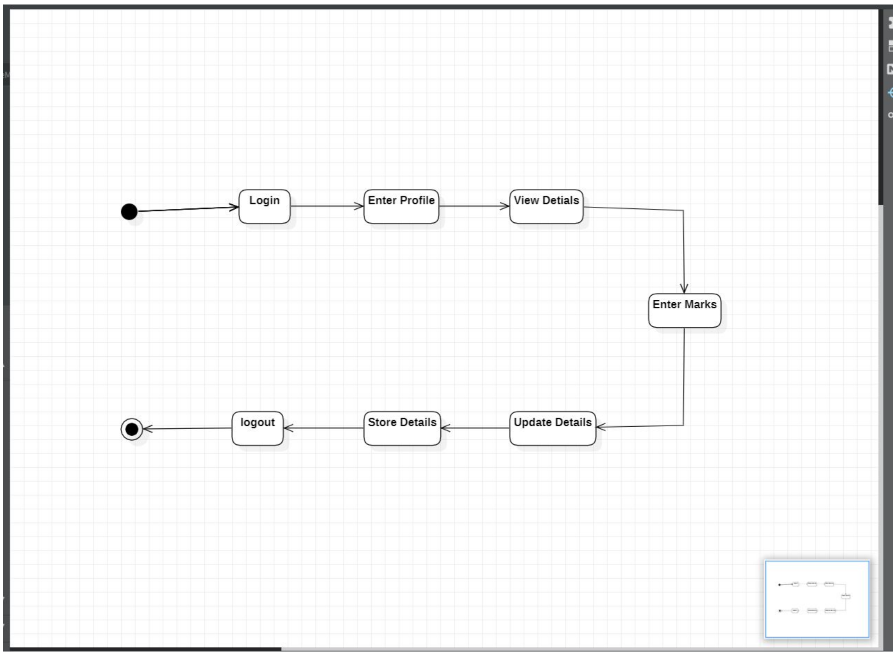
SEQUENCE DIAGRAM :



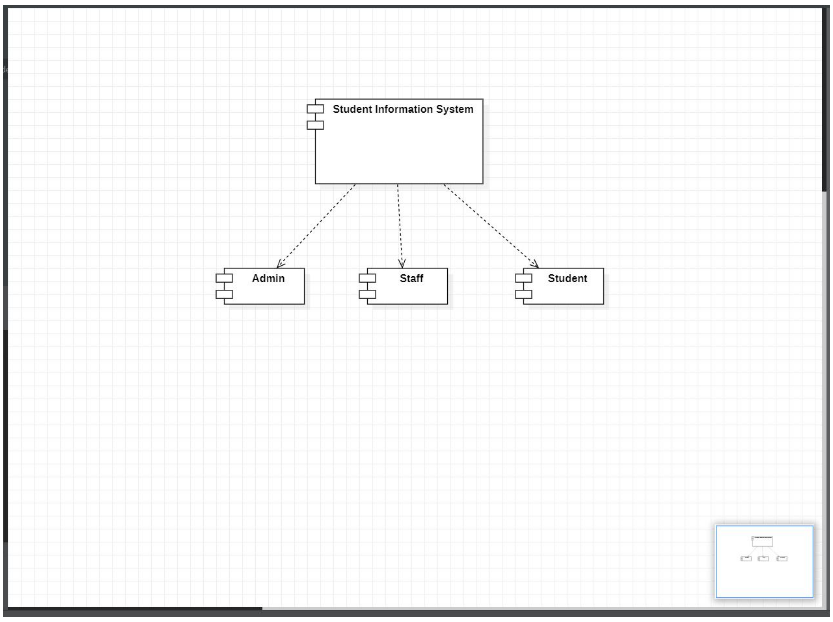
COLLABORATION DIAGRAM :



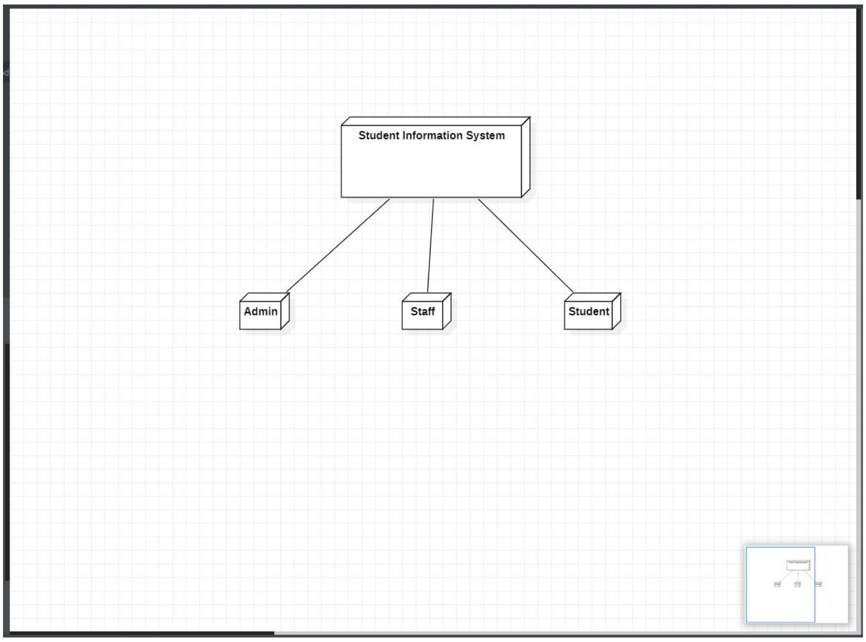
STATE- CHART DIAGRAM



COMPONENT DIAGRAM



DEPLOYMENT DIAGRAM



LESSON: 07

EXAM REGISTRATION SYSTEM

AIM:

To create a system to perform the Exam Registration system by using Star UML

PROBLEM ANALYSIS AND PROJECT PLANNING:

In the Exam registration system, the main process is a student have to login the database then the database verifies that particular username and password then student select the subject and next the process taken to the fees payment processing. There completing the processing of fees payment and finally acknowledgement giving to the student and controller of examiner.

PROBLEM STATEMENT:

Exam Registration system.is used in the effective dispatch of registration form to all of the students. This system adopts a comprehensive approach to minimize the manual work and schedule resources, time in a cogent manner. The core of the system is to get the online registration form (with details such as Userid, Password etc.,) filled by the student whose testament is verified for its genuineness by the Exam Registration System with respect to the already existing information in the database. After all the necessary criteria have been met, the original information is added to the database and the hall ticket is sent to the student.

USECASE DIAGRAM:

The Exam Registration use cases in our system are:

- Login - The student enters his userid and password to login.
- View exam details - The student views the details about the exam schedule which contains Date, time, etc...
- Select subject - The student selects the subject which he / she want subjects.
- Fee processing - All the details should be viewed by both the student and the controller to verify whether all the entered details are correct.
- Confirmation processing - The selected subject's exam registration is confirmed.

- Acknowledgment - The exam fees should be paid by the student to get the hall ticket from the exam controller.

ACTIVITY DIAGRAM:

The Activity diagram shows the activity of the process. An activity diagram is a variation or a Special case of a state machine in which the states or activity representing the performance of Operation and transition are triggered by the completion of operation. The propose is to provide view Of close and what is going inside a use case.

CLASS DIAGRAM:

The class diagram, also referred to as object modelling is the main static analysis diagram. The main task of object modelling is to graphically show what each object will do in the problem domain. The problem domain describes the structure and the relationships among objects. The Exam Registration System class diagram consists of three classes of registration system.

DOCUMENTATION OF CLASS DIAGRAM:

This class diagram has three classes student, Exam registration Database.

- Student - logins the Exam Registration page and select the required data fields.
- Exam registration Database - verify the login and Select the details for exam register
- Amount - Payment should be processed after the confirmation of Student.
- Acknowledgement - Acknowledgement forward controller of examiner, student and management

SEQUENCE DIAGRAM:

The sequence and collaboration diagram represents that the student enter the information to get the hall ticket and the exam controller issues the hall ticket after verifying the necessary itemsand this data are stored in the database.

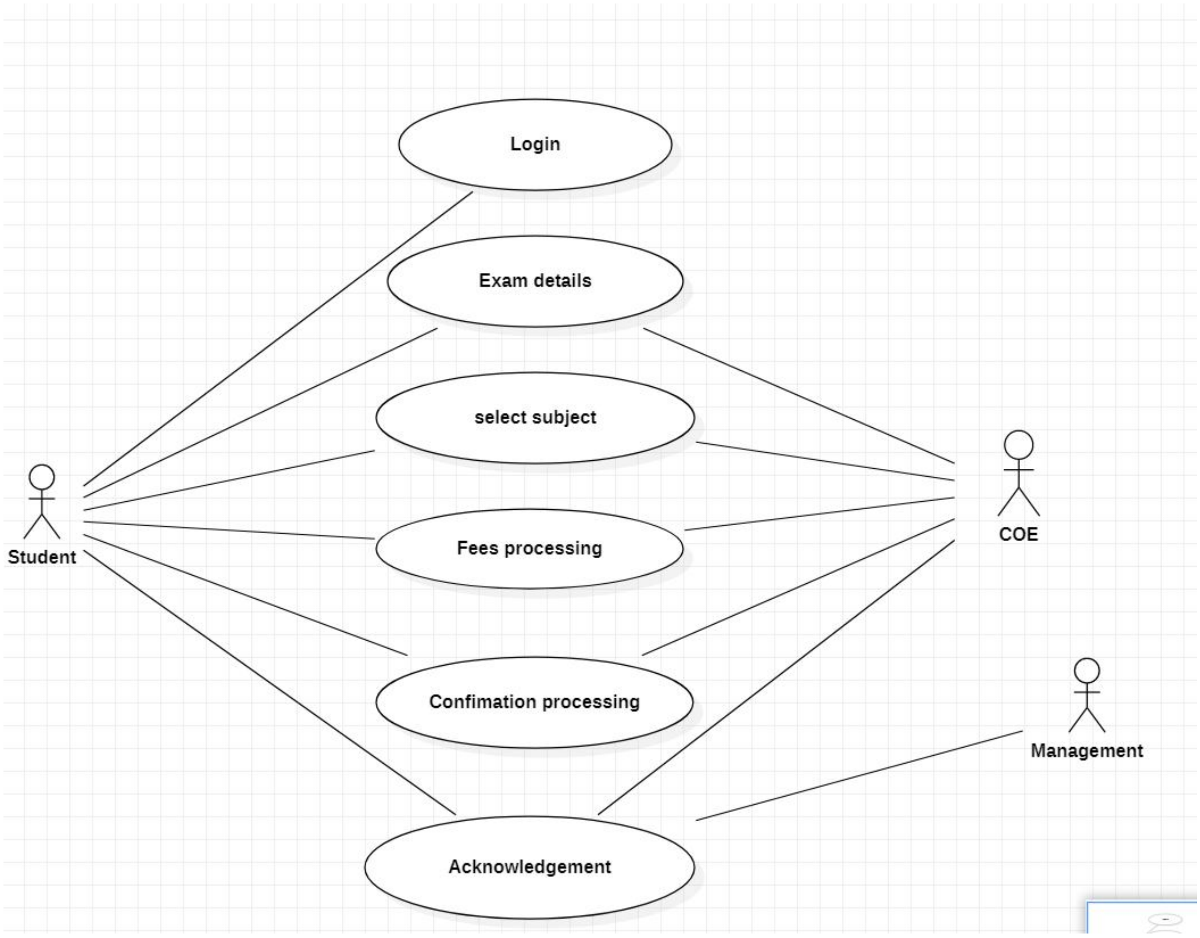
DEPLOYMENT DIAGRAM:

Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.

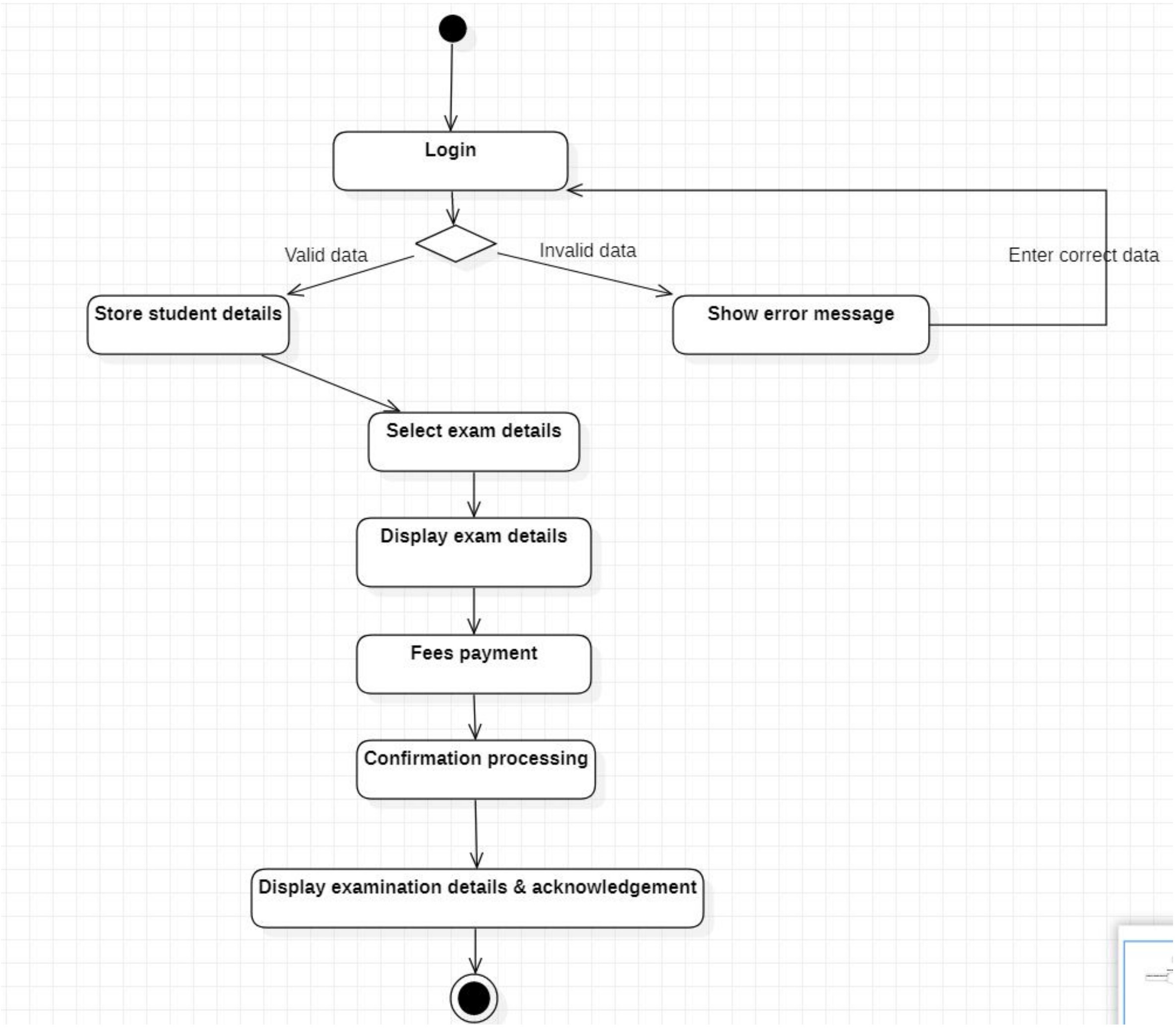
COMPONENT DIAGRAM:

Component diagrams are used to visualize the organization and relationships among components in a system.

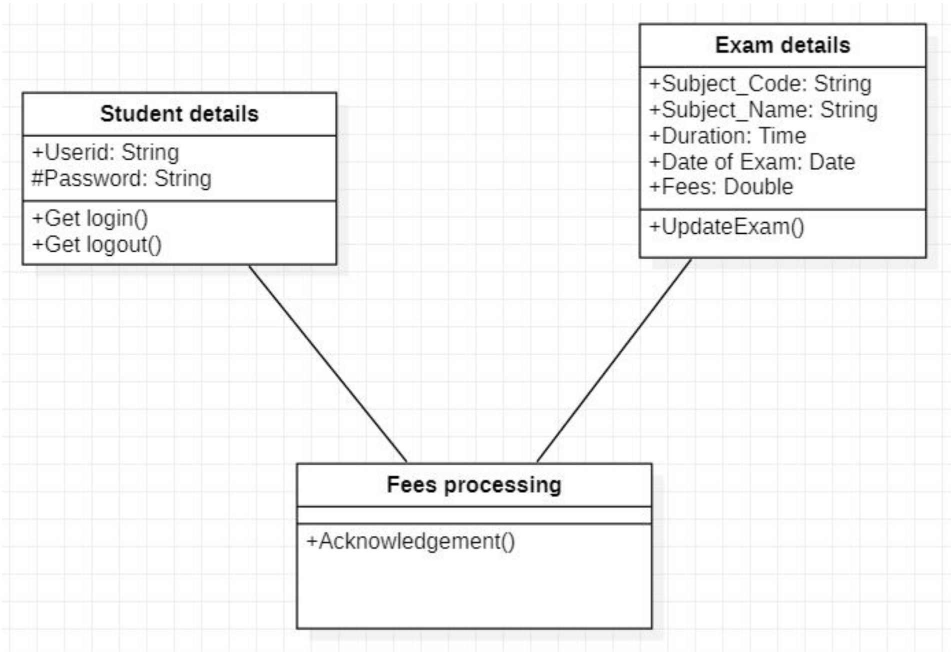
USE CASE DIAGRAM:



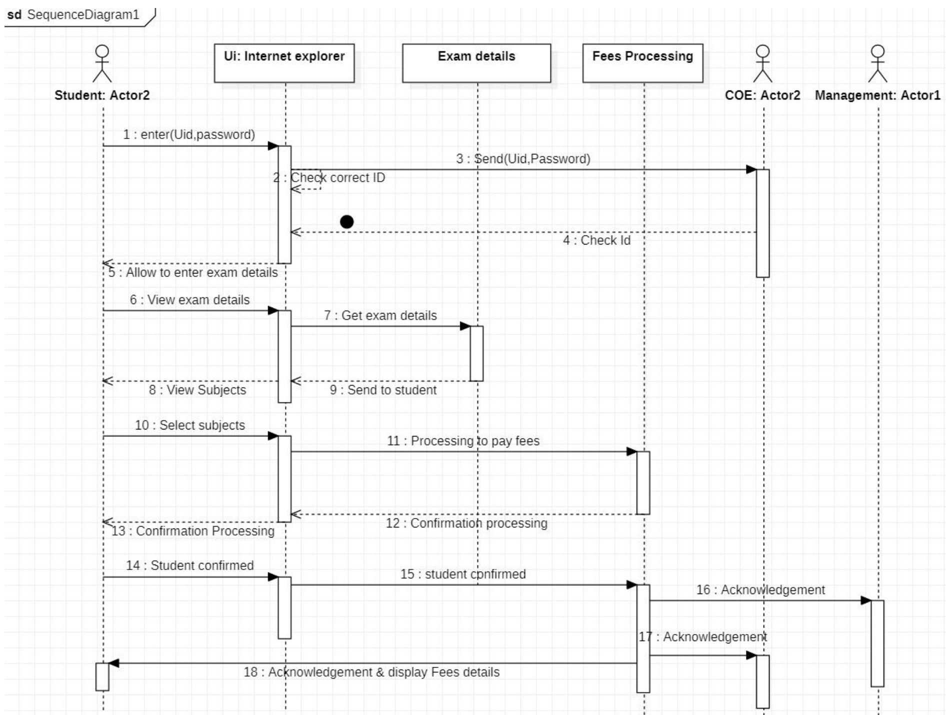
ACTIVITY DIAGRAM:



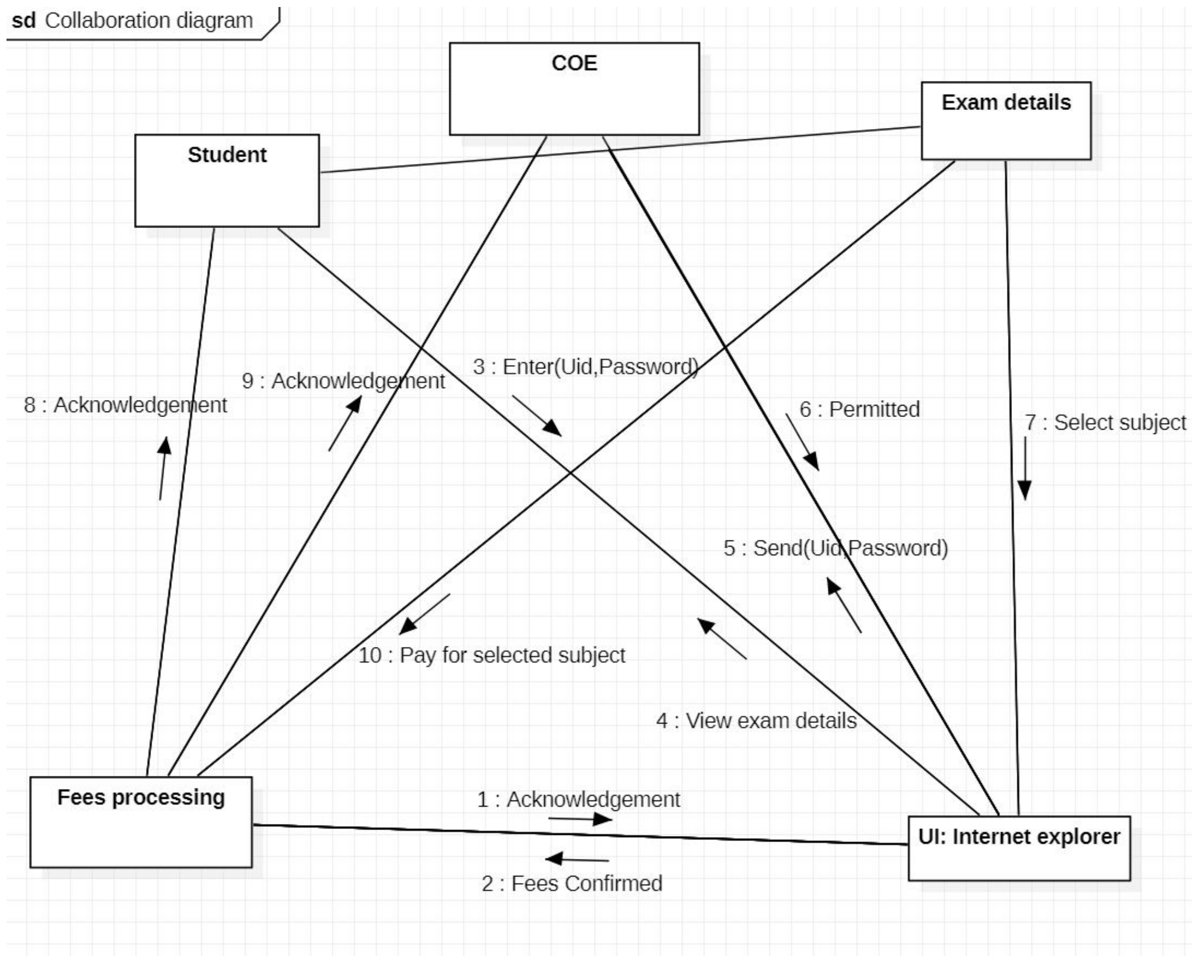
CLASS DIAGRAM:



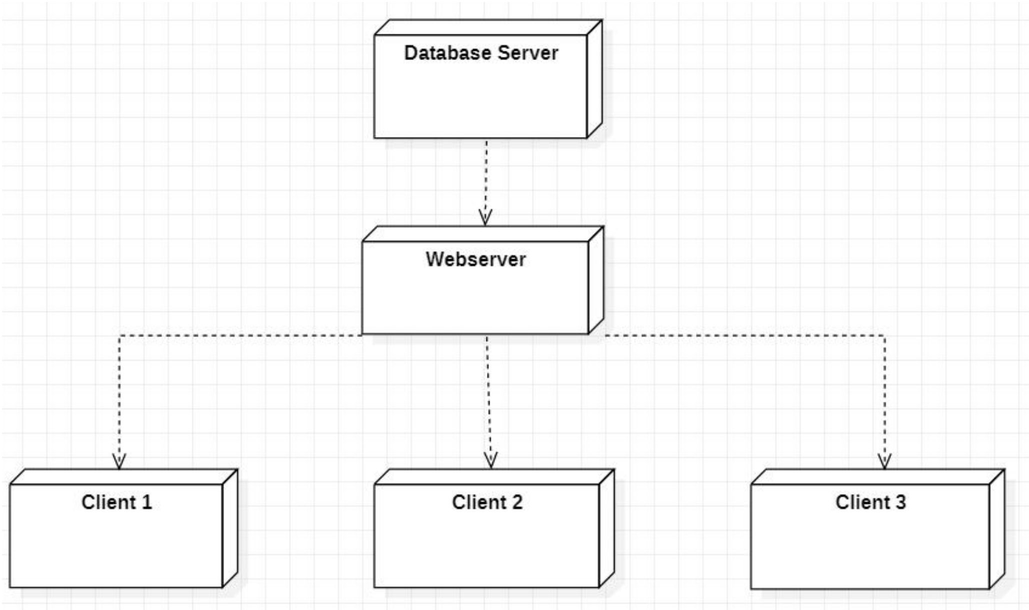
SEQUENCE DIAGRAM:



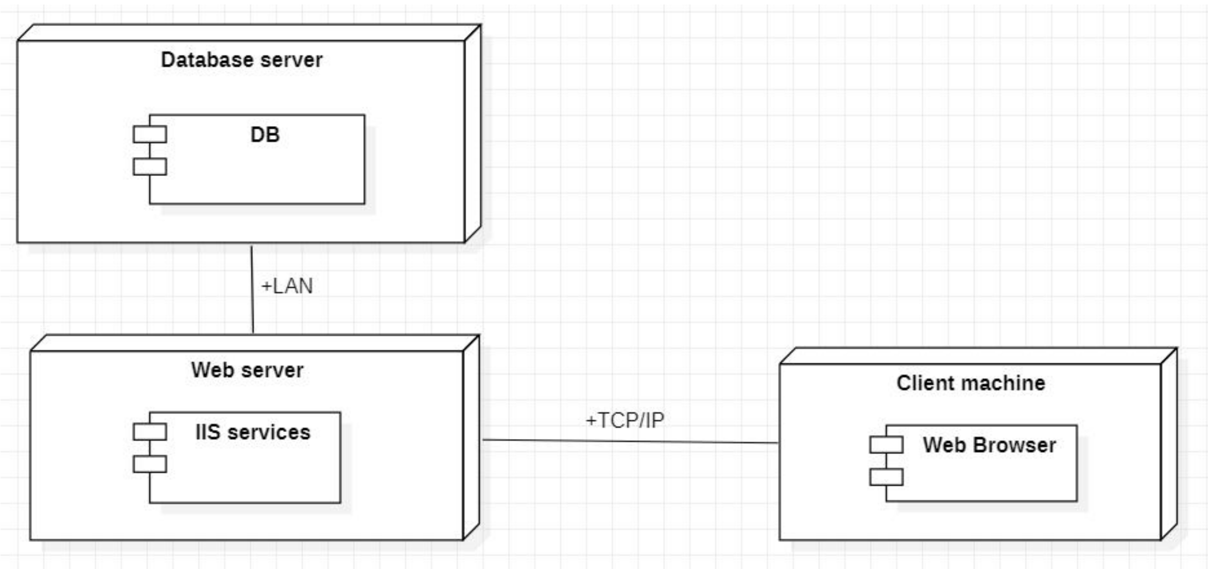
COLLABORATION DIAGRAM:



DEPLOYMENT DIAGRAM:



COMPONENT DIAGRAM:



LESSON - 08
E-TICKETING

AIM :

To draw the diagrams [use case, class, activity, sequence, collaboration, state-chart, component and deployment] for the E-Ticketing.

PROBLEM ANALYSIS AND PROJECT PLANNING:

In the E-Ticketing system the main process is a applicant have to login the database then the database verifies that particular username and password then the user must fill the details about their personal details then selecting the flight and the database books the ticket then send it to the applicant then searching the flight or else cancelling the process.

PROBLEM STATEMENT:

The E-Ticketing system is the initial requirement to develop the project about the mechanism of the E-ticketing system what the process do at all.

- The requirement are analyzed and refined which enables the end users to efficiently use the E-ticketing system.
- The complete project is developed after the whole project analysis explaining about scope and project statement is prepared.
- The main scope for this project is the applicant should reserve for the flight ticket.
- First the applicant wants to login to the database after that the person wants to fill their details.
- Then the database will search for ticket or else the person will cancelled the ticket if he/she no need.

USE CASE DIAGRAM:

A use case is a methodology used in system analysis to identify, clarify, and organize system requirements. The use case is made up of a set of possible sequences of interactions between systems and users in a particular environment and related to a particular goal. It is represented using ellipse.

Actor is any external entity that makes use of the system being modelled. It is represented using stick figure.

The use cases are the activities performed by actors.

The actors in this use case diagram are Applicant - logins the E-Ticketing and filling the required data fields.

- E-Ticketing Database-verify the login and filling the details and selected applicant details are stored in it. The use cases in this use case diagram are Login - applicant enter their username and password to enter in to the E- Ticketing form.
- Administrator-administrator can access the data and solve the enquiry and issues of the applicant.

ACTIVITY DIAGRAM :

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modelling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control. An activity is shown as a rounded box containing the name of the operation.

DOCUMENTATION OF ACTIVITY DIAGRAM :

This activity diagram describes the behaviour of the system.

- First state is login where the applicant login to the E- Ticketing system.
- The next state is filling details the applicant are used to fill the form.
- Then applicant used to select the flight.
- The applicant appears for book ticket and search details from E-Ticketing

CLASS DIAGRAM:

A class diagram in the unified modelling language (UML) is a type of static structure diagram that describes the structure of a system by showing the system’s classes, their attributes, and the relationships between the classes. It is represented using a rectangle with three compartments. Top compartment have the class name, middle compartment the attributes and the bottom compartment with operations.

DOCUMENTATION OF CLASS DIAGRAM :

This class diagram has Five classes applicant, E-Ticketing Database.

- Applicant - logins the E-Ticketing and filling the required data fields.
- E-Ticketing Database-verify the login and filling the details and selected
- Administrator-Administrator can access the details of the applicant and provide the needful.
- Amount-Payment should be processed after the confirmation of flight ticket.
- Airlines-Airlines are used to book the ticket and processed in the E-Ticketing Database.

CLASS	ATTRIBUTE	METHODS
Applicants	PassportCode, First Name, Last Name, Address, Nationality, Phone, Sex, Passport Number, ResidentPermit	Login(),Filling Details()
Airlines	Flight Name, Flight No, Booking Date, Destination, Departure place, Departure Date	Selecting Flight(), Book Ticket(), Cancel Ticket()
Amount		Payment()
E-Ticketing Database	Print	Ticket Booking(), Verify Ticket()
Administrator	Issue	Helpline(), Enquiry()

SEQUENCE DIAGRAM

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction. Diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. There are two dimensions.

- Vertical dimension-represent time.
- Horizontal dimension-represent different objects.
- Now the E-Ticketing Database verify the filling Details.
- And then the E-Ticketing Database display the ticket information.
- In case of any sudden change of the plan the applicant can cancel the ticket.

DOCUMENTATION OF SEQUENCE DIAGRAM:

This sequence diagram describes the sequence of steps to show

- Applicant are used to login the form and then it verify the username and password
- If the password and username are correct then applicants are used to login the filling details
- Applicants are used to selecting the flights and book the tickets.
- E-Ticketing Database- verify the login and filling the details and selected applicant details are stored in

COLLABORATION DIAGRAM :

A collaboration diagram, also called a communication diagram or interaction diagram. A sophisticated modelling tool can easily convert a collaboration diagram into a sequence diagram and the vice versa. A collaboration diagram resembles a flowchart that portrays the roles, functionality and behaviour of individual objects as well as the overall operation of the system in real time.

DOCUMENTATION OF COLLABORATION DIAGRAM :

This collaboration diagram is to show how the applicant login and register in the E-Ticketing system. Here the sequence is numbered according to the flow of execution.

This collaboration diagram is to show the selection process of the applicant for the ticket booking. The flow of execution of this selection process is represented using the numbers.

STATE CHART DIAGRAM:

The purpose of state chart diagram is to understand the algorithm involved in performing a method. It is also called as state diagram. A state is represented as a round box, which may contain one or more compartments. An initial state is represented as small dot. A final state is represented as circle surrounding a small dot.

This state diagram describes the behavior of the system.

- First state is Register where the applicant register to the E-Ticketing system.
- The next state is login where the applicant can provide the details.
- The next state is filling details the applicant are used to fill the form.
- Then applicant used to selecting the flight.
- The applicant appears for book ticket and search details from ETicketing.

COMPONENT DIAGRAM:

The component diagram's main purpose is to show the structural relationships between the components of a system. It is represented by boxed figure. Dependencies are represented by communication association. The main component in this component diagram is E-Ticketing systems. And Login, Filling Details and selecting flights applicants are the components comes under the main component.

DEPLOYMENT DIAGRAM:

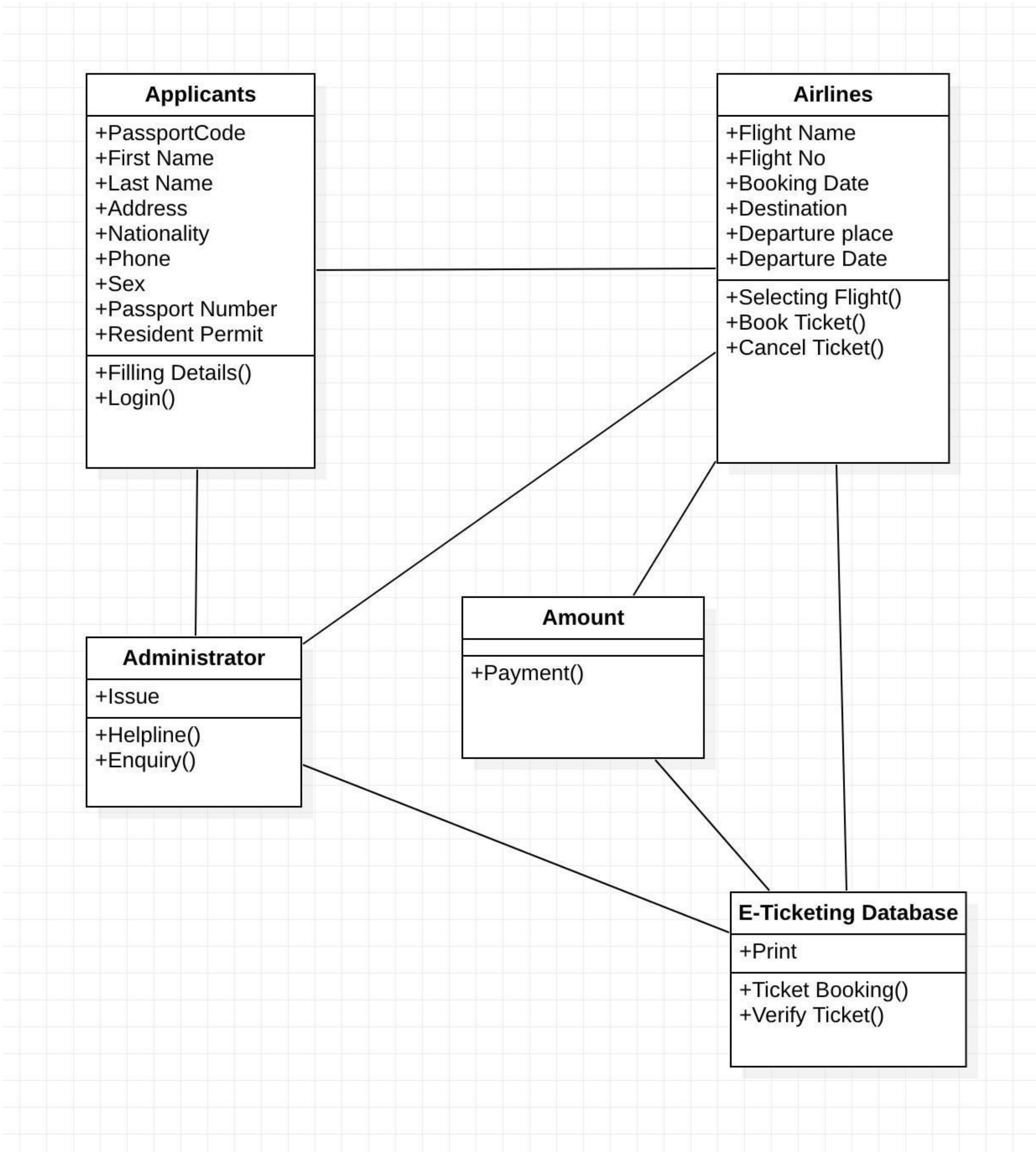
A deployment diagram in the unified modeling language serves to model the physical deployment of artifacts on deployment targets. Deployment diagrams show "the allocation of artifacts to nodes according to the Deployments defined between them. It is represented by 3dimentional box. Dependencies are represented by communication association.

DOCUMENTATION OF DEPLOYMENT DIAGRAM:

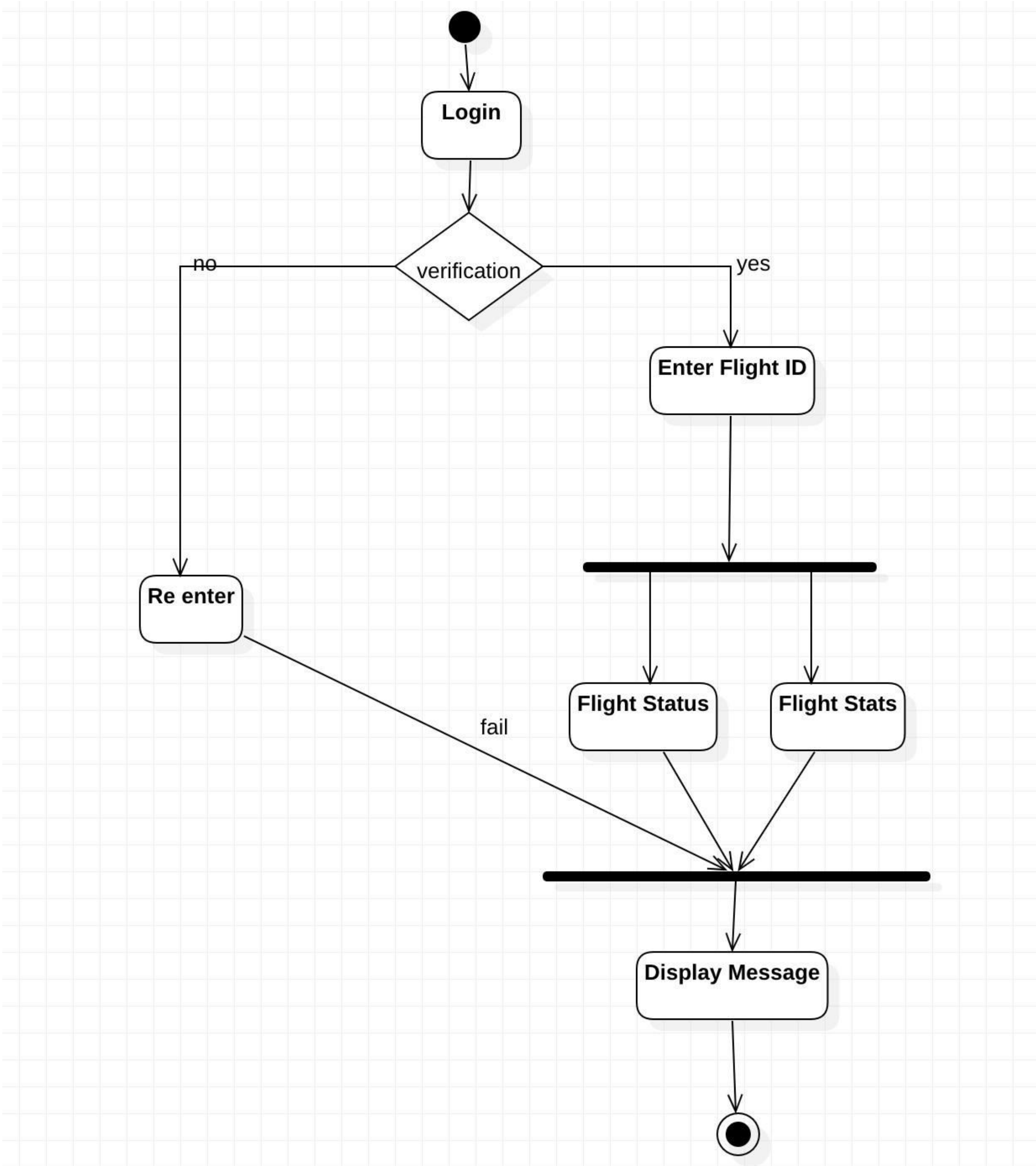
The processor in this deployment diagram is the E-Ticketing system which is the main part and the devices are the Flight ticket web server, Application server and Database server, Ticket Generator and printer which are the some of the main activities performed in the system.

USE CASE DIAGRAM:

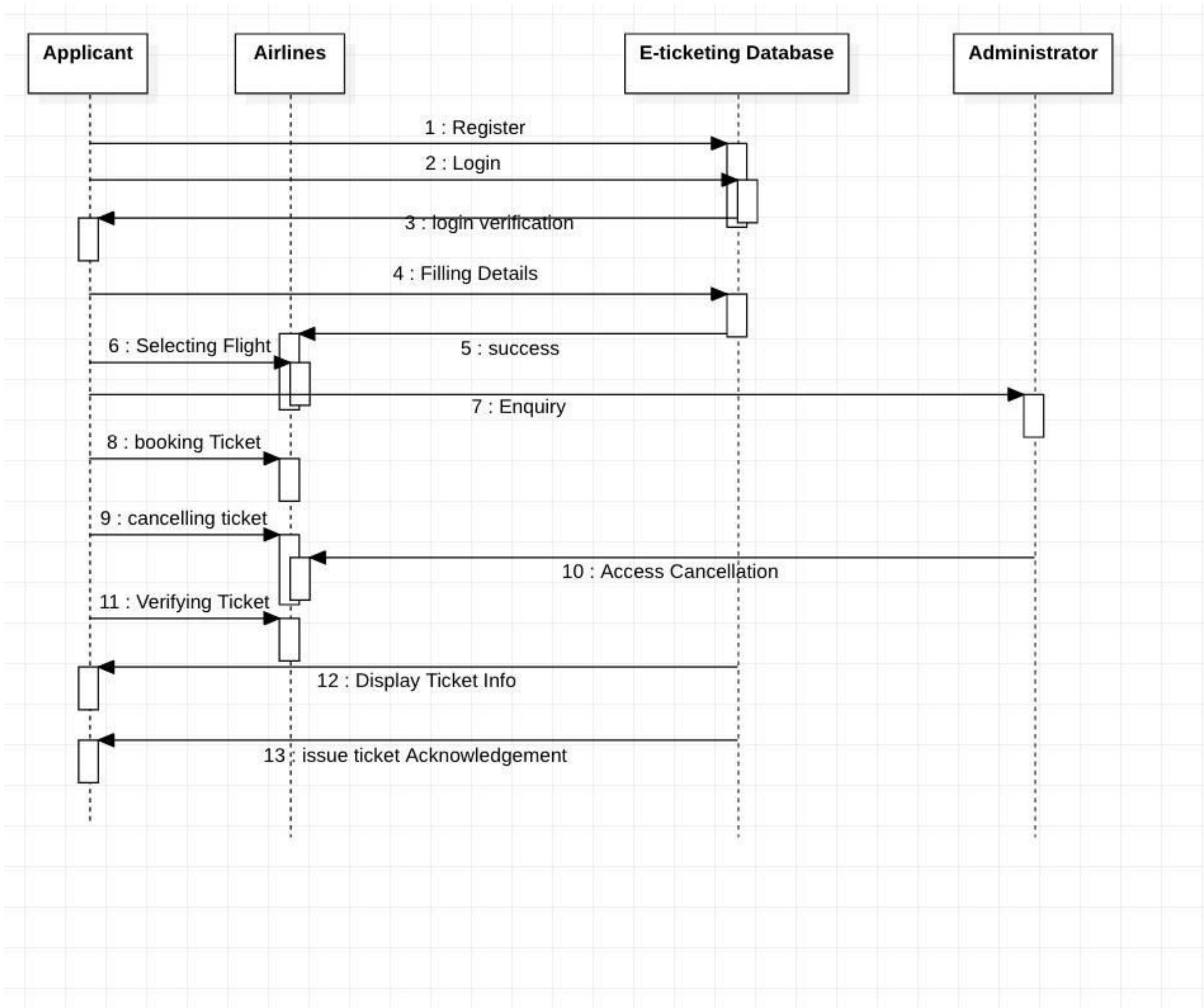
CLASS DIAGRAM:



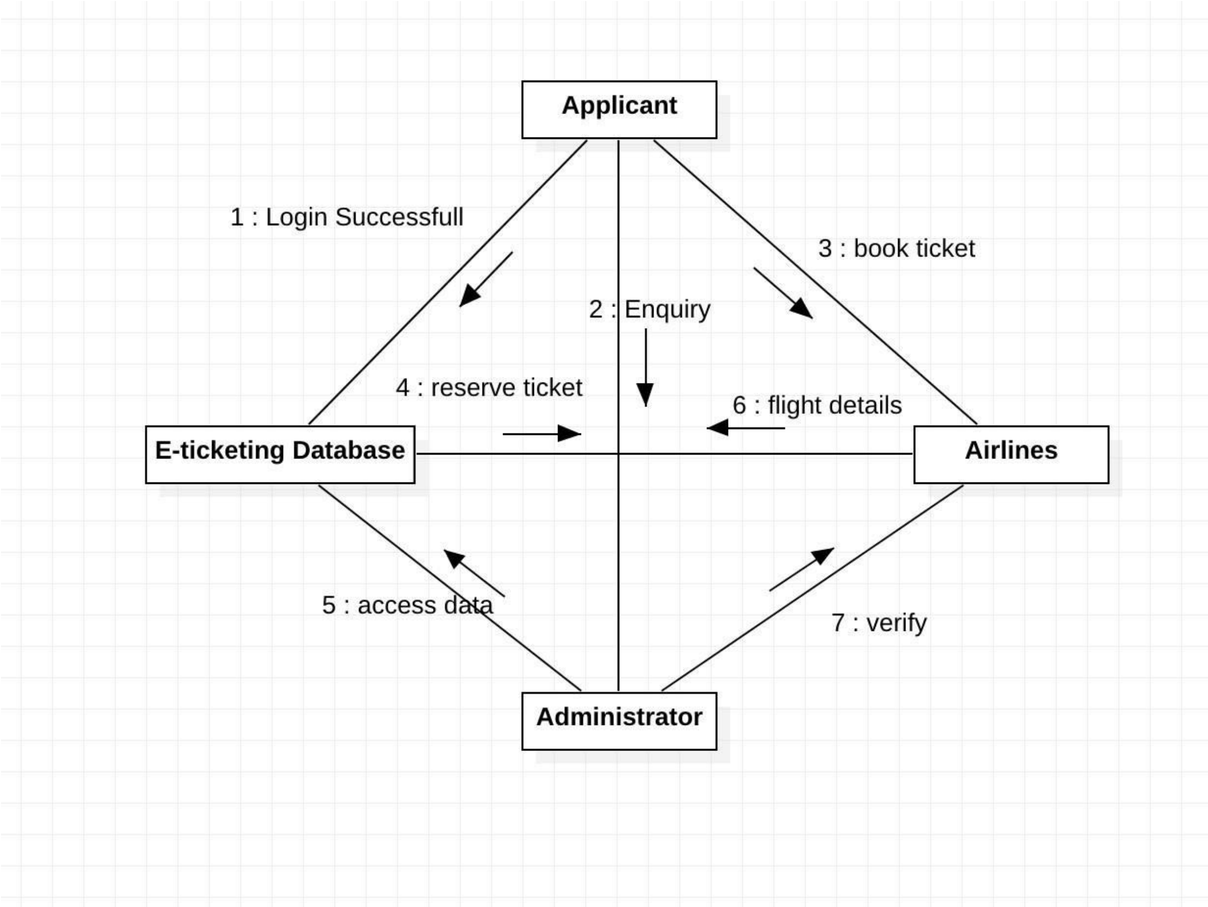
ACTIVITY DIAGRAM:



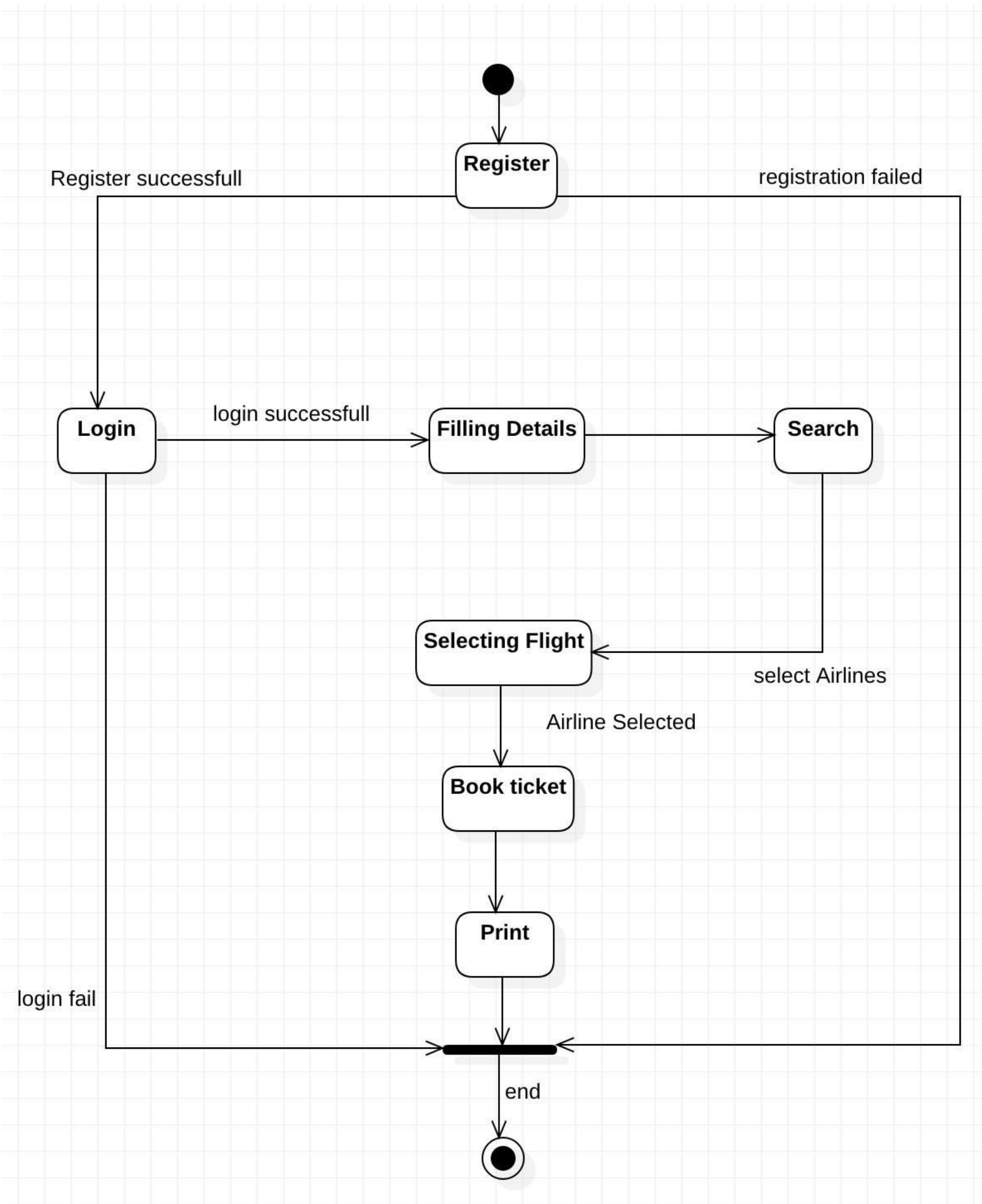
SEQUENCE DIAGRAM:



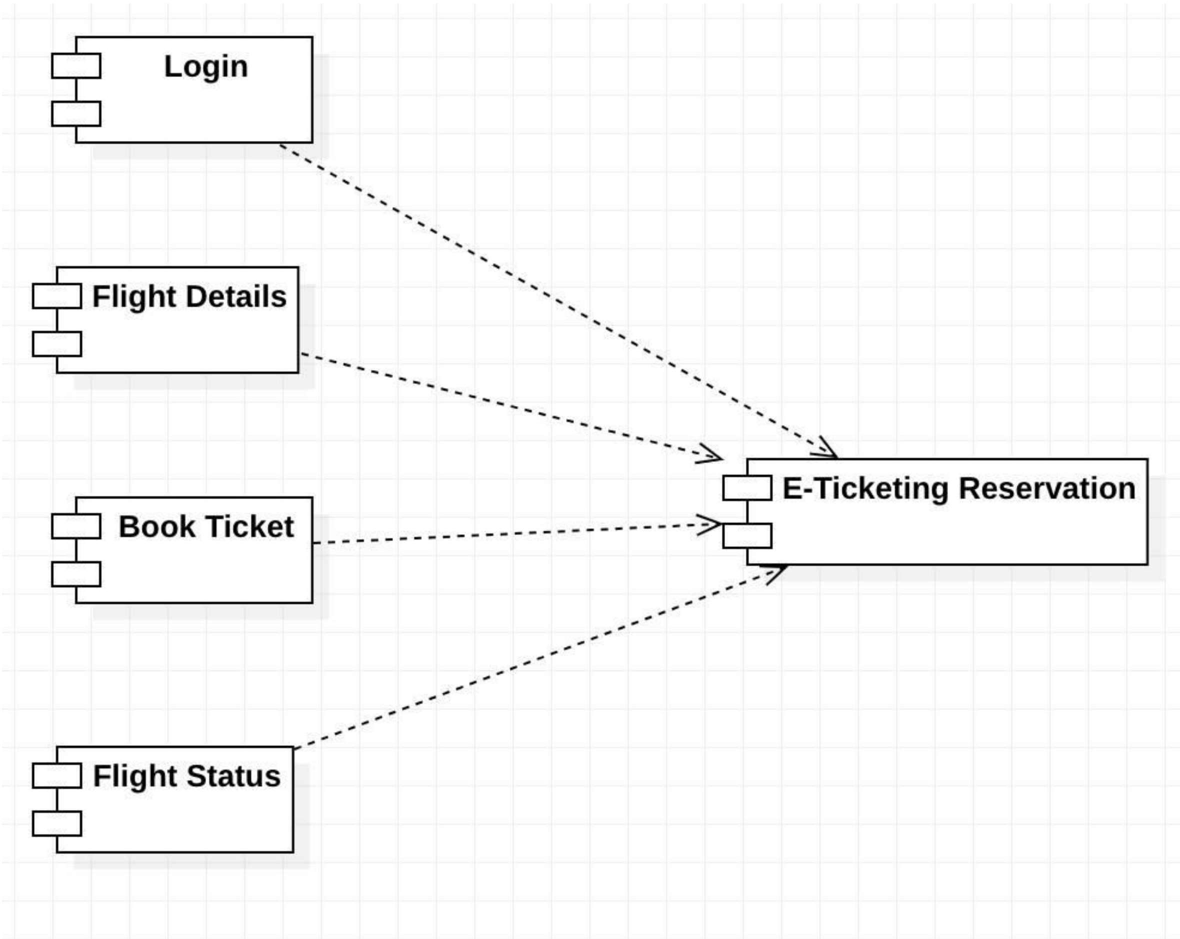
COLLABORATION DIAGRAM:



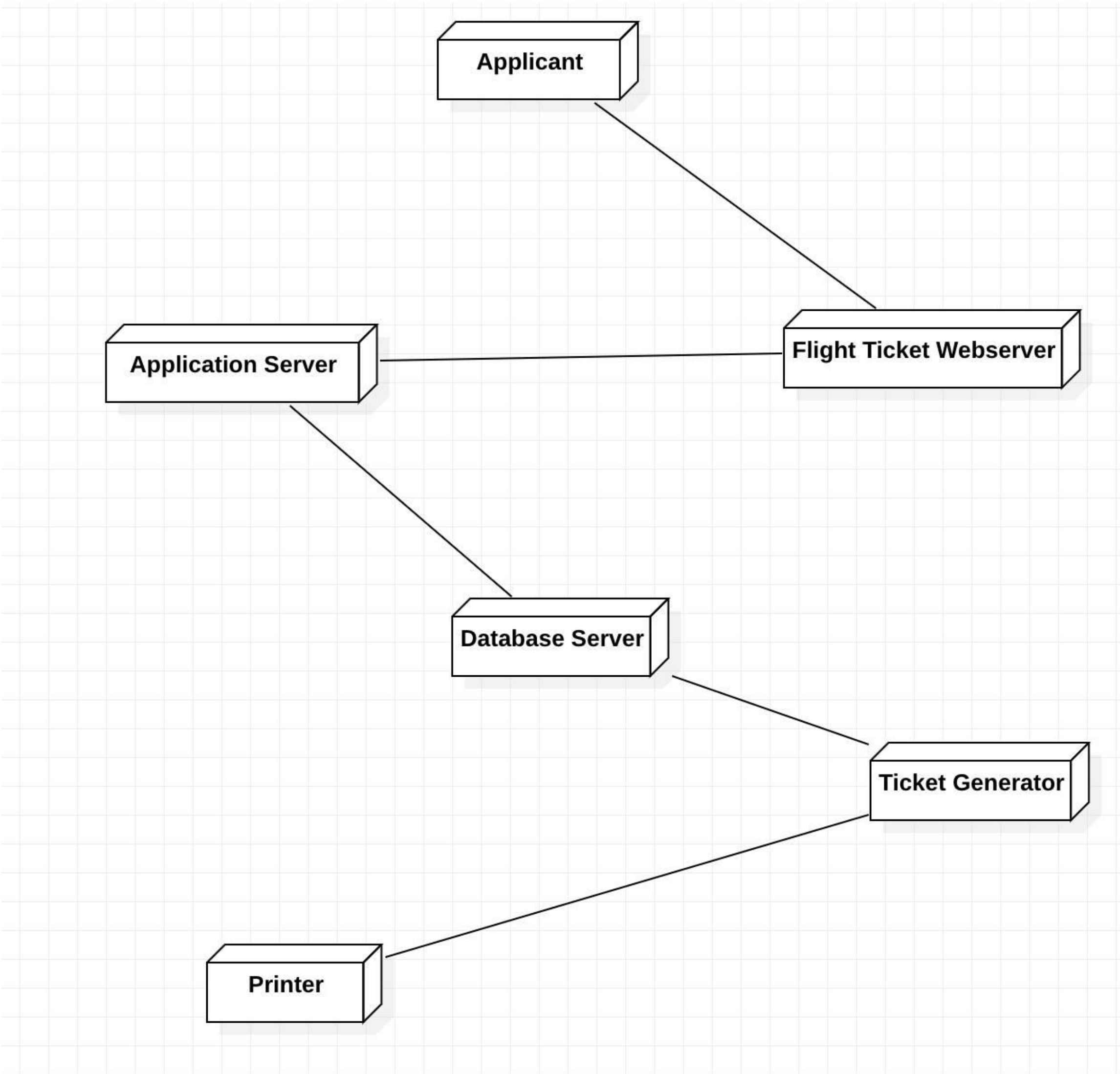
STATE CHART DIAGRAM:



COMPONENT DIAGRAM:



DEPLOYMENT DIAGRAM:



LESSON: 09
E-BOOK MANAGEMENT SYSTEM

AIM:

To draw the diagrams [use case, class, activity, sequence, collaboration, state-chart, component and deployment] for the E- Book Management System.

PROJECT DESCRIPTION:

E-Book process is well organized online buying and selling of books. This system is well developed in various resources, for example Amazon site deals more about e-booking concept. This process has various issues in the basics of maintenance of database and updating in sites, and virus problem in pdf books, so we have many issues in this process. The process of e-books is fully based on online, and the process for this mainly interaction between buyer and seller, buyer who enter the site for purchase of book will use search engine for book to purchase, the search engine will mainly focused on the database process, it used to search book for the buyer who mentioned the book name, author name, edition, publication details in the site, so that the search engine will show many books. There will be a payment option and option for pdf file or hardcopy delivery to home, the user should decide whether he want which one. Whether he choice hardcopy means, full detail address, driving license no, and then he should login with his username and password, and then payment through atm debit or credit card applicable.

USECASE DIAGRAM:

The E-book use cases in our system are:

1. Register 2. Login 3. Searchbook

4. Download 5. Payment 6. Publisher

7. Update

Actors involved: 1. Visitor 2. Paid User 3. Author 4. Administrator

CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBUTES	OPERATIONS
Paid User	User id, name, password, email id, phone no, security questions	download(), login(), search(), register()
Administrator	Name, password, email id, admin id	update(), record()
Author	User id, password, publisher id, book name	publish()
Book	Book id, book name, author, price	update(), add()
Visitor	User name, email id	search book(), read book()
Download	User id, book id, amount	Search download()

ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modelling Language, activity diagrams can be used to describe the business and operational step- by-step workflows of components in a system. An activity diagram shows the overall flow of control. An activity is shown as a rounded box containing the name of the operation.

DOCUMENTATION OF ACTIVITY DIAGRAM:

The activity diagram shows the activity of the process here first login is done when the user is valid then the welcome page appears. Here fork is used where two transaction line may be got search book and online reading. Search book can be used to search book and online reading can allow user to learn online and when any of these two process is selected a join is used where download occurs, in this download of book is done then finally cost of book is paid online.

SEQUENCE DIAGRAM:

A sequence diagram represents the sequence and interactions of a given USE-CASE or scenario. Sequence diagrams can capture most of the information about the system. Most object to object interactions and operations are considered events and events include signals, inputs, decisions, interrupts, transitions and actions to or from users or external devices.

An event also is considered to be any action by an object that sends information. The event line represents a message sent from one object to another, in which the “form” object is requesting an operation be performed by the “to” object. The “to” object performs the operation using a method that the class contains.

It is also represented by the order in which things occur and how the objects in the system send message to one another.

DOCUMENTATION OF SEQUENCE DIAGRAM:

VISITORS:

1) This first diagram is the representation of a visitor visiting the e-book website.
2) First the visitor seeks for registration and enters all the necessary details and then clicks register.
3) Now that the visitor is registered he can search for books.
4) The system checks for the availability of the books, if it is available then the book is displayed to the visitor.

PAID USERS:

1) The second diagram shows the sequence of the paid user .
2) The paid user seeks for the login. After login the user searches for the book, if the book is available then the system will display the book to the user.
3) If the user wants to buy the book, the user has to click the download link , then the user will be redirected to the payment page.
4) If the payment is completed then the download will start.

COLLABORATION DIAGRAM:

A collaboration diagram, also called a communication diagram or interaction diagram. A sophisticated modelling tool can easily convert a collaboration diagram into a sequence diagram and the vice versa. A collaboration diagram resembles a flowchart that portrays the roles, functionality and behaviour of individual objects as well as the overall operation of the system in real time.

The two collaboration diagram one for Paid user and another for visitor are given below.

STATE-CHART DIAGRAM:

The purpose of state chart diagram is to understand the algorithm involved in performing a method. It is also called as state diagram. A state is represented as a round box, which may contain one or more compartments. An initial state is represented as small dot. A final state is represented as circle surrounding a small dot.

DOCUMENTATION OF STATE-CHART DIAGRAM:

The diagrams show first login to the system and view the books and search for required book is done and then required book is downloaded and amount paid in online. Finally logout from the system.

COMPONENT DIAGRAM:

Component diagrams are used to visualize the organization and relationships among components in a system.

DOCUMENTATION OF COMPONENT DIAGRAM:

The component diagram's main purpose is to show the structural relationships between the components of a system. It is represented by boxed figure. Dependencies are represented by communication association. The main component in this component diagram is E-Book management systems. And Register, Login, Search book, Payment and Download are the components comes under the main component.

DEPLOYMENT DIAGRAM:

A deployment diagram in the unified modelling language serves to model the physical deployment of artifacts on deployment targets. Deployment diagrams show "the allocation of artifacts to nodes according to the Deployments defined between them. It is represented by 3dimentional box. Dependencies are represented by communication association.

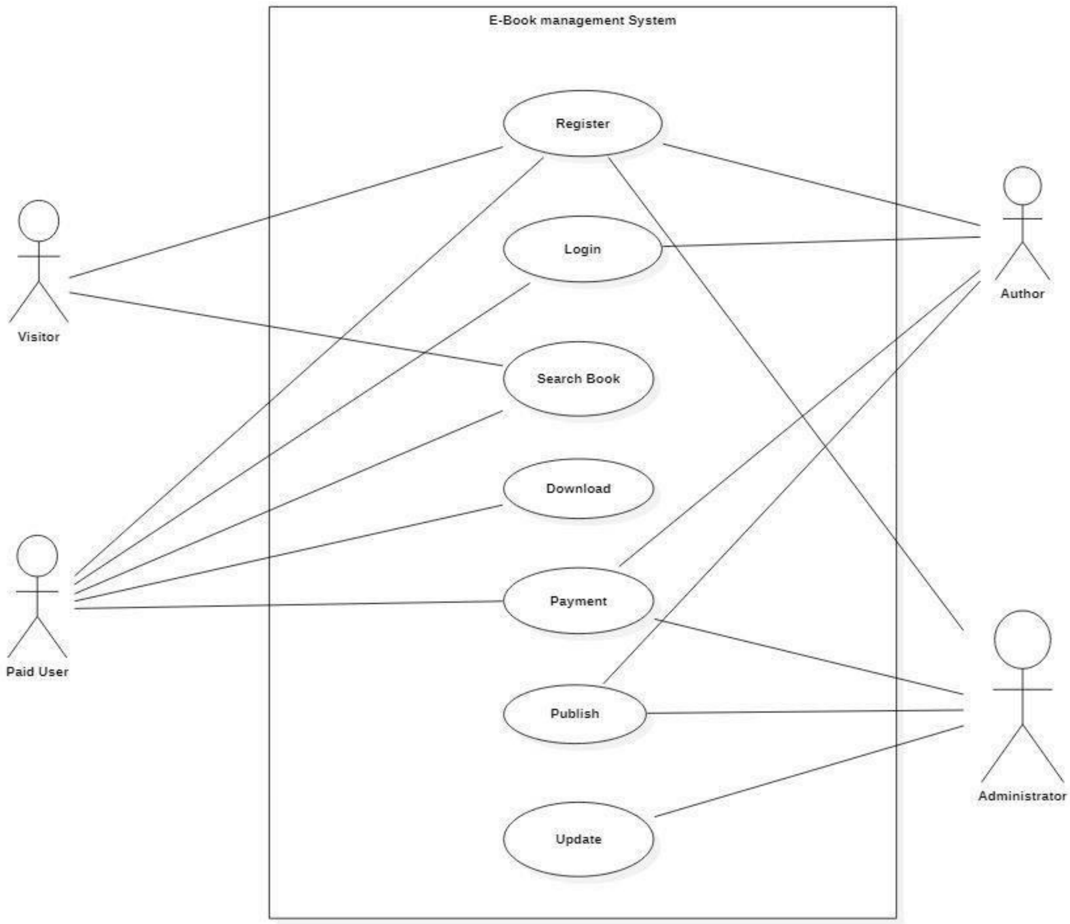
Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.

DOCUMENTATION OF DEPLOYMENT DIAGRAM

The processor in this deployment diagram is the E-Book management system which is the main part and the devices are the Database server, Web server for Clients which are the some of the main activities performed in the system.

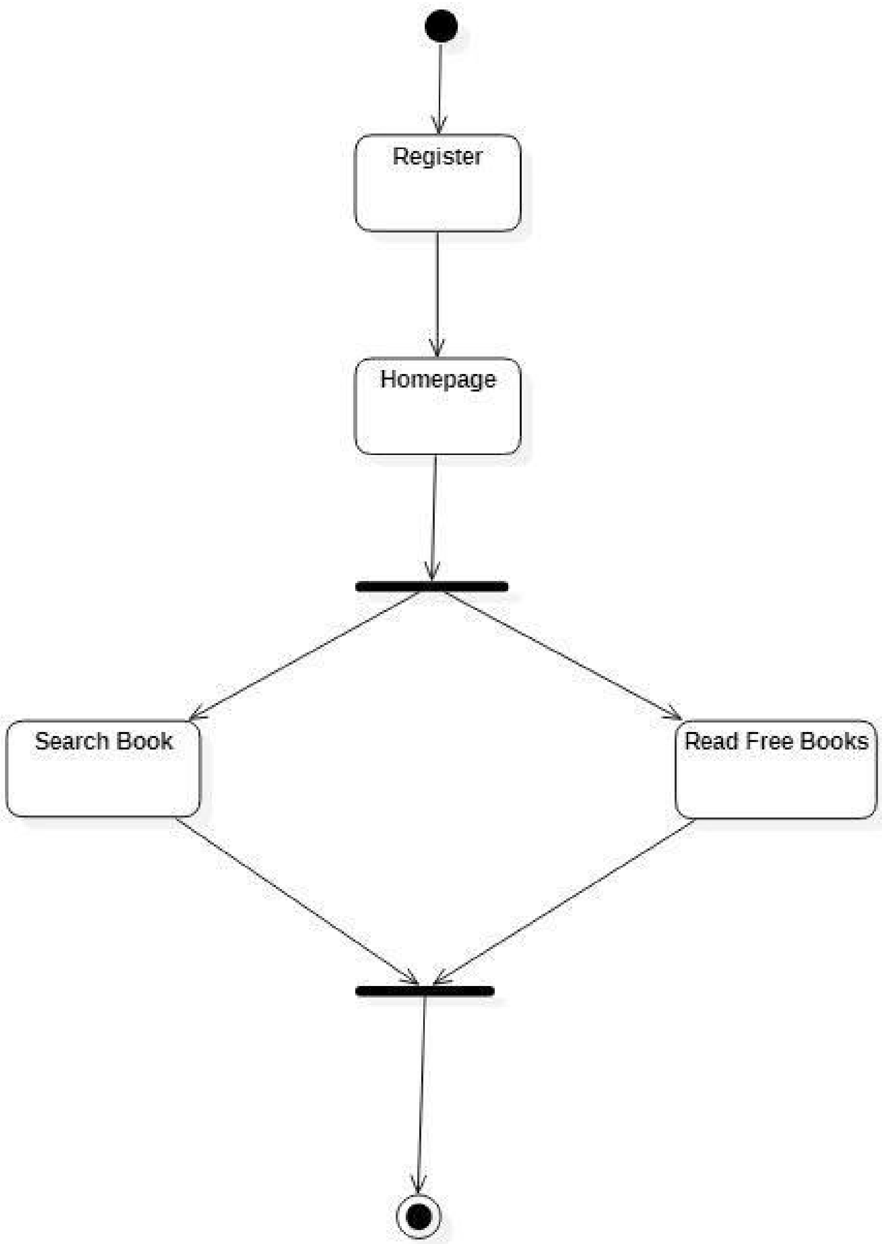
DIAGRAMS

USE CASE DIAGRAM:

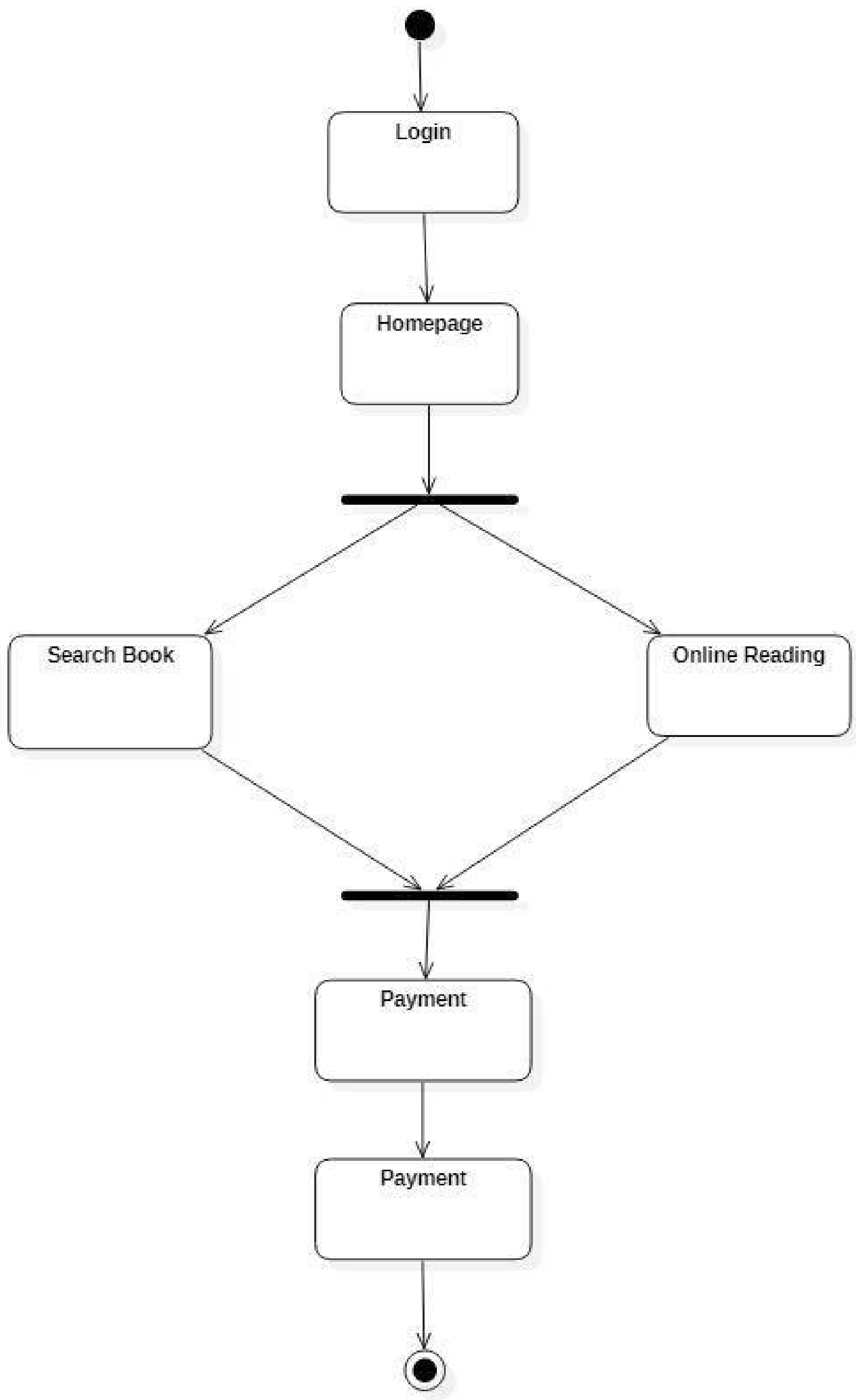


ACTIVITY DIAGRAM:

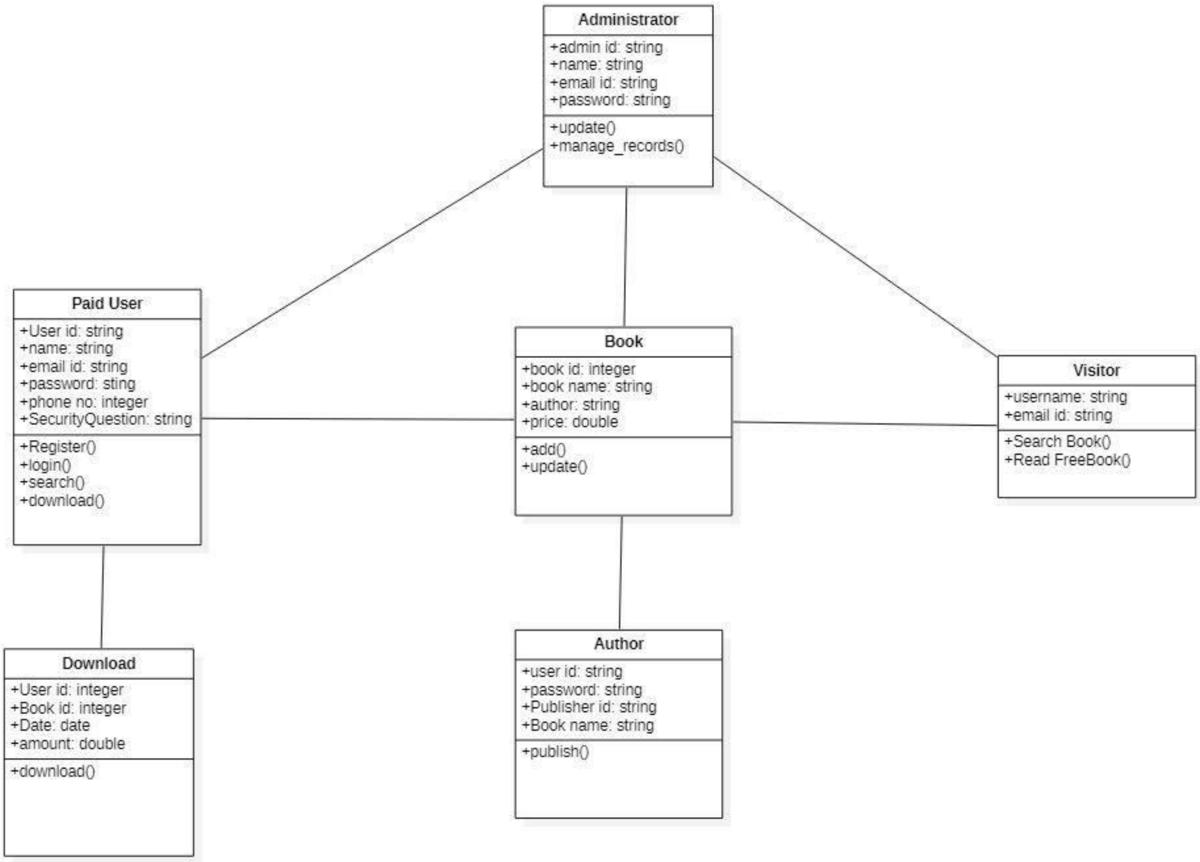
1) VISITORS ACTIVITY:



2) PAID MEMBER ACTIVITY:

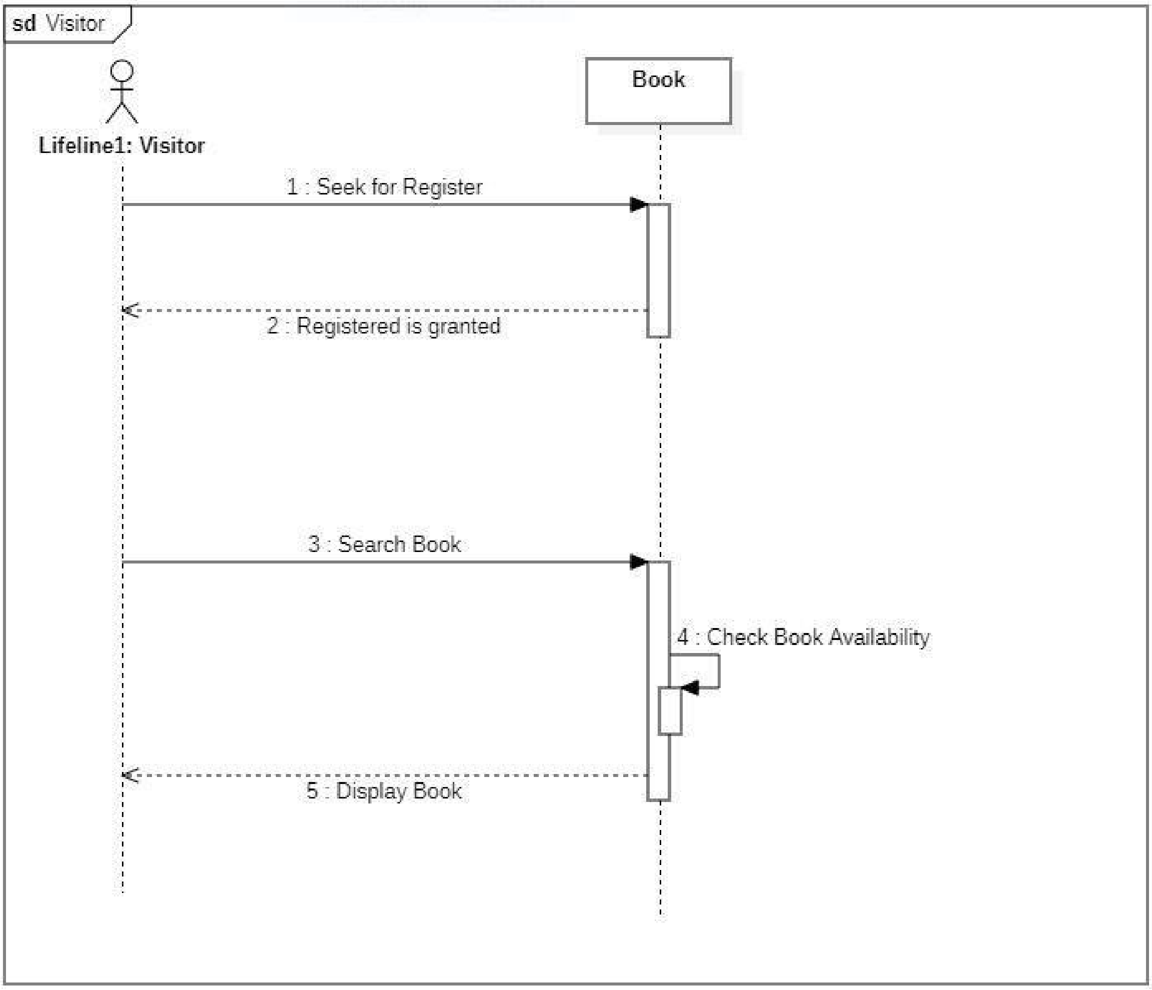


CLASS DIAGRAM:

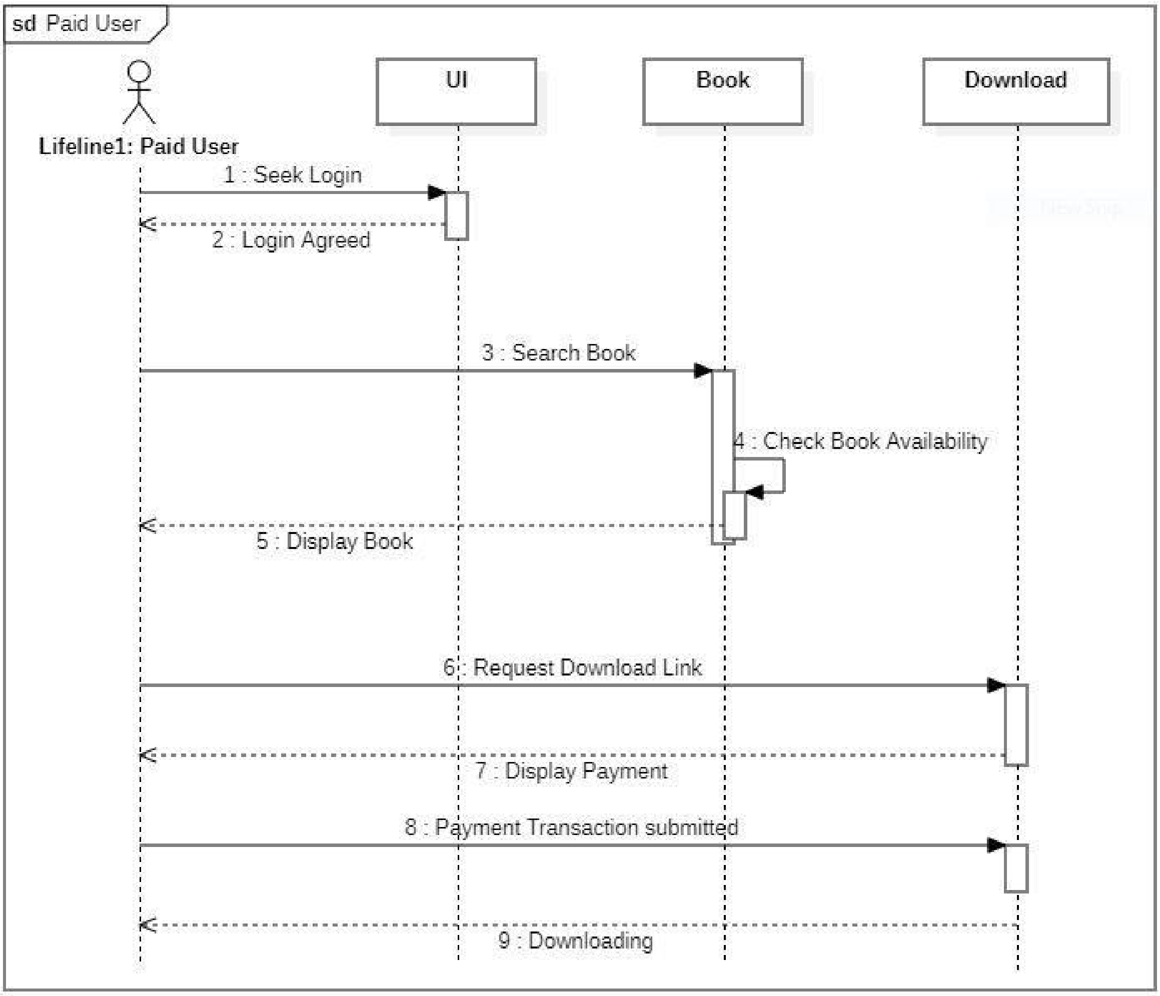


SEQUENCE DIAGRAM:

1) VISITORS:



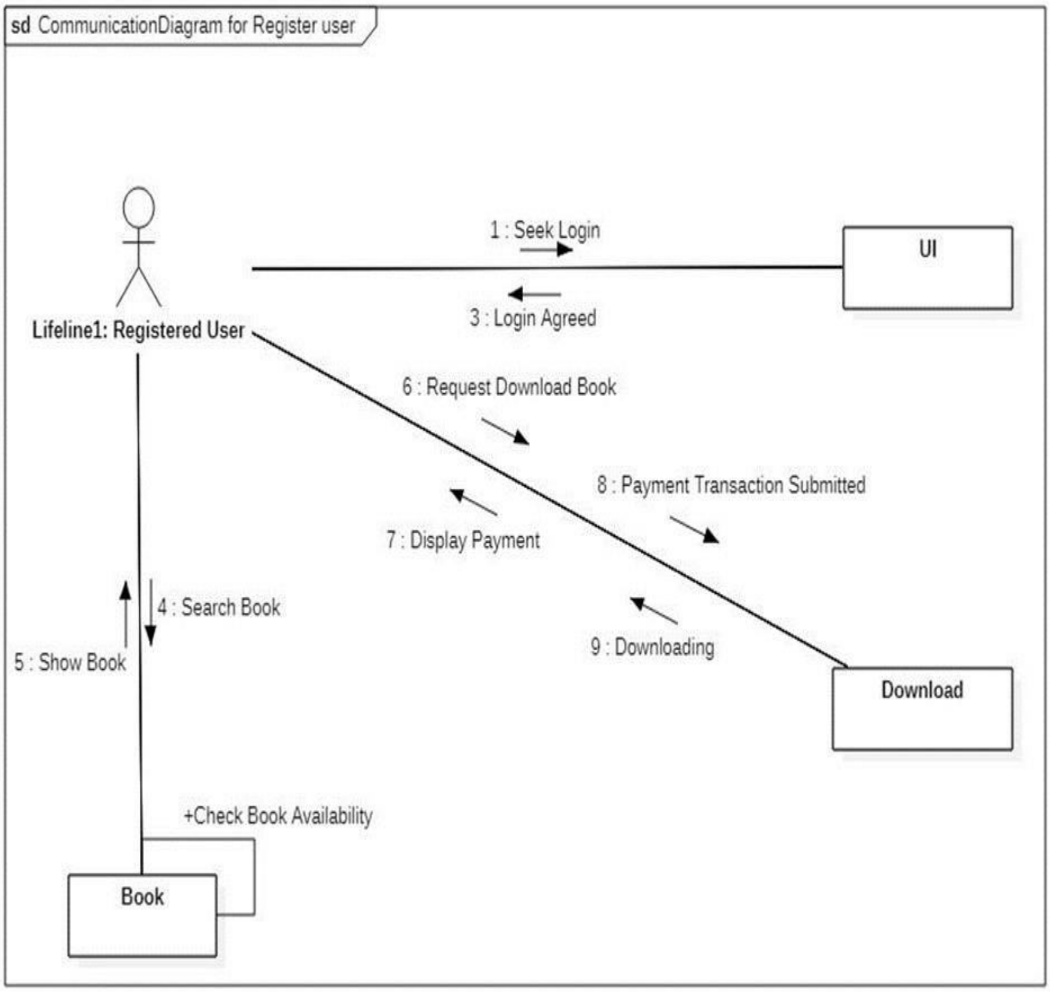
2) PAID MEMBERS:



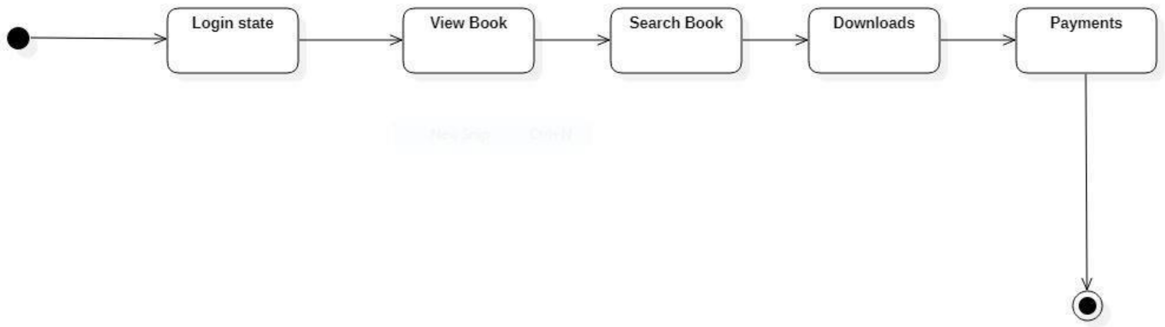
COLLABARATION DIAGRAM:

1) VISITORS:

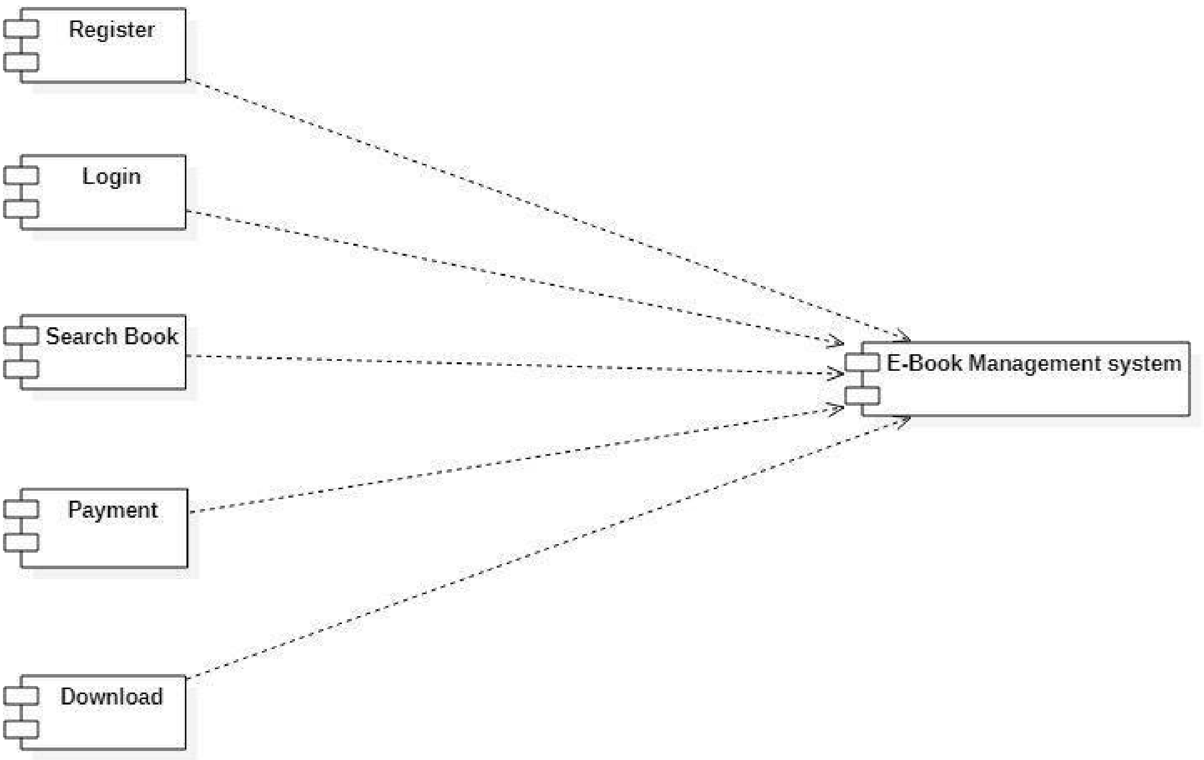
2) PAID MEMBERS:



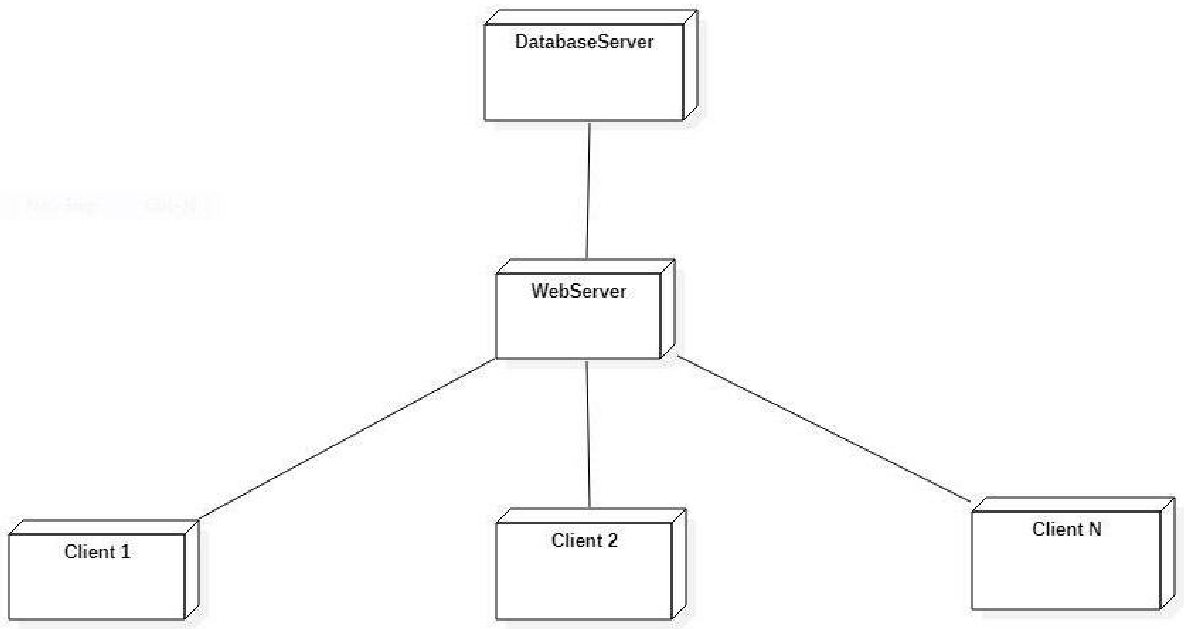
STATE CHART DIAGRAM:



COMPONENT DIAGRAM:



DEPLOYMENT DIAGRAM:



LESSON: 10
ONLINE GAS BOOKING SYSTEM

AIM:

To develop a project for ONLINE GAS BOOKING SYSTEM by using Star UML.

PROJECT DESCRIPTION:

- The objective of this project is to create the system where the customer can easily book their LPG gas cylinder through online system and agency can track the record of its customer and the delivery of the cylinder.

  The system will help the customers by providing a simple user interactive interface for booking the gas through online which will save their time and money.

- It also gives the agencies ease by helping them make the booking process faster and easier to maintain.

- There are various steps to book a gas like issuing an entry book, to travel agency from that to go to the delivery centre, our system makes this whole process at one place. It gives every user a simple and secure system by authorizing the user before entering the system.

· This is helpful to the agency to get all the desired data through so many simple steps without going through manual records.

· The system will display the user the number of the gas they booked through online with detail description as there should be limited time after which new gas can be booked.

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USE CASE DIAGRAM:

UML provides use case diagram notation to illustrate the names of use case and author relationship between them.

The gas booking use cases in this system are:

1. New connection	2. Add booking	3. Delivery bookings	4. Login/logout
5. Update profile	6. Generate OTP	7. Refill booking	8. Payment

Actors Involved: 1. Agency 2. Customer

CLASS DIAGRAM:

A class diagram is drawn as rectangle box with three compartments or components separated by horizontal lines. The top compartment holds the class name and middle compartment holds the attribute and bottom compartment holds list of operation.

CLASSES	ATTRIBUTES	METHODS
Agency	user id, user name, user email, user dob, user address	adduser() edituser() deleteuser() searchuser()
Customer	customer id, customer name, mobile no, email id, address, username, password	add()edit() delete() select()
Permission	Permission id, permission role, permission title, permission module, permission description	add permission() edit permission() delete permission()
Connection	Connection id, connection name, connection type, description, customer id	add() edit() delete() search()
Booking	Booking id, description, type, date	add() edit() search()
Payment	Payment id, amount, date, customer id	add()

ACTIVITY DIAGRAM:

The activity diagram shows the activity of the process. An activity diagram is a variation or a special case of a state machine in which the states or activity representing the performance of operation and transition are triggered by the completion of operation. The purpose is to provide view of close and what is going on inside a use case.

DOCUMENTATION OF ACTIVITY DIAGRAM:

This activity diagram describes the behaviour of the system.

- First state is login where the customer login to the Gas Booking System.
- Second state it will check the user level and permission and will provide the OTP.
- The next state it will access the gas agency and manage the booking.
- The customer will process the payment and get the acknowledgement
- After completion customer should logout from the gas booking system.

SEQUENCE DIAGRAM:

A sequence diagram shows an interaction arranged in time sequence. It shows object participating in interaction by their lifeline by the message they exchange arranged in time sequence. Vertical dimension represent time and horizontal dimension represent object.

DOCUMENTATION OF SEQUENCE DIAGRAM:

This sequence diagram describes the sequence of steps to show

- Customer are used to login the gas booking agency database then it verifies the username and password of the customer.
- If the password and username are correct then customer are used to login the gas management and book their refill.

COLLOBORATION DIAGRAM:

Communication diagram that illustrates that object interact on a graph or network format in which object can be placed in a diagram. In collaboration diagram the object can be placed in anywhere on the diagram. The collaboration comes from sequence diagram.

DOCUMENTATION OF COLLOBORATION DIAGRAM:

This collaboration diagram is to show how the customer login and register in the gas booking database system. Here the sequence is numbered according to the flow of execution.

This collaboration diagram is to show the selection process of the customer booking the gas. The flow of execution of this selection process is represented using the number.

STATE-CHART DIAGRAM:

The state chart diagram in UML diagram defines the states available in the system. It has various states in the system. It has one starting state and one ending state. State chart diagram is a curved rectangle box diagram. The various states in online gas booking system are registration, enter details, gas booking, payment, generate receipt and logout.

DOCUMENTATION OF STATE-CHART DIAGRAM:

This state diagram describes the behaviour of the system.

- First state is Register where the customer register and login to the gas booking database.
- The next state is where the customer enter the details and book the gas
- The next state is where the gas booking agency accepts the customer request and provide to gas booking payment.
- Then the database generate the receipt and will provided to the customer.

COMPONENT DIAGRAM:

The component diagram's main purpose is to show the structural relationships between the components of a system. It is represented by boxed figure. Dependencies are represented by communication association. The main component in this component diagram gas booking system. Gas, connection, booking and payments are the components comes under the main component.

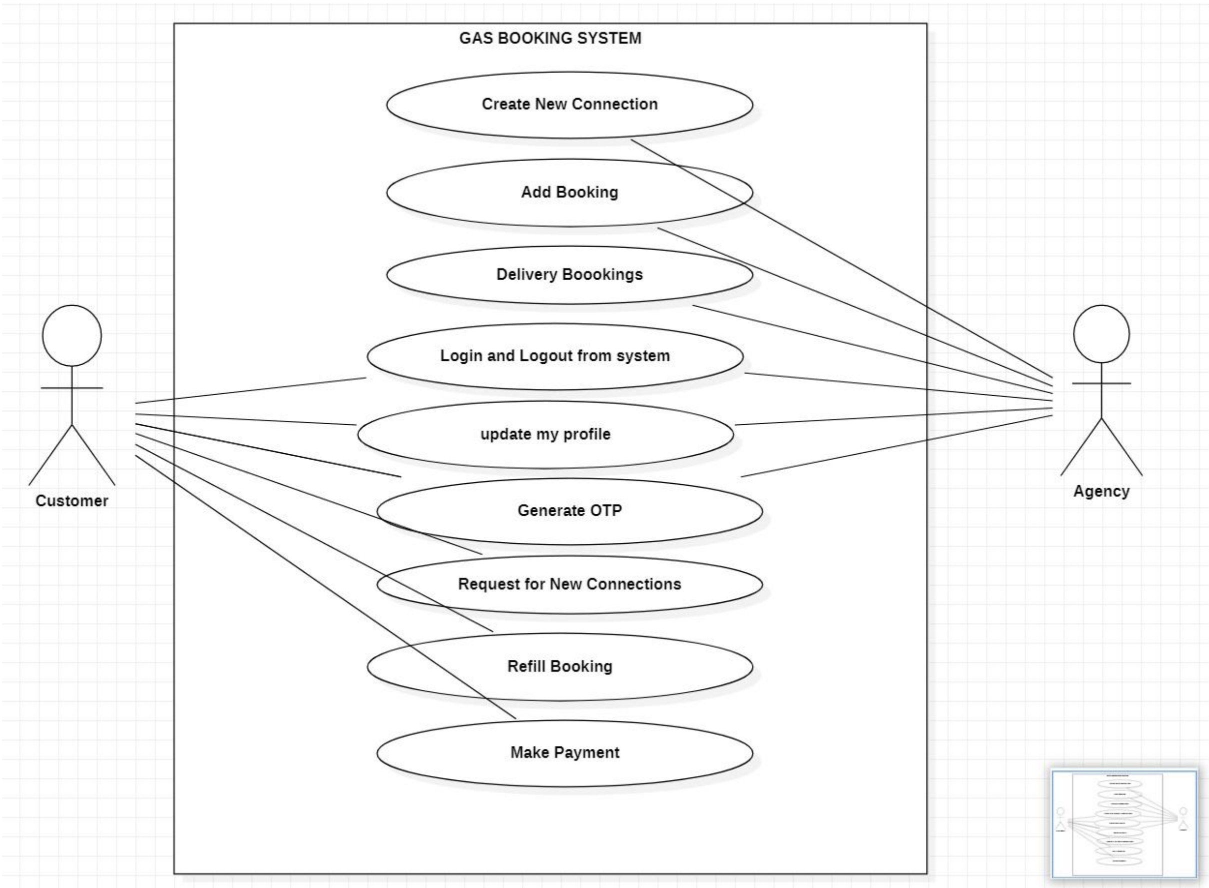
DEPLOYMENT DIAGRAM:

It is a graph of nodes connected by communication association. A deployment diagram in the unified modelling serves to model the physical deployment of artefacts on deployment targets.

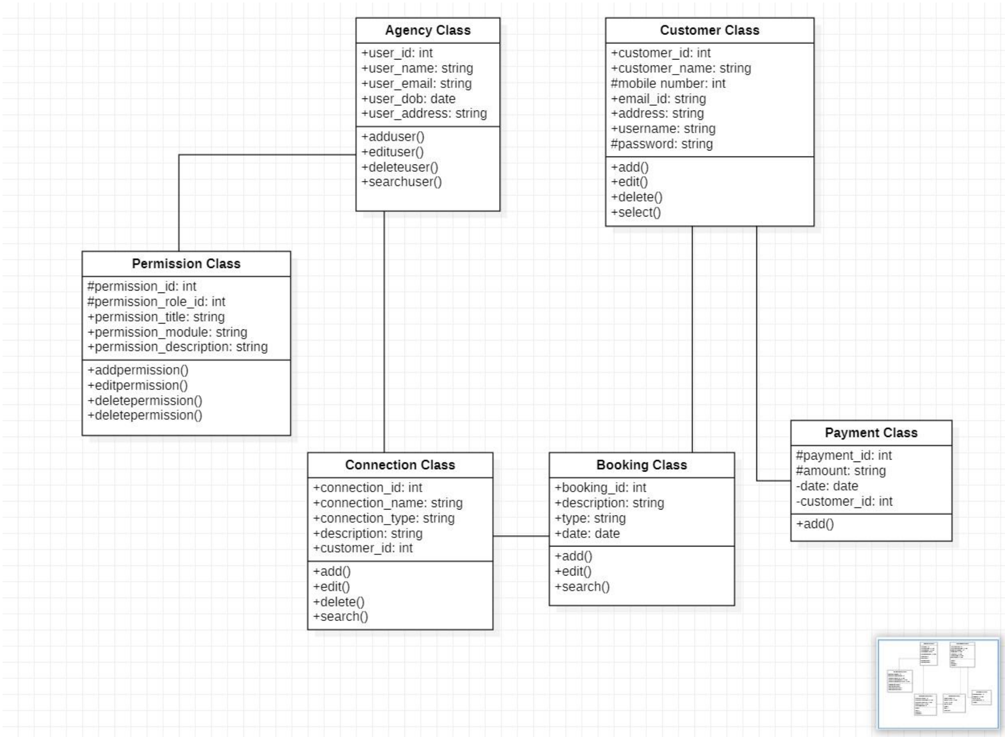
DOCUMENTATION OF DEPLOYMENT DIAGRAM:

The processor in this deployment diagram is the Gas Booking system which is the main part and the devices are the Gas Agency Web Server, Booking Management Server and Database Payload, Unique Pin Generator which conforms the booking confirmation and provides the booking status are the some of the main activities performed in the system.

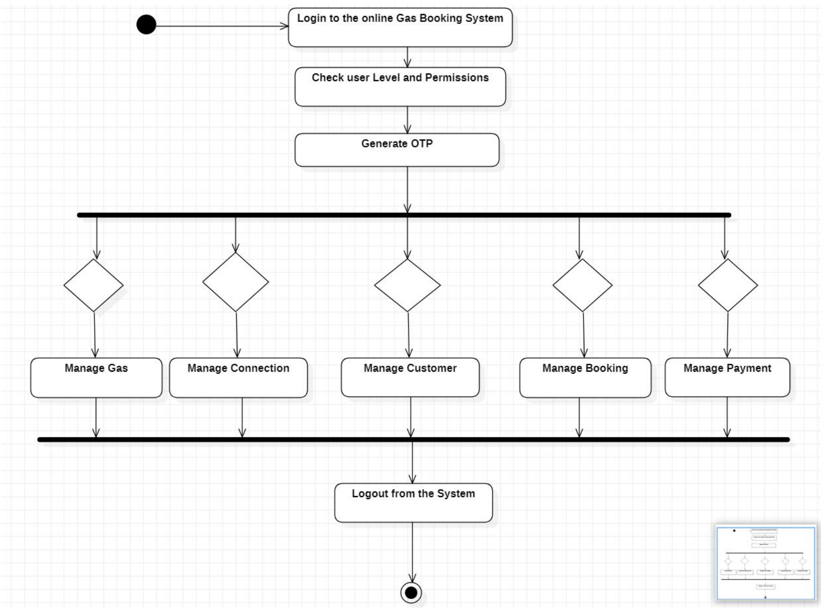
USE CASE DIAGRAM:



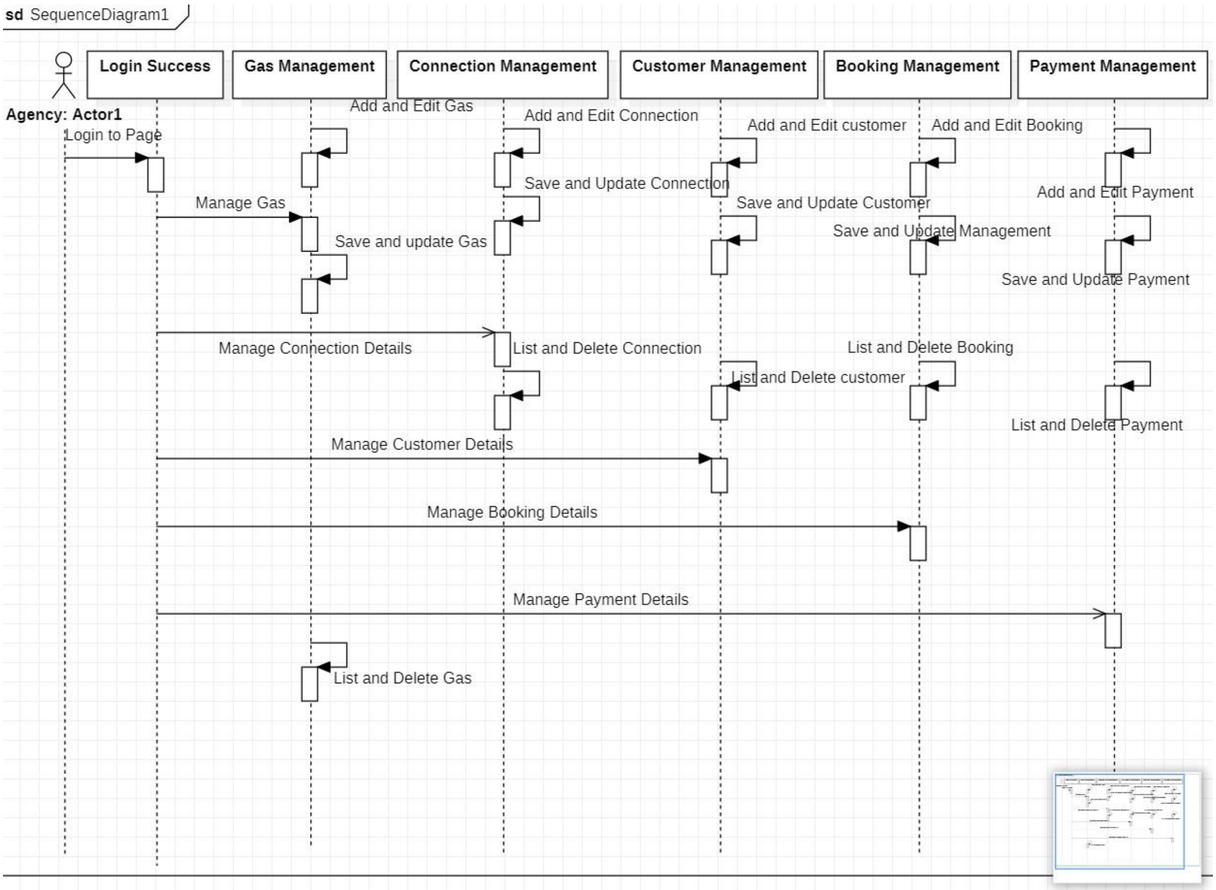
CLASS DIAGRAM:



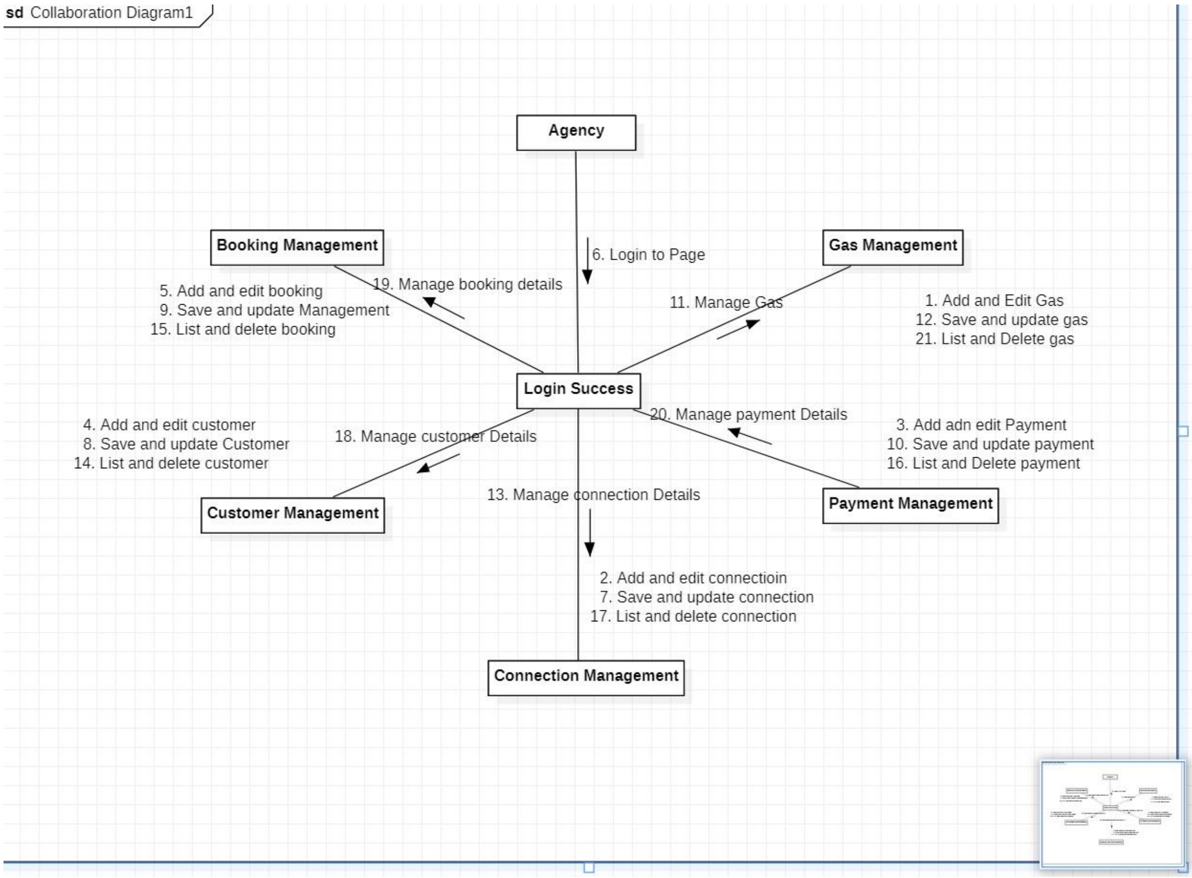
ACTIVITY DIAGRAM:



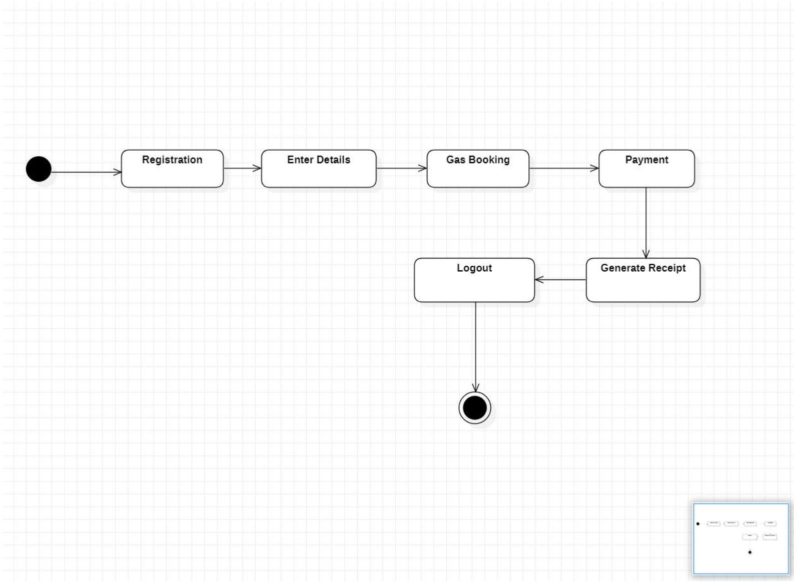
SEQUENCE DIAGRAM:



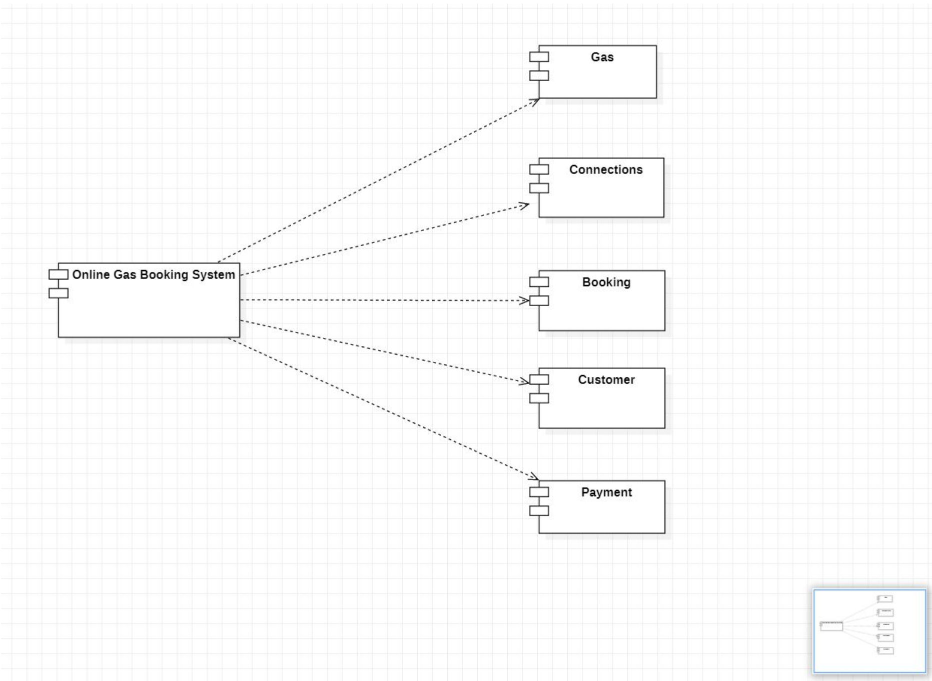
COLLABORATION DIAGRAM:



STATE-CHART DIAGRAM:



COMPONENT DIAGRAM:



DEPLOYMENT DIAGRAM: